

Routing densities and representation contraction for Thorp sweeps

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Abstract

For $N = 2^d$ cards, we prove that an absolute number of coordinate sweeps of the Thorp shuffle brings the full permutation law to total-variation distance tending to zero from uniform. The mixing time therefore has optimal order $\Theta(\log N)$ in physical shuffles.

1 Introduction

An absolute number of coordinate sweeps suffices to mix the full Thorp permutation. Since a sweep consists of d physical shuffles on $N = 2^d$ cards, this gives the optimal order $\Theta(d)$ for the mixing time. A single sweep already sends each card to a uniform position, but the joint positions retain dependencies because all cards use the same random switches. We bound these dependencies through the density of ordered partial permutations, then use representation multiplicities to pass from partial information to the full deck.

Write $N = 2^d$ with $d \geq 1$. The positions are the binary words in $\{0, 1\}^d$. A coordinate layer independently swaps or leaves unchanged the cards on every edge in one coordinate direction, with probability $1/2$ each. A sweep visits all d directions in order. In the rotating coordinate frame this is exactly d physical Thorp shuffles; the precise physical map is given in Section 2. Let T_N be the average action of one sweep. Its adjoint is the average action of a sweep in the reverse order. Thus $K_N = T_N^* T_N$ is the positive operator represented by a sweep followed by its reflection, using fresh independent switches.

For $0 \leq k \leq N$, let $\mathcal{X}_{N,k}$ be the set of ordered lists of k distinct positions, let $(N)_k = N!/(N-k)!$, and let $u_{N,k}$ be its uniform probability measure. For $x, y \in \mathcal{X}_{N,k}$ define $R_x(y) = (N)_k \Pr_{K_N}(x \mapsto y)$. Thus R_x is a density relative to a uniform injection. For a partition $\lambda \vdash N$, write D_λ for its irreducible dimension and $T_N(\lambda)$ for the sweep's average matrix.

Theorem 1.1 (Routing density and representation contraction). *There are absolute constants $\eta, c, C > 0$ such that, with $a = 1/1000$,*

$$\log \mathbb{E}_{y \sim u_{N,k}} R_x(y)^{1+a} \leq Ck(k/N)^\eta \quad (1 \leq k \leq N),$$

uniformly over every starting injection x . For every sufficiently large dyadic N and every nontrivial irreducible representation,

$$T_N(\lambda) = 0 \quad \text{or} \quad \|T_N(\lambda)\|_{\text{op}} \leq D_\lambda^{-c}.$$

Consequently an absolute number of forward sweeps has worst-start total-variation distance from uniform tending to zero as $d \rightarrow \infty$.

The density estimate also measures how much information about the starting tuple remains. Let P_x be its endpoint law under K_N , and let $H(P_x)$ be its Shannon entropy, with natural logarithms. Jensen’s inequality under P_x gives

$$\log(N)_k - H(P_x) = \text{KL}(P_x \| u_{N,k}) \leq a^{-1} \log \mathbb{E}_{u_{N,k}} R_x^{1+a} \leq (C/a)k(k/N)^\eta.$$

Thus the entropy deficit per tracked card tends to zero when $k/N \rightarrow 0$. At full occupancy the deficit is $O(N)$ after the reflected pair of sweeps, compared with the initial deficit $\log N!$.

The same saving has a representation-theoretic use. A shape whose first row is long occurs many times in a moderately larger injection module. A density cutoff has bounded Hilbert–Schmidt norm on that module, and these repeated copies force its norm on the shape to be small. Larger dimensions need a different use of the network, described below. All traces in the paper are unnormalized; in particular $\|A\|_{\text{HS}}^2 = \text{Tr}(A^*A)$.

Let q_d denote the law of one physical shuffle, let U_{S_N} be the uniform law on S_N , and use $\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_g |\mu(g) - \nu(g)|$. Define $t_{\text{mix}}(d)$ as the least number of physical shuffles at which this distance is at most $1/4$; by translation the distance is independent of the starting permutation. The lower obstruction counts random bits. Each physical shuffle uses $N/2$ bits, so after t shuffles at most $2^{tN/2}$ permutations can occur. The exact bound

$$\|q_d^{*t} - U_{S_N}\|_{\text{TV}} \geq 1 - 2^{tN/2}/N!$$

implies a lower bound $2d - O(1)$ at distance $1/4$. Together with Theorem 1.1, this gives the optimal order $\Theta(d)$ in physical time. The theorem does not specify a smallest mixing constant or a cutoff profile.

1.1 Network ancestry and previous mixing bounds

Thorp introduced the shuffle in his study of shuffling and Faro [30]. For power-of-two decks, Morris’s 2005 preprint proved the first polynomial bound, $O(d^{44})$, using evolving sets and a chameleon process [15]. Montenegro and Tetali’s spectral-profile and evolving-set analysis gave $O(d^{29})$ [14, Theorems 6.14–6.15]. Morris then used entropy contraction to prove $O((\log N)^4)$ for every even deck size [16], and sharpened the contraction over a sweep to obtain $O(d^3)$ when $N = 2^d$ [17, Lemma 7 and Theorem 8]. These bounds concern the full permutation. Our argument bounds the joint routing densities and converts them into matrix contraction on each irreducible representation.

The reflected butterfly has the topology of the classical Beneš network [2]. Beneš’s rearrangeability result concerns which permutations can be routed. Morris already studied the fair-switch sweep–reflection law under the name *zigzag shuffle* [15, Section 4]. The expanded experiment has $2d$ layers. Its two central layers average the same subgroup, so their product has the law of one such average and gives the usual $2d - 1$ -layer endpoint description. We retain both layers when counting internal routing choices.

Gelman and Ta-Shma analyzed this random network through its two-child recursion [11, Section 4, Proposition 1]. Their Theorem 4 and Lemma 5 give total-variation error at most $k(k-1)/(2N)$ for the image of an unordered k -subset. Our estimate concerns ordered injections and a higher density moment, including dense tuples. The alternating-color calculation below is the routing structure behind the recursion; the extra work is to control its cycle counts under the actual random switch law.

Partial-permutation questions also arise in cryptography. Morris, Rogaway and Stegers bounded the joint positions of prescribed cards in their analysis of small-domain encryption [18, 19]. Czumaj and Vöcking proved that, for every fixed $0 < c < 1$, the joint positions of any prescribed $\lfloor cN \rfloor$ cards mix in $O((\log N)^2)$ physical layers by a non-Markovian coupling [5]. Their full-permutation application pads the network with empty cards. Czumaj later obtained

logarithmic-depth partial-permutation bounds for other switching networks satisfying an expansion condition [4, Theorems 3.4–3.6]. The fixed tracked fraction, padding, and choice of network distinguish these results from mixing the unchanged full-deck Thorp chain.

The passage from Fourier estimates to total variation follows the finite-group method of Diaconis and Shahshahani [6]. For a conjugacy-invariant law, the Fourier matrices are scalar and can be estimated through character ratios. A directed coordinate sweep gives matrix products of projections, so its singular values and its nonnormal powers must both be controlled. The inverse-dimension summation used in the final step is a special case of Liebeck–Shalev [13, Proposition 2.5 and Theorem 2.6]; we include an elementary proof of the estimates needed here. Sharp versions allowing the exponent to decrease with N are now known [29, Proposition 1.8]. Recent character bounds also yield cutoff and its profile, relative to the appropriate parity coset, for nontrivial conjugacy-class walks with a positive fraction of fixed points [20, 28]. Those scalar estimates do not provide the matrix contraction required for a coordinate sweep.

Young branching, hook lengths, and Schur orthogonality connect the ordered routing densities to individual Fourier blocks [26, 27, 31]. Later we use the commuting position and label actions on injections [7, Section 3.2] and the associated transposition-content formula for coordinate deletion [1, Lemma 3.1 and Theorem 3.5]. These algebraic identities identify the harmonic spaces; the probabilistic work is to bound the loss of harmonic level when the network splits.

1.2 The first proof and the further estimates

Splitting the reflected network at its outer coordinate leaves two independent smaller networks. Prescribing the endpoints of some cards puts two matchings on those cards: one records common input switches, the other common output switches. Their union consists of alternating paths and cycles. Each component has two possible assignments to the children, and each cycle creates one extra factor of two in the route count. Section 3 gives the exact density identity and separates uniform auxiliary colorings from the actual switch law. This distinction identifies the probability measure under which a discarded-density tail must be bounded.

Section 4 completes the first proof. We mark several cycles in distinguished slots, expose their routes, and compare the number of choices of their starts with the probability that their last paths close. A contact bound controls the switches already known at those closures. Factorial moments then bound cycles at each height. Conditional estimates over alternating height bands sum the bounds without assuming independence between heights.

The density estimate and the multiplicity cutoff cover shapes with long first rows and those with very large dimensions. To handle the remaining dimensions, replace fixed-size central subnetworks by uniform permutations, allowing identity on a small specified set of blocks. The modified density moments have arbitrarily small cost per card. The original positive central operator is bounded by a sum of these modified operators and an exponentially small remainder. This completes the dimension-power estimate. Its Fourier consequence uses submultiplicativity and Schatten norms; a power of $T_N^* T_N$ is never identified with a power of the generally nonnormal T_N .

Section 5 gives an independent certificate argument. It uses reciprocal cycle lengths to retain explicit density costs for dense tracked sets, including the margins needed at densities $7/25$ and 1 . Section 6 instead proves an exponential cost per tracked card by encoding a permutation while omitting the bits forced by its closing routes. Its pointwise consequence applies to both $T_N^* T_N$ and $T_N T_N^*$, for every injection size, after a fixed number of compositions. Section 6.6 gives a separate certificate and entropy-window proof of that density bound.

The first density theorem already implies an e^{Ck} moment bound. The later density arguments give different mechanisms and lead to pointwise smoothing and lower-diagram

estimates. In particular, Section 7 proves a contact-cancellation bound uniform in the number of tracked cards. Summing out a tuple slot annihilates a representation at its first injection level; a weighted forest count turns this cancellation into sparse-level contraction.

The remaining methods retain representation data beyond dimension alone. Put $k = N - \lambda_1$, $\beta = (\lambda_2, \lambda_3, \dots)$, and $h_N(k) = k \log(eN/k)$, with $h_N(0) = 0$. The dimension D_β detects complexity below the first row that the level k alone cannot measure. A fixed trace moment controls the combined mass of singular values, and can therefore be used before the final amplification to the regular representation. Table 1 records these outputs; its constants are absolute and need not agree between rows.

Estimate	Information retained
Cycle encoding, Section 6	$H^u(x, y) \leq e^{Ck}/(N)_k$ for either positive orientation; $\ T_N(\lambda)\ ^2 \leq e^{Ck} D_\beta^{-a}$.
Contact cancellation, Section 7	$\ T_N(\lambda)\ ^2 \leq \min\{1, (Cdk/N)^{k/2}\}$, uniformly for $1 \leq k < N$.
Casimir split, Section 8	$\ K_N(\lambda)\ \leq e^{-ah_N(k)}$ and $D_\lambda \text{Tr } K_N(\lambda)^{65} \leq e^{Ch_N(k)}$.
Harmonic restriction, Section 9	$\ T_N(\lambda)\ \leq e^{-c\{h_N(k) + \log D_\beta\}}$ and $\text{Tr}_{\mathcal{X}_{N,k}} K_N^{p_0} \leq e^{Ch_N(k)}$.
Adaptive alphabets, Section 10	$\text{Tr}_{V_\lambda}(T_N T_N^*)^4 \leq e^{-c \log D_\lambda + Ch_N(k)}$, paired with level contraction.
Core cutoff, Section 11	Exponentially small overlap with product types of the midpoint subcube subgroups whose product dimension is at most $D_\lambda^{0.50003}$.

Table 1: Further density and representation estimates. All traces are unnormalized. Here u, p_0 are absolute integers and H is either $T_N^* T_N$ or $T_N T_N^*$.

The Casimir proof compares the transposition sum with its two child sums; the sum defect and squared imbalance require different matrix estimates. The harmonic proof bounds the loss caused by removing slots, then transfers cycle moments to either lower-diagram dimension or tuple trace. These arguments share the elementary tensor-swap estimate, but use different level recursions. The alphabet proof changes its tensor realization as descendant diagrams shrink and pays the resulting multiplicity costs across the tree. The core-cutoff proof chooses between a diagram bound and a bound using the actual trace at the current recursion frontier. Its cutoff is a projection onto small product dimensions. Each route supplies its own final passage to the full permutation law using the common Fourier preliminaries.

The companion papers use several of these interfaces while retaining their own arguments. The compatibility-entropy paper uses the shared-switch contact bound in Lemma 3.2 in a sparse path second-moment estimate [21, Lemma 3.1]. Prescribed-type signed tensor moments and one-sided projection-overlap bounds are developed separately in [25, Theorems 1.1 and 3.2]. The row–column geometry paper combines Theorem 6.1 and Proposition 6.3 with a coherent-overlap argument to obtain a dimension-weighted fixed moment [24, Theorem 10.1]. The random-subspace paper applies Corollary 6.2, in both positive orientations, in its Proposition 2.1; complex random-plane tests then give a regular trace at most $1 + N^{-10}$ [23, Theorem 1.1]. An independent conditional-marginal proof gives the explicit bound of $1600d$ physical shuffles for total-variation distance tending to zero [22, Theorem 1.1].

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2 The physical chain, Fourier norms and dimension estimates

This section fixes the common normalization. Every later passage from a sweep estimate to mixing uses physical time measured here. We use n for a generic dyadic block size, including the full-deck size N introduced above; the operators T_n and K_n are the corresponding sweep and reflected sweep on that block.

2.1 Binary positions and sweep operators

Number positions by $x = (x_1, \dots, x_d) \in \mathbb{F}_2^d$, most significant bit first. One shuffle sends

$$x \mapsto (x_2, \dots, x_d, x_1 + \xi_{x_2, \dots, x_d}),$$

where the 2^{d-1} bits ξ are independent and fair. Write R for the cyclic rotation and h_t for the pair switches at physical time t , so the time- t permutation is $Rh_t \cdots Rh_1$. Factoring out R^t conjugates the switches into the successive coordinate directions. This deterministic left multiplication preserves distance to uniform. At time d , $R^d = I$, and one sweep has exactly the law of d physical shuffles.

Permutations send each input position to its output. A product gh applies h first. For measures, $\mu * \nu$ denotes the law of gh for independent $g \sim \mu, h \sim \nu$. In a unitary representation ρ_λ , define

$$\widehat{\mu}(\lambda) = \sum_g \mu(g) \rho_\lambda(g), \quad \widehat{\mu * \nu} = \widehat{\mu} \widehat{\nu}.$$

A fair switch layer averages the subgroup generated by its disjoint transpositions, and hence is an orthogonal projection Π_i . With chronological coordinate order $1, \dots, d$, put

$$T_n = \Pi_d \cdots \Pi_1, \quad K_n = T_n^* T_n, \quad n = 2^d.$$

At the one-position recursive base, the empty product gives $T_1 = K_1 = I$.

Lemma 2.1 (Annihilated shapes). *The sign representation is annihilated by every nonempty fair layer. For $n \geq 2$, every irreducible of S_n indexed by a shape with more than $n/2$ rows is annihilated by a sweep.*

Proof. The sign average contains a fair transposition. For the second assertion, by Young's rule, the permutation module induced from the trivial representation of $(S_2)^{n/2}$ has only shapes dominating $(2^{n/2})$, hence at most $n/2$ rows. $\square$

Lemma 2.2 (Finite-size norm gap). *For every fixed dyadic $n \geq 2$, $\|T_n(\lambda)\|_{\text{op}} < 1$ on every nontrivial irreducible.*

Proof. Equality on a unit vector would force equality at every orthogonal projection in the product. The vector would be fixed by all coordinate-pair transpositions. The edges of the cube form a connected graph, and its edge transpositions generate S_n , contradicting nontrivial irreducibility. $\square$

Lemma 2.3 (Support obstruction). *For every integer $t \geq 0$,*

$$\|q_d^{*t} - U_{S_n}\|_{\text{TV}} \geq 1 - 2^{tn/2}/n!.$$

Consequently $t_{\text{mix}}(d) \geq \lceil (2/n) \log_2(3n!/4) \rceil = 2d - O(1)$.

Proof. At most $2^{tn/2}$ coin strings are available, so the law is supported on at most that many permutations. Comparing this support set with its uniform mass proves the first assertion. Distance at most $1/4$ requires support mass at least $3/4$, giving the exact integer lower bound. Stirling's formula gives the final expression. $\square$

2.2 Plancherel and powers of a nonnormal sweep

All matrix traces and Schatten norms are unnormalized. Thus $\|A\|_{S^p}^p = \text{Tr } |A|^p$, with $S^2 = \text{HS}$ and $S^\infty = \text{op}$. If D_λ is the irreducible dimension, finite-group orthogonality gives

$$|G| \sum_g |\mu(g)|^2 = \sum_\lambda D_\lambda \|\hat{\mu}(\lambda)\|_{\text{HS}}^2.$$

It applies to signed measures and subprobabilities as well as probabilities. Together with Cauchy–Schwarz, it gives the finite-group Fourier upper bound used by Diaconis and Shahshahani [6, Section 2, Lemma 2; Section 3, Lemma 14]. For a measure v of mass m ,

$$\|v - mU_G\|_1^2 \leq \sum_{\lambda \neq 1} D_\lambda \|\hat{v}(\lambda)\|_{\text{HS}}^2. \quad (1)$$

For a probability the left side is $4\|v - U_G\|_{\text{TV}}^2$. If $0 \leq v \leq \mu$ and μ is a probability, then

$$\|\mu - U_G\|_{\text{TV}} \leq 1 - m + \frac{1}{2}\|v - mU_G\|_1. \quad (2)$$

Both follow from Cauchy–Schwarz and the triangle inequality, respectively.

Lemma 2.4 (Safe use of sweep powers). *For integers $r \geq s \geq 1$ and any matrix T , setting $Q = T^*T$,*

$$\|T^r\|_{\text{HS}}^2 \leq \|Q\|_{\text{op}}^{r-s} \text{Tr } Q^s.$$

Proof. Schatten Hölder gives $\|T^s\|_{\text{HS}} \leq \|T\|_{S^{2s}}^s$, and multiplying the remaining $r - s$ factors costs at most $\|T\|_{\text{op}}^{r-s}$. Squaring proves the claim. No normality of T is used. $\square$

2.3 Injection multiplicities and inverse-degree sums

We use the standard indexing of complex irreducibles of S_n by partitions $\lambda \vdash n$, with $D_\lambda = f^\lambda$ equal to the number of standard tableaux [26, 27]. Write $k = n - \lambda_1$ and $\bar{\lambda} = (\lambda_2, \lambda_3, \dots)$. Young branching and Frobenius reciprocity show that the representation on ordered distinct r -tuples contains λ with multiplicity

$$f^{\lambda/(n-r)}.$$

At $r = k$ this is $f^{\bar{\lambda}}$, and λ does not occur on fewer slots. If $k \leq r \leq n - k$, the remaining first-row boxes are separated from the tail, giving

$$m_\lambda(r) = \binom{r}{k} f^{\bar{\lambda}}, \quad D_\lambda \leq \binom{n}{k} f^{\bar{\lambda}}. \quad (3)$$

The inequality follows by choosing the entries below the first row and forgetting cross-row tableau constraints. The same hook formula gives the useful quantitative refinement

$$D_\lambda \geq \binom{n}{k} f^{\bar{\lambda}} \exp\left(-\frac{k}{n - 2k + 1}\right) \quad (0 \leq k \leq n/2). \quad (4)$$

Indeed the hook inflation along the top row, relative to $(n - k)!$, is

$$\prod_{j \leq \lambda_2} \left(1 + \frac{\lambda'_j - 1}{n - k - j + 1} \right).$$

The denominators are at least $n - 2k + 1$ and $\sum_j (\lambda'_j - 1) = k$. Taking logarithms bounds this product by $\exp(k/(n - 2k + 1))$. The remaining hooks are exactly those of $\bar{\lambda}$. In particular for $k \leq .14n$ the loss is $\exp(O(k/n))$, which also supplies the weaker $\exp(O(k \log n/n + k/n))$ loss when that form is convenient.

The inverse-degree conclusions below are special cases of Liebeck–Shalev [13, Proposition 2.5 and Theorem 2.6]. We include an elementary proof of the estimates needed here, together with a lower bound for individual dimensions.

Lemma 2.5 (Degree sums). *Uniformly in n , $\sum_{\lambda \vdash n} D_\lambda^{-1}$ is bounded, and*

$$\sum_{\lambda \neq (n), (1^n)} D_\lambda^{-1} \rightarrow 0.$$

If $\ell = n - \max(\lambda_1, \lambda'_1)$, then

$$\log D_\lambda \geq (\log 2)\sqrt{\ell}/2, \quad \sup_n \sum_{\lambda \vdash n} D_\lambda^{-32} < \infty.$$

Proof. Transpose if necessary so that the first row has length $r = \max(\lambda_1, \lambda'_1)$. If $r \leq n/8$, the hook formula gives $D_\lambda \geq n!/(2r)^n$, exponential in n . If $n/8 < r \leq 3n/4$, retain the first row and $\lfloor r/4 \rfloor$ further boxes. Their tableaux extend to the whole diagram. A first row of length r and a tail of size $j \leq r$ have at least

$$\binom{r+j}{j} \frac{r-j+1}{r+1}$$

tableaux: ballot interleavings of the first row with any fixed standard order of the tail respect all column inequalities. This is exponential in n in the present range. There are $\exp(O(\sqrt{n}))$ partitions, so these ranges contribute $o(1)$ to the inverse-degree sum. If $r > 3n/4$, write $j = n - r$. The same ballot bound is at least a fixed multiple of $\binom{n}{j}$. There are at most $2p(j)$ possible shapes up to transpose, and $\binom{n}{j} \geq 4^j$ for $j < n/4$. Since $p(j) = \exp(O(\sqrt{j}))$ and each fixed positive j gives a divergent binomial coefficient, dominated convergence proves the first assertion.

For the square-root estimate, put $b = \lambda_2$ and $h = \lambda'_1$, so $b(h - 1) \geq \ell$. If $h - 1 \geq \sqrt{\ell}$, a hook with row and column length h has at least 2^{h-1} tableaux. Otherwise $b \geq \sqrt{\ell}$ and the two-row rectangle of width b has the Catalan number of tableaux, at least 2^{b-1} . Subdiagram tableaux extend; handling $\ell = 0$ separately proves the displayed bound. Finally $p(j) \leq e^{3\sqrt{j}}$ follows by evaluating the partition generating product at $e^{-1/\sqrt{j}}$. Summing $2e^{3\sqrt{\ell}}e^{-16(\log 2)\sqrt{\ell}}$ proves the inverse-32 bound. $\square$

3 The routing density and repeated irreducible copies

A sweep followed by its reflection has a recursive description: an outer layer of fair switches, two independent half-sized networks, and another outer layer. The architecture is the Beneš network [2]. Its two-child probabilistic recursion was studied in [11, Section 4, Proposition 1]; we derive below the partial-permutation density identity needed here. The goal is to express a partial-permutation density in terms of the cycles created by its routing constraints. We then prove a truncation lemma that converts a density bound into contraction on repeated copies of an irreducible representation.

3.1 Alternating colors and multiplicity

Fix an ordered list of k input positions and a candidate list of k distinct output positions. At one node, make a graph on its tracked paths, joining two when they use the same outer input switch or the same outer output switch. Its degree is at most two. Input and output edges alternate; parallel edges form a cycle. If s paths are present, a is the number of edges, and C is the number of cycles, the number of proper assignments of paths to the two children is

$$2^{s-a+C}.$$

Indeed a path component has one more vertex than edge, whereas a cycle has equally many. Each component admits two child assignments, obtained by exchanging the colors. Thus each cycle supplies one factor of two beyond 2^{s-a} , as illustrated in Figure 1.

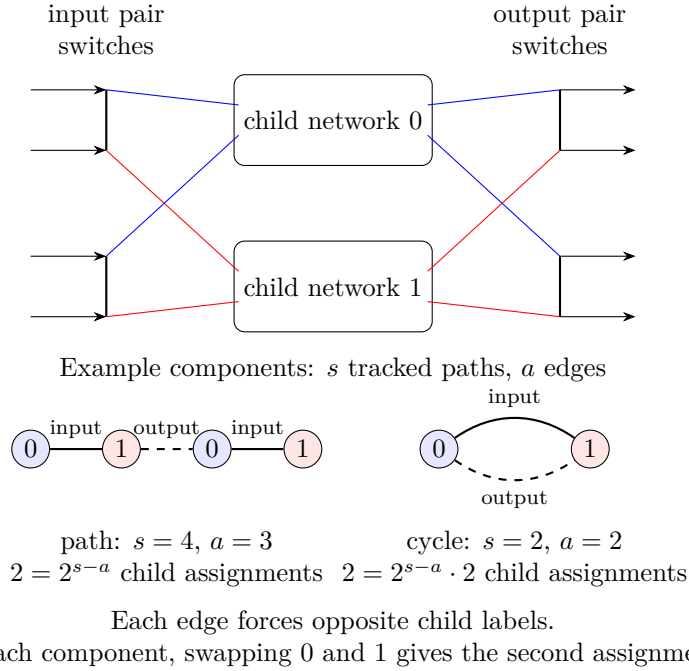


Figure 1: A recursive network node (top) and two components of a routing-constraint graph (bottom). Each graph vertex represents a tracked path, and each input or output edge forces its endpoints into opposite children. Both displayed components have two child assignments, but the cycle contributes an extra factor 2 relative to 2^{s-a} . The four paths in the network panel are schematic; each child contains half the positions.

Each assignment prescribes $2s - a$ independent outer coins, hence costs $2^{-(2s-a)}$. Their combined factor is 2^{-s+C} . Recursing through the nodes and choosing proper colorings uniformly gives a density relative to the uniform injection with excess exponent

$$X = \sum_{j=1}^d C_j + \log_2 \frac{(N)_k}{N^k}, \quad (5)$$

where C_j totals tracked cycles at height j . We distinguish the auxiliary coloring law from the actual switch law. For fixed input and output injections x, y , let ω be a legal complete tree of child colorings. At a node v , write s_v, e_v, C_v for its path, edge and cycle counts, and put $a_v = 2^{s_v-e_v+C_v}$. The auxiliary weight and actual routing-event weight are respectively

$$q_{xy}(\omega) = \prod_v a_v^{-1}, \quad w_{xy}(\omega) = \prod_v 2^{-(2s_v-e_v)}, \quad \frac{w_{xy}(\omega)}{q_{xy}(\omega)} = N^{-k} 2^{\sum_v C_v}.$$

The last identity uses $\sum_v s_v = kd$. At the leaves the routing is determined. Summing unused switch coins contributes one, so $w_{xy}(\omega)$ is exactly the probability of the corresponding endpoint-and-coloring event. Thus

$$K_N(x, y) = \sum_{\omega} w_{xy}(\omega) = \frac{1}{(N)_k} \mathbb{E}_{q_{xy}} 2^X.$$

For the certificate alternative in Section 5, one may truncate the routing event itself. Define $A(x, y) = \sum_{\omega} w_{xy}(\omega) \mathbf{1}_{\{X \leq L\}}$ and $E = K_N - A$. Then $A(x, y) \leq 2^L / (N)_k$, while

$$\sum_y E(x, y) = \sum_{y, \omega} w_{xy}(\omega) \mathbf{1}_{\{X > L\}} = \mathbb{P}_{\text{actual}, x}(X > L). \quad (6)$$

Indeed (y, ω) partitions the actual switch experiment. The tail estimates below therefore bound discarded mass under the actual network law, without identifying that law with the auxiliary uniform-coloring law. The factor $(N)_k / N^k$ is essential: the endpoints form an injection, not a tuple sampled with replacement.

Lemma 3.1 (Truncation and repeated irreducible copies). *Let a finite group act transitively on a set of size M , and let Q be the symmetric Markov kernel induced on its permutation module by a probability measure on that group. Suppose $Q = A + E$ entrywise, where $A, E \geq 0$, every entry of A is at most B/M , and the row and column sums of E are at most p . If an irreducible representation occurs with multiplicity $m > 0$, then*

$$\|Q(\lambda)\|_{\text{op}} \leq \sqrt{B/m} + p.$$

It is not necessary that A preserve the irreducible subspaces.

Proof. Schur's test gives $\|E\|_{\text{op}} \leq p$. Since the row sums of A are at most one, $\|A\|_{\text{HS}}^2 \leq (B/M) \sum_{x, y} A(x, y) \leq B$. Choose one maximizing unit vector for $Q(\lambda)$ in each of its m orthogonal equivalent copies. They are orthonormal and each image under A has norm at least $\|Q(\lambda)\|_{\text{op}} - p$. Their squared image norms sum to at most $\|A\|_{\text{HS}}^2$. This proves the claim. $\square$

The first proof in Section 4 instead truncates the endpoint density: it retains an entry when $(N)_k K_N(x, y) \leq z$. Its density moment bounds the discarded row mass by Markov's inequality, and symmetry bounds the column mass. Both cutoffs use Lemma 3.1; the letter Q in that abstract lemma denotes any kernel satisfying its hypotheses.

For the routing-event cutoff above, reversal preserves the cutoff event, so a uniform discarded-row bound also bounds columns. Lemma 3.1 therefore gives $2^{L/2} / \sqrt{m} + p$.

3.2 How often can selected paths meet?

The next deterministic fact will control the information revealed about a closing route by other paths. Its logarithm is to base two.

Lemma 3.2 (Butterfly encounters). *Any h distinct routed paths through a butterfly that updates each of d binary coordinates once have at most $h \log_2 h / 2$ shared switches in total, with $0 \log_2 0 = 0$. Their contact graph has maximum degree at most d . Counting both path endpoints at each shared switch gives at most $h \log_2 h$ contacts.*

Proof. Split by the coordinate edited last. If the two classes have sizes $a, h - a$, that layer contributes at most $\min(a, h - a)$ meetings. Apply induction to the earlier subnetworks and use

$$2 \min(a, h - a) \leq h \log_2 h - a \log_2 a - (h - a) \log_2 (h - a).$$

This is the binary entropy inequality $H_2(x) \geq 2 \min(x, 1-x)$, where $H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x)$. Empty terms are zero. Summing over the recursive splits proves the assertion. Distinct paths meet at most once: after a common switch they differ in a coordinate that is never edited again. A path meets at most one other path at each of its d switches, proving the degree assertion. $\square$

4 The factorial-moment proof

We now prove Theorem 1.1. The cycle estimate controls sparse injections and very large representation dimensions. The remaining dimensions require a second density estimate, obtained after replacing fixed-size middle blocks by uniform permutations. We prove both density assertions before making the representation split.

For the recursive description number the coordinates so that the palindrome uses layers $d, \dots, 1, 1, \dots, d$. A node of size 2^t varies in its low t coordinates. This reversal of coordinate names does not change its law up to conjugation. Thus $K_N = T_N^* T_N$, consistently with Section 1, and $\|K_N(\lambda)\| = \|T_N(\lambda)\|^2$.

4.1 Closing routes and factorial cycle moments

An alternating cycle in the network can be followed by going through one child and returning through the other. After one child map is fixed, this produces a permutation containing fresh butterfly layers between fixed bijections. The following lemma counts several cycles of such a permutation at once. Its distinguished slots let us sum over all choices of cycle lengths without losing a factorial.

Lemma 4.1 (Factorial moments with distinguished cycle slots). *On a finite set take a permutation given by independent butterflies of block size $n = 2^r$ in parallel, composed before and after with arbitrary fixed permutations (we allow identifications of the sets at each stage). Fix a subset of size at most k , and count cycles lying entirely inside it, with F_j the number of length j . For nonnegative integers a_j (finitely supported) and $A = \sum a_j$,*

$$\mathbb{E} \prod_j (F_j)_{a_j} \leq (k/\sqrt{n})^A \prod_j j^{-a_j/2}. \quad (7)$$

Here $(F)_a = F(F-1) \cdots (F-a+1)$ and $(F)_0 = 1$.

Proof. Order the sought cycles by nondecreasing length, with slots distinguished, and choose a starting vertex for each (at most k^A choices). Following a cycle involves querying successive butterfly paths, with the last one required to close the cycle. Demand that these are the specified lengths with no repetitions before closure, are disjoint cycles, and are within the subset; otherwise call this failure. Each closing path has a specified input and output by the time it needs to be queried; conditional on the previous path exposures its probability is zero or $2^u/n$, where u counts switches on that route already exposed (a route through a single butterfly to a given compatible endpoint is unique). If $\mathcal{G}$ is the exploration just before a closing query, its required input and output are $\mathcal{G}$ -measurable, and

$$\mathbb{E}[\mathbf{1}_{\{\text{closure}\}} n 2^{-u} \mid \mathcal{G}] \leq 1,$$

with value zero for an incompatible route. Ordinary nonclosing queries insert no weight. Iterating these conditional identities, while requiring all the distinctness and length conditions, gives

$$\mathbb{E}[\mathbf{1}_{\{\text{success}\}} n^A 2^{-U}] \leq 1$$

for each choice of starts, with U the sum of those counts on success. This follows also by sequentially forcing the closing paths when possible, which changes measure by the indicated factors while other path steps cost no weight.

It remains to bound how much the previously exposed paths can help the closing routes. For a realized collection in specified slots, rotate its starts independently uniformly. U counts meetings of selected butterfly paths, namely when the closing path uses a switch met previously. Weight each selected path in a cycle of length j by $1/j$. For meetings between different cycles the expected charge per meeting over rotations is the smaller of the weights (or bounded by it), since cycles are processed by length. Within one cycle it is at most $2/j$. For any subset of s selected paths the total meetings between them in a butterfly layer sequence is at most $sr/2$. Within paths of one cycle of length j , Lemma 3.2 gives the stronger bound $j \log_2 j/2$. For several independent butterfly blocks, apply the lemma in each block and use $\sum_i j_i \log j_i \leq j \log j$ when $\sum_i j_i = j$. Now integrate the subset bound $sr/2$ over thresholds on the weights: this gives $Ar/2$ for the sum charging each meeting the minimum weight. The additional within-cycle charges give at most $\sum_j a_j \log_2(j)/2$. By Jensen, the sum over rotations ($\prod_j j^{a_j}$ of them) of 2^{-U} for each collection is thus at least

$$\prod_j j^{a_j} n^{-A/2} \prod_j j^{-a_j/2}.$$

To make the counting explicit, let $\mathfrak{C}(\omega)$ be the set of ordered collections of distinct cycles of the realized permutation, with a_j distinguished slots of length j . Its cardinality is $\prod_j (F_j(\omega))_{a_j}$. Each collection has exactly $\prod_j j^{a_j}$ choices of starts. Sum the weighted success inequality over all start choices, then interchange that sum and the expectation. The preceding lower bound for the total rotation weight gives

$$k^A \geq \mathbb{E} \sum_{\mathcal{C} \in \mathfrak{C}(\omega)} \sum_{\text{starts of } \mathcal{C}} n^A 2^{-U} \geq n^{A/2} \prod_j j^{a_j/2} \mathbb{E} \prod_j (F_j)_{a_j}.$$

Dividing proves (7). $\square$

For later use, if $q \geq 1$ and $J \geq 1$ is an integer, cycles longer than J contribute at most k/J . Expanding $q^{\sum_{j \leq J} F_j}$ in factorial moments and using $\sum_{j \leq J} j^{-1/2} \leq 2\sqrt{J}$ gives

$$\log \mathbb{E} q^{\sum_j F_j} \leq (k/J) \log q + 2qk\sqrt{J}/\sqrt{n}. \quad (8)$$

4.2 The two density estimates

Lemma 4.2 (Palindromic row-density moments). *Let $N = 2^d$, $1 \leq k \leq N$, and $a = 1/1000$. For a kernel Q on ordered k -injections define*

$$M(Q, k) = \sup_{x \in \mathcal{X}_{N,k}} \mathbb{E}_{y \sim u_{N,k}} [(N)_k Q(x, y)]^{1+a}.$$

There are absolute $C, \eta > 0$ with the following properties; below M denotes $M(Q, k)$ for the kernel in question.

- for K_N , $\log M \leq Ck(k/N)^\eta$, for some absolute $C < \infty$ and $\eta > 0$;
- consider the modified palindrome obtained by replacing the central subnetworks of size $S = 2^L$ by independent uniform permutations, except in at most b specified such blocks where they are replaced by identity. For every $\xi > 0$, there is L_0 such that for each fixed $L \geq L_0$ there are $\epsilon > 0$ and $d_0 \geq L$ for which $M(Q, N) \leq e^{\xi N}$ whenever $d \geq d_0$ and $bS/N \leq \epsilon$.

Proof. For $N = 1$, the only injection has one position and every kernel in the statement is identity, so $M = 1$. Assume henceforth that $d \geq 1$. To see how to estimate these integrals, expand the transition probability by splitting off the outermost layers of each block, recursively. At a given node of size 2^t , let h be the number of marked paths (the paths of the tuple) there. Given their endpoints at that node, write b_x, b_y for the numbers of occupied direction- t pairs at the two ends. Each term assigns each path to one of the two children, with opposite assignments for a pair coming from the same direction- t edge on either end. Its probability factor for the outer layers is $2^{-b_x - b_y}$. On the labels of paths the two matchings (allowing unmatched vertices) giving these constraints form alternating chains and even cycles, counting parallel matching edges as a cycle as well. If there are z cycles the number of assignments is at most $2^{b_x + b_y - h + z}$, by counting components. Each assignment fixes the child endpoints, and has as further factors the child probabilities. Apply

$$\left(\sum_i p_i\right)^{1+a} \leq (\#i)^a \sum_i p_i^{1+a}$$

at each expansion. On summing over terminal tuple images y , keeping one power of each term as the actual routing probability, the remaining factors at expanded nodes give $2^{a(-h+z)}$. Thus, for any initial tuple,

$$\mathbb{E}_{y \sim u_{N,k}} R_x(y)^{1+a} \leq [(N)_k / N^k]^a \mathbb{E} 2^a \sum_{t=1}^d Z_t, \quad (9)$$

where Z_t totals the cycles at level t of the actual routing (unused paths can be filled in via the actual switches). For the modified palindrome with $k = N$, stop after levels $d, \dots, L+1$. Instead the bound is

$$[N! (S/N)^N (S!)^{-N/S+b}]^a \mathbb{E} 2^a \sum_{t>L} Z_t. \quad (10)$$

Indeed all paths are then used and the extra central probability factors are uniform $(S!)^{-1}$ or identity.

Fresh layers at one height. Lemma 4.1 concerns fresh butterfly layers, whereas cycle counts at different network heights share coins. Label paths at the central cross section and read the two butterflies outward, calling them X and Y . Conditioning on X fixes the marked central labels.

Fix a height t . Let Ω_0, Ω_1 be the unions of the bit-zero and bit-one halves in direction t of all size- 2^t nodes. Reading the first $t-1$ layers of X gives bijections X_0, X_1 from these halves to a common set of switch columns. The input matching, viewed from Ω_0 to Ω_1 , is $I = X_1^{-1} X_0$. The corresponding output matching is $Y_1^{-1} Y_0$. Hence following an output edge and then an input edge gives the permutation

$$\pi_t = I^{-1} Y_1^{-1} Y_0 = X_0^{-1} X_1 Y_1^{-1} Y_0 \quad \text{on } \Omega_0. \quad (11)$$

Each full alternating cycle gives one cycle of π_t . The tracked count Z_t includes only those cycles whose vertices on both sides are marked. In particular it is bounded by the number of cycles of π_t contained in the marked subset of Ω_0 , a fixed set of size at most k under the conditioning on X . Switches at layer t change neither matching.

Now expose the first $s < t-1$ outward layers of Y , and, for this single-height estimate, the remaining maps on Ω_1 . The map Y_1 is then fixed. On Ω_0 , write $Y_0 = BU$, where U is the exposed map through height s and B consists of fresh parallel butterflies in coordinates $s+1, \dots, t-1$, each of size $n = 2^{t-1-s}$. Thus $\pi_t = (I^{-1} Y_1^{-1}) BU$ has exactly the form of Lemma 4.1. Its bound is uniform in the exposed maps, so it remains valid after averaging the extra exposure of Y_1 .

Summing heights with their conditioning.

Here and below group levels above some integer $H \geq 1$ into slabs

$$l < t \leq 2l, \quad l = H, 2H, 4H, \dots$$

cut off at d . Put $\mathcal{F}_s = \sigma(X, \text{the first } s \text{ outward layers of } Y)$ and $B_l = \sum_{l < t \leq \min(2l, d)} Z_t$. The band variable B_l is $\mathcal{F}_{2l}$ -measurable (with $\mathcal{F}_s = \mathcal{F}_d$ for $s > d$). Split the slabs into their two alternating families. In either family, the preceding slab ends at $l/2$, so its cycle count is $\mathcal{F}_{\lfloor l/2 \rfloor}$ -measurable.

Conditioned on that field, the single-level estimate above has a fresh butterfly of size $n \geq 2^{l/2}$. Hölder within the slab uses at most l factors. For the exponential moment with exponent $2a$, take $q = 2^{2al}$ and $J = \lceil 2^{l/16} \rceil$ in (8). Uniformly in the conditioning,

$$\log \mathbb{E}[q^{Z_t} \mid \mathcal{F}_{\lfloor l/2 \rfloor}] \leq Ck2^{-c_0 l}. \quad (12)$$

The two terms on the right of (8) decay exponentially because $2a < 1/16$. Thus, after Hölder,

$$\mathbb{E}[2^{2aB_l} \mid \mathcal{F}_{\lfloor l/2 \rfloor}] \leq \exp(Ck2^{-c_0 l}).$$

For example, if l_* is the last slab in one family and V is the sum of its earlier bands, the tower property gives

$$\mathbb{E}[2^{2a(V+B_{l_*})} \mid X] = \mathbb{E}\left[2^{2aV} \mathbb{E}[2^{2aB_{l_*}} \mid \mathcal{F}_{\lfloor l_*/2 \rfloor}] \mid X\right] \leq e^{Ck2^{-c_0 l_*}} \mathbb{E}[2^{2aV} \mid X].$$

Iterate backwards through each family and apply Cauchy–Schwarz to the two families. The geometric sum of their costs proves

$$\log \mathbb{E}[2^a \sum_{t \geq H} Z_t \mid X] \leq Ck2^{-c_1 H}. \quad (13)$$

The same reasoning holds with a replaced by $2a$: even the resulting single-level exponent $4a$ is smaller than $1/16$. We will use this explicit doubling when separating low and high levels.

Low heights and the sparse density bound. The high-height cost decays with its starting height. At low heights, sparse occupancy supplies the gain instead. For the low levels of (9), partition the central labels into size $R = 2^H$ blocks and let V_D count marked positions in such a block D . (Use $H \leq d$.) All low-level cycles are internal to such blocks and

$$\sum_{t \leq H} Z_t \leq H \sum_D V_D 1_{V_D \geq 2}.$$

For every set of central positions of size v the probability under X they are all marked is at most p^v , $p = k/N$. Indeed the upper product-of-marginals bound for every subset holds initially (trace the marked inputs towards the center). It is preserved under a single fair switch: for sets with one endpoint use averaging of the two bounds; with both endpoints use the old product, bounded by the product with both marginals replaced by their mean. After the layers of X each position has marginal p .

Write $w = 2^{aH}$. In each block use

$$w^{V_D 1_{V_D \geq 2}} \leq 1 + \sum_{i=2}^R \binom{V_D}{i} w^i.$$

Expanding the product and using the joint inclusion bounds yields

$$\log \mathbb{E} 2^{a \sum_{t \leq H} Z_t} \leq (N/R) \sum_{i=2}^R \binom{R}{i} (pw)^i \leq C(N/R)(Rpw)^2$$

when Rpw is bounded. Choose $H = \max(1, \lfloor \log_2(1/p)/4 \rfloor)$; then $H \leq d$. For the low-height estimate with exponent $2a$, replace w by 2^{2aH} . The product Rpw stays bounded, and the resulting logarithmic cost is at most

$$CkpR2^{4aH} \leq C'kp^{(3-4a)/4}.$$

The high-height estimate (13) also holds at exponent $2a$, with cost at most $Ck2^{-c_1H}$. Low heights still depend on Y , so use Cauchy–Schwarz between the two height ranges, then average over X . Their costs are at most Ckp^η for some absolute $\eta > 0$. The normalization in (9) is at most one, proving the first claim. If there are no heights above H , only the low-height bound is needed.

Uniform central blocks and the dense density bound. For (10), label paths at the input boundary of the central size- S blocks. Fix the input-side outer butterfly X , and include the uniform or identity central maps at the beginning of Y . Use the same slabs with $H = L$. We distinguish their first members, $l = L$ and $l = 2L$, from the later ones.

At a later slab, $l \geq 4L$, condition on X , the central maps, and the ordinary output layers through height $\lfloor l/2 \rfloor$. The previous same-parity slab is measurable in this field. All fresh coordinates used in (12) lie strictly above the central blocks, so that estimate applies unchanged with $k = N$.

A first slab has no preceding member in its parity family. We therefore need its moment only conditional on X , without fixing the central permutations. For a height t in this slab expose the bit-one map as in (11). In each good bit-zero central block, realize its uniform permutation as FU , where U is an independent uniform permutation and F is an independent fresh size- S butterfly. This has the original uniform law. Condition on U , but leave F fresh. In a bad block take U to be identity; the original central map there is identity. In a comparison experiment insert a fresh F in that block too, keeping all good-block maps and all later switches the same.

Here is why this insertion costs at most bS cycles. Let B be the union of the bad block inputs in the fresh-map coordinates. The original and comparison maps agree on every input outside B . After any fixed premap U , the affected domain is $U^{-1}B$, still of size at most bS ; a fixed postmap does not enlarge that domain. Every cycle of the original relative permutation disjoint from $U^{-1}B$ retains all its arrows and is therefore a cycle of the comparison permutation. At most bS disjoint cycles meet that domain. Thus, writing Z'_t for the full comparison cycle count,

$$Z_t \leq Z'_t + bS.$$

This comparison holds for every realization of the inserted switches; no cycle is selected for subsequent conditioning.

In the comparison, the central butterflies and the ordinary output layers together form fresh parallel butterflies in coordinates $1, \dots, t-1$, between fixed maps. Lemma 4.1 therefore bounds their cycle moment. With $q = 2^{2al}$, the same choice $J = \lceil 2^{l/16} \rceil$ as before gives, after averaging the auxiliary maps,

$$\log \mathbb{E}[q^{Z_t} \mid X] \leq CN2^{-c_0l} + bS \log q \quad (l = L \text{ or } 2L).$$

Hölder within this first slab costs $O(albS)$ in addition to its decaying term. Integrate the later slabs backwards in each parity family using their lower-height conditioning, finish with this first-slab bound, and apply Cauchy–Schwarz between the families. The result is

$$\log \mathbb{E}2^{a \sum_{t>L} Z_t} \leq CN2^{-c_1L} + CaLbS.$$

By Stirling, the logarithm of the normalization in square brackets in (10) is at most $O(\log N + N \log S/S) + b \log(S!)$. Given $\xi > 0$, first take L large enough that the terms $N2^{-c_1L}$ and

$N \log S/S$ have sufficiently small coefficients. With that L fixed, choose ϵ so that $bS/N \leq \epsilon$ controls both bad-block terms, and finally take d large enough to absorb $O(\log N)$. This proves the second claim in the stated order of choices. $\square$

4.3 Multiplicity amplification and the middle dimension range

We now use the branching multiplicity from (3): the ordered k -injection module contains $\lambda \vdash N$ with multiplicity $m = f^{\lambda/(N-k)}$, and for $k = N$ this is D_λ . The density bounds just proved will make truncation effective on these repeated copies.

For either symmetric kernel Q above, retain an entry when its density is at most z . Markov's inequality gives discarded row mass at most Mz^{-a} , and symmetry gives the same column bound. The retained entries are at most $z/(N)_k$. Applying Lemma 3.1 with $B = z$ therefore gives, on an irreducible occurring with multiplicity $m > 0$,

$$\|Q(\lambda)\| \leq Mz^{-a} + \sqrt{z/m}. \quad (14)$$

In particular if $\log M \leq (a/4) \log m$, take $z = \sqrt{m}$ to get at most $2m^{-a/4}$.

First consider $j = N - \lambda_1$ with $1 \leq j \leq \delta N$, where a sufficiently small absolute $\delta > 0$ will be used. Choose a fixed $u_0 > 0$ small enough that $\eta(1 - u_0) > u_0$ and $u_0 < 1$, and take $k = \lceil j(N/j)^{u_0} \rceil$. For small δ this gives $N - k \geq j$, so the skew tableau of shape $\lambda/(N - k)$ has its remaining first row independent of the entire tail shape $\rho = (\lambda_2, \dots)$. Consequently

$$m = \binom{k}{j} f^\rho, \quad D_\lambda \leq \binom{N}{j} f^\rho.$$

Thus $\log m \geq u_0 j \log(N/j) + \log f^\rho$, at least a fixed positive fraction of $\log D_\lambda$ (using the elementary binomial bounds). The first density bound proved above gives $\log M \leq C' j(j/N)^{\eta(1-u_0)-u_0}$, hence at most $(a/4) \log m$ by reducing δ . By (14), and m tending uniformly to infinity here as $d \rightarrow \infty$, we have $\|K_N(\lambda)\| \leq D_\lambda^{-c_2}$ for some constant $c_2 > 0$.

By Lemma 2.1, shapes with more than $N/2$ rows are annihilated by T_N and hence by K_N .

All remaining shapes other than the trivial one and those already treated have $j \geq \delta N$ and

$$\log D_\lambda \geq hN$$

with $h > 0$ fixed, for large N . Here are details. Both first row and column are then bounded by $(1 - \delta)N$ (take $\delta < 1/2$). If both are smaller than $N/10$, hook lengths are at most $N/5$ and the bound follows from Stirling. Otherwise transpose if necessary (which does not change dimension) so the row has length $s \geq N/10$. Retain only it and a subdiagram of the tail of size $v = \lfloor \min(\delta/2, 1/100)N \rfloor$, which can be obtained by removing corners. All tableaux on this diagram have extensions. Fill its first v positions in the top row first, then interleave the remaining row and a fixed tableau order on the tail; this gives $\binom{s}{v}$ possibilities, yielding the asserted exponential lower bound.

Take a fixed large A_0 so that when $\log D_\lambda \geq A_0 N$, the first density bound on all N positions gives $\log M \leq (a/4) \log D_\lambda$. Then (14) with $Q = K_N$ again gives $\|K_N(\lambda)\| \leq D_\lambda^{-c_3}$ for an absolute $c_3 > 0$, for large N .

The only remaining range is $hN \leq \log D_\lambda \leq A_0 N$. Here the full-deck moment cost CN is comparable with the log dimension and need not fit the truncation threshold. We use the modified central networks to reduce that cost before invoking multiplicity. Choose L fixed sufficiently large and then $\epsilon > 0$ sufficiently small so the second of the two assertions about the $(1 + a)$ -moment above gives $\log M \leq ahN/4$ whenever $b \leq \epsilon N/S$, for large N . We justify the comparison in positive semidefinite order. Let O be the product of the outer butterfly

layers, and let C be the parallel product of the central size- S palindrome operators. Thus, on the representation under consideration,

$$K_N = O^*CO.$$

By Lemma 2.2, each central positive contraction has norm at most a fixed $\theta < 1$ on every nontrivial irreducible of S_S ; increase θ to a number in $(0, 1)$ if necessary.

Restrict to $(S_S)^{N/S}$ and decompose into tensor-product irreducibles, with their multiplicity spaces. If a component is nontrivial on more than $\epsilon N/S$ blocks, the tensor product defining C has norm at most $\theta^{\epsilon N/S}$. This accounts for the first term below.

For $E \subseteq [N/S]$, let P_E act as identity in blocks in E and as the projection to invariants in the other blocks. On a component whose nontrivial factors lie in a set F with $|F| \leq \epsilon N/S$, P_F is identity. Consequently $\sum_{|E| \leq \epsilon N/S} P_E$ dominates identity on all these components and therefore dominates C there. Its sandwich

$$Q_E = O^*P_EO$$

is precisely the modified network with identity in E and uniform central permutations elsewhere. Since O is a contraction, these two component cases give

$$0 \leq K_N(\lambda) \leq \theta^{\epsilon N/S} I + \sum_{E: |E| \leq \epsilon N/S} Q_E(\lambda)$$

where Q_E is exactly the modified palindrome with identity in E and uniform central permutations elsewhere. Each summand by (14), $k = N$, is at most $2D_\lambda^{-a/4}$ in norm. The log number of summands is at most $(N/S)[- \epsilon \log \epsilon - (1 - \epsilon) \log(1 - \epsilon)]$ by the binomial theorem ($\epsilon < 1/2$), so we may require also that this be $\leq ahN/8$. Since $hN \leq \log D_\lambda \leq A_0N$ and all parameters now are fixed absolutely, this proves $\|K_N(\lambda)\| \leq D_\lambda^{-c_4}$ for some absolute $c_4 > 0$ and sufficiently large N .

We have consequently, on every nontrivial shape not killed, $\|T_N(\lambda)\| \leq D_\lambda^{-c_*}$ for an absolute $c_* > 0$. Plancherel on the finite group and Cauchy-Schwarz bound four times the squared total variation distance after r butterflies by

$$\sum_{\lambda \neq (N)} D_\lambda \|T_N(\lambda)^r\|_{\text{HS}}^2,$$

independently of initial permutation by translation. For r forward sweeps,

$$\|T_N(\lambda)^r\|_{\text{HS}}^2 \leq D_\lambda \|T_N(\lambda)^r\|_{\text{op}}^2 \leq D_\lambda \|T_N(\lambda)\|_{\text{op}}^{2r}.$$

Thus, keeping only shapes not killed, the sum is at most $\sum D_\lambda^{2-2rc_*}$.

This last bound tends to zero for sufficiently large fixed r . Medium or larger dimensions with $\log D_\lambda \geq hN$ have at most 2^N terms (bound partitions by compositions). For the small- j range, $D_\lambda \geq \binom{N-j}{j}$: again fill j positions of the first row first and then interleave tail and remaining row. There are at most 2^j shapes per j . Thus, with $s_* = 2rc_* - 2 > 0$, this range contributes at most

$$\sum_{1 \leq j \leq \delta N} 2^j (j/(N-j))^{s_* j}$$

tending to zero (choose s_* large for a geometric summable majorant and each fixed term vanishes). Increase r also so that $s_* h > \log 2$. Hence rd steps suffice for the required threshold for all sufficiently large d .

5 Certificate tails and numerical dense bounds

The factorial proof is complete. This independent route keeps information that its central-block comparison does not: explicit bounds on the exponential density cost at fixed tracked-card densities. We first count fixed certificates for many short cycles, then estimate reciprocal cycle lengths and pass to all representation shapes. Here $n = 2^d$ is the full-deck size and r is the number of tracked cards. Thus the exponent in (5) is used with $N = n$ and $k = r$. Logarithms in this section are to base two unless $\ln$ is written.

5.1 A fixed-certificate cycle tail

Proposition 5.1 (High-height cycle tail). *There are absolute $b, \delta > 0$ and an integer $J_0 \geq 1$ such that, uniformly in the tracked input injection of size r and for every integer $J \geq J_0$,*

$$\log_2 \mathbb{E} 2^{b \sum_{j \geq J} C_j} \leq O(r 2^{-bJ}), \quad (15)$$

$$\log_2 \mathbb{E} 2^{b \sum_j C_j} \leq O(r(r/n)^\delta). \quad (16)$$

The first bound also holds with expectation conditional on all input-side switches and all output switches of heights at most s , for every integer $0 \leq s \leq d$ with $2s < J$, uniformly in the fixed switch values and with the same absolute implied constant.

Proof. The estimates are immediate for $r = 0$, so assume $r \geq 1$. Let $\mathcal{F}_t$ be the sigma-field generated by all input-side switches and the output switches of heights at most t . The arrivals at every node are determined by the input switches. At a height- j node the two child maps determine its cycles; the outer output switch only exchanges the two positions in an output pair and does not change their pairing. Thus C_j is $\mathcal{F}_{j-1}$ -measurable.

Fix a nonnegative integer t with $2t < j$ and condition on $\mathcal{F}_t$. In each height- j node expose one child's remaining switches. In the other child the fixed low layers are followed by independent parallel output butterflies of height $j - 1 - t$. Since $j - 1 - t \geq \lfloor j/2 \rfloor$, at least $m = \lfloor j/2 \rfloor - 1$ final output layers remain fresh. This includes $t = (j - 1)/2$ when j is odd. The two child bijections identify alternating cycles with cycles of their relative permutation. Explore such a cycle through the fresh child and back through the known inverse of the other. Its closing target is specified, and the closing route through its remaining butterfly is unique.

Suppose $C_j \geq s \geq r 2^{-\eta j}$, where $\eta > 0$ is a sufficiently small absolute constant. At least $s/2$ of some s counted cycles have length at most $2r/s$. Among these choose $u = \max(1, \lfloor s/100 \rfloor)$ cycles uniformly without replacement and choose a uniform cyclic start on each. A fresh-layer switch of a closing path meets at most one path of another cycle. The conditional chance of selecting that other short cycle is at most $(u - 1)/(s/2 - 1)$, less than $1/49$ when $u > 1$, and zero when $u = 1$. Here we bound interference against all other selected cycles, including those later in the canonical exploration order; restricting to already explored cycles only decreases it. Thus mean cross-cycle interference on a closing path is less than $j/30$. Within its own cycle, Lemma 3.2 bounds the mean number of earlier encountered switches by $\log_2(2r/s) \leq 1 + \eta j$. For large j , the expected total previously exposed switches on all closing paths is less than $uj/4$.

Consequently each realized network in this event admits a certificate of starts and lengths with at most $uj/4$ previously exposed closing switches. This is an existence assertion before taking a union bound; we do not condition the coin law on the chosen favorable cycles. There are at most

$$\binom{r}{u} (2r/s)^u$$

fixed certificates. Order their starts canonically and explore each fixed certificate in that order. Define its success event to require that all prescribed cycles close and that at least $K = \lceil u(j/4 - 2) \rceil$ closing queries are fresh. The preceding low-interference existence argument supplies a certificate in this event for every network being counted. At every fresh closing query its required bit is already determined, so the query succeeds with conditional probability $1/2$, irrespective of intervening nonclosing queries. Abort if a closing bit fails or the certificate becomes invalid, and stop on the K th successful fresh closing query. The probability of reaching this stopping threshold without failure is at most $2^{-K} \leq 2^{-u(j/4-2)}$. This bounds the stated success event even when unsuccessful explorations make fewer than K fresh queries. The base-two logarithm of the certificate count is at most

$$u\{\log_2(200er/s) + \log_2(2r/s)\} \leq u(O(1) + 2\eta j).$$

Since $u \geq s/200$, reducing η gives $\mathbb{P}(C_j \geq s \mid \mathcal{F}_t) \leq 2^{-cjs}$, uniformly in the conditioning whenever $2t < j$.

Split the moment tail at $r2^{-\eta j}$. The deterministic part costs $O(jr2^{-\eta j})$ and the remaining geometric tail is summable at any sufficiently small fixed exponent. Decreasing that exponent gives absolute $b', A > 0$ and J_0 such that, for every $j \geq J_0$ and $2t < j$,

$$\log_2 \mathbb{E}[2^{b'jC_j} \mid \mathcal{F}_t] \leq Ar2^{-b'j}. \quad (17)$$

We retain the conditioning while summing over heights. Fix integers $J \geq J_0$ and $0 \leq s \leq d$ with $2s < J$. If $J > d$, the sum is empty. Otherwise put $L_q = 2^q J$ and, for each $q \geq 0$ with $L_q \leq d$, define

$$I_q = \{L_q, \dots, \min(2L_q - 1, d)\}, \quad B_q = \sum_{j \in I_q} C_j, \\ \ell_q = |I_q|, \quad t_q = \left\lfloor \frac{L_q - 1}{2} \right\rfloor.$$

These bands partition the integer heights from J to d . We have $s \leq t_q$, $2t_q < L_q$, and $\ell_q \leq L_q$. Choose $0 < b \leq b'/2$. Conditional Hölder within a band and $2b\ell_q \leq b'j$ for every $j \in I_q$ give

$$\begin{aligned} \log_2 \mathbb{E}[2^{2bB_q} \mid \mathcal{F}_{t_q}] &\leq \frac{1}{\ell_q} \sum_{j \in I_q} \log_2 \mathbb{E}[2^{2b\ell_q C_j} \mid \mathcal{F}_{t_q}] \\ &\leq \frac{Ar}{\ell_q} \sum_{j \in I_q} 2^{-b'j} \leq Ar2^{-b'L_q}. \end{aligned}$$

Split the band indices into their two parity classes, and let $V_q = \sum_{p < q, p \equiv q \pmod{2}} B_p$. For $q \geq 2$, every earlier band in the same class ends at or below $L_q/2 - 1 \leq t_q$. Since C_j is $\mathcal{F}_{j-1}$ -measurable, V_q is $\mathcal{F}_{t_q}$ -measurable; for $q = 0, 1$ it is zero. Also $\mathcal{F}_s \subseteq \mathcal{F}_{t_q}$ for every band. The conditional tower step is therefore

$$\begin{aligned} \mathbb{E}[2^{2b(V_q + B_q)} \mid \mathcal{F}_s] &= \mathbb{E}\left[2^{2bV_q} \mathbb{E}[2^{2bB_q} \mid \mathcal{F}_{t_q}] \mid \mathcal{F}_s\right] \\ &\leq 2^{Ar2^{-b'L_q}} \mathbb{E}[2^{2bV_q} \mid \mathcal{F}_s]. \end{aligned}$$

Integrating the highest band first and then descending gives, for $\varepsilon \in \{0, 1\}$,

$$\mathbb{E}\left[2^{2b \sum_{q \equiv \varepsilon \pmod{2}} B_q} \mid \mathcal{F}_s\right] \leq 2^{Ar \sum_{q \equiv \varepsilon \pmod{2}} 2^{-b'L_q}}.$$

All band sums here are finite. Conditional Cauchy–Schwarz between the two classes now yields

$$\log_2 \mathbb{E} \left[2^{b \sum_{j \geq J} C_j} \middle| \mathcal{F}_s \right] \leq \frac{Ar}{2} \sum_{q \geq 0} 2^{-b'2^q J} \leq A'r 2^{-bJ}$$

with an absolute A' . This proves the asserted conditional bound; averaging it proves (15). No independence between different heights or bands has been assumed.

For (16), put $\rho = r/n$ and take $J = \lfloor .1 \log_2(1/\rho) \rfloor$, or a fixed sufficiently large minimum. Below this height dominate cycles by shared input pairs. A law on occupancy bits is called strongly Rayleigh if its multiaffine generating polynomial has no zero when all variables have positive imaginary part. The deterministic occupancy law has a monomial generating polynomial. Fair transpositions preserve this property, and strongly Rayleigh laws are negatively associated: products of increasing functions on disjoint supports have expectation at most the product of their expectations [3, Theorems 4.9 and 4.20]. This is a fixed input-layer process, with no additional path conditioning. Indicators that both sites in a disjoint pair are marked are increasing functions on disjoint supports and inherit negative association. Their exponential moment is bounded by the product of the Bernoulli moments. A site's marginal before height j is at most $2^j \rho$, and the two pre-switch input groups are independent because that coordinate has not yet been used. The expected number of shared pairs is therefore at most $r 2^j \rho$. Hölder over the low heights introduces an exponent $O(J)$; decreasing the fixed b if necessary, its cost is a positive power of ρ times r .

The low-height dominating factor depends only on the input switches. Conditioning on $\mathcal{F}_0$, pull that factor outside the expectation and apply the uniform conditional high-height bound just proved with $s = 0$. Averaging over the input switches combines the two costs and gives (16) for some absolute $\delta > 0$. This final estimate uses the input randomness. If ρ is bounded below, the fixed low heights cost $O(r)$, which is consistent with (16) after changing the constant. $\square$

5.2 Dense tuple densities

The sparse moment estimate alone does not provide a useful exponent at fixed density. Here a bound on reciprocal cycle lengths gives a smaller explicit density exponent. Define

$$h_j = \max \left\{ \mathbb{E}(1/L) : L \in \mathbb{Z}_{\geq 1}, \mathbb{P}(L = l) \leq l/2^{j-1} \right\},$$

and

$$H(\rho) = \max \left\{ \frac{1}{2} \sum_{j \geq 1} h_j x_j : 0 \leq x_j \leq 1, \sum_{j \geq 1} 2^{-j} x_j \leq \rho \right\}.$$

Proposition 5.2 (Dense truncation). *For every $\rho_0 \in (0, 1]$ and $\epsilon > 0$, there are $c > 0$ and n_0 such that, for every dyadic $n \geq n_0$ and every integer r with $\rho = r/n \in [\rho_0, 1]$, the density exponent (5) exceeds*

$$r \left[H(\rho) + \frac{-\rho - (1 - \rho) \ln(1 - \rho)}{\rho \ln 2} + \epsilon \right]$$

with probability at most $\exp(-cr)$, uniformly in the input injection. The constants depend only on ρ_0, ϵ , and the expression is interpreted continuously at $\rho = 1$.

Proof. We first estimate the reciprocal length of a cycle in the relative permutation of the two children at a height- j node. Put $M = 2^{j-1}$. Fix one child's map and the other child's input-side map, leaving its final output butterfly fresh. Start at a uniform boundary column. For a proposed cycle length l , expose the first $l - 1$ edges. The closing route is unique

if compatible, and if v of its switches have already been exposed, its conditional closing probability is $2^v/M$. Hence

$$\mathbb{E}[\mathbf{1}_{\{L=l\}}2^{-v}] \leq M^{-1}.$$

For a fixed realized cycle, rotating its start makes the closing path uniform among its l paths. Lemma 3.2 bounds their shared switches by $l \log_2 l/2$, so the mean of v over starts is at most $\log_2 l$. Jensen's inequality gives a mean weight at least $1/l$. Averaging over configurations proves $\mathbb{P}(L = l) \leq l/M$ for the uniform start.

Now average all the child-interior randomness. Each child law is invariant under conjugation by an XOR translation of its boundary columns. The relative permutation has the same invariance, and these translations act transitively. Therefore, for every specified column i , its cycle length L_i satisfies $\mathbb{E}(1/L_i) \leq h_j$. This assertion uses the unconditioned child interiors; it is not asserted after the maps fixed in the preceding exploration have been revealed.

Let D_v be the set of shared input columns at a height- j node v : both positions of such a column carry tracked cards. Write $L_{v,i}$ for the full cycle length at that node and set

$$S_j = \sum_{v: \text{height}(v)=j} |D_v|, \quad Z_v = \sum_{i \in D_v} \frac{1}{L_{v,i}}, \quad Y_j = \sum_{v: \text{height}(v)=j} Z_v.$$

A tracked cycle uses every column of one full relative-permutation cycle. It contributes one to Y_j ; cycles only partly contained in D_v add nonnegative terms. Thus $C_j \leq Y_j$. Let $\mathcal{A}_j$ record the arrivals at this height and the boundary input switches of its nodes. These data depend only on coins outside the child interiors. The reciprocal-length estimate therefore gives

$$C_j \leq Y_j, \quad \mathbb{E}(Y_j \mid \mathcal{A}_j) \leq h_j S_j. \quad (18)$$

The means of the shared counts satisfy the budget defining H . Use the coordinate convention of Section 4, so the input sweep visits $d, \dots, 1$. Let A_j count unordered pairs of tracked original inputs that agree in coordinates $1, \dots, j-1$ and differ in coordinate j . The least differing coordinate assigns every pair to exactly one A_j . Before direction j , the two paths remain in different subcubes, so their already updated coordinates are independent uniforms. They meet at direction j with probability $2^{-(d-j)}$. Consequently, with $x_j = 2\mathbb{E}S_j/r$,

$$\mathbb{E}S_j = A_j 2^{-(d-j)}, \quad 0 \leq x_j \leq 1, \quad \sum_j 2^{-j} x_j = \frac{2 \sum_j A_j}{rn} = \frac{r-1}{n} \leq \rho.$$

Here $S_j \leq r/2$ gives $x_j \leq 1$. Taking expectations in (18) now yields $\sum_j h_j \mathbb{E}S_j \leq rH(\rho)$.

We next turn this mean bound into a tail bound, without conditioning several heights simultaneously. Fix a large height cutoff J . At each $j \leq J$, the disjoint-pair occupancy indicators are negatively associated by the fixed-input-layer argument above. Their sum S_j , whose mean is at most $r/2$, has the Bernoulli-product upper exponential moment. For every fixed $t > 0$, Chernoff's bound therefore gives

$$\mathbb{P}\{S_j > \mathbb{E}S_j + tr\} \leq e^{-ct^r}.$$

Conditional on $\mathcal{A}_j$, the variables Z_v in different nodes are independent, and $0 \leq Z_v \leq 2^{j-1}$. There are $n/2^j$ such nodes. Hoeffding's inequality and (18) give

$$\mathbb{P}\{Y_j > h_j S_j + tr \mid \mathcal{A}_j\} \leq \exp\{-ct^2 r^2 / (n2^j)\} \leq \exp\{-ct^2 \rho_0 r / 2^j\}.$$

The second inequality uses $r/n \geq \rho_0$. A union bound over the fixed set $j \leq J$, taking t small in terms of J, ϵ , controls their total excess over $\sum_{j \leq J} h_j \mathbb{E}S_j$. By (15), choosing J large

enough in terms of ϵ controls $\sum_{j>J} C_j$ at the remaining additive slack with probability $e^{-c\epsilon r}$. Neither this choice nor the low-height constants depend on the individual density $\rho \in [\rho_0, 1]$. Enlarging n_0 absorbs the finite union factor and shows that $\sum_j C_j$ exceeds $rH(\rho) + \epsilon r/2$ with probability at most $e^{-c\epsilon r}$.

Finally Stirling's formula, with $0! = 1$ at the endpoint, gives

$$\log_2((n)_r/n^r) = r \frac{-\rho - (1-\rho)\ln(1-\rho)}{\rho \ln 2} + O(\log n).$$

The error is uniform for $1 \leq r \leq n$. Since $r \geq \rho_0 n$, it is at most $\epsilon r/2$ for $n \geq n_0(\rho_0, \epsilon)$. Adding this deterministic normalization proves the proposition. $\square$

The defining linear optimization fills the shortest allowed cycle lengths first. With $M = 2^{j-1}$ and $q(q+1) \leq 2M < (q+1)(q+2)$,

$$h_j = \frac{q}{M} + \frac{1 - q(q+1)/(2M)}{q+1}, \quad h_j \leq \sqrt{2/M}. \quad (19)$$

The allocation ratios $2^{j-1}h_j$ increase with j . Indeed $Mh(M)$ is the maximum of $\sum_l z_l/l$ subject to $0 \leq z_l \leq l$ and $\sum_l z_l = M$; increasing M permits adding positive-valued mass. Thus the budget for H is assigned at the highest indices first, with at most one partially filled index.

Here are explicit arithmetic bounds, with enough detail to reproduce the finite check. Formula (19) simplifies to $h_j = q/(2M) + 1/(q+1)$. The inequality $h_j \leq \sqrt{2/M}$ follows by putting $x = 2M$: the convex quadratic $(q + x/(q+1))^2 - 4x$ is nonpositive at the two endpoints $q(q+1)$ and $(q+1)(q+2)$, hence throughout the interval. Therefore

$$\frac{1}{2} \sum_{j>20} h_j \leq \frac{1+\sqrt{2}}{1024} < \frac{169}{70 \cdot 1024}, \quad \sum_{j>20} 2^{-j} = 2^{-20}.$$

Subtract this last budget, allocate the first twenty levels greedily, and add the rational value bound. Exact rational arithmetic gives

$$H(1/8) < .791402, \quad H(13/50) < 1.098069, \quad H(1) < 1.958069.$$

These imply the stated useful bounds .80, 1.11 and 1.97. For the logarithms in the entropy correction, use

$$-\ln(1-x) \leq \sum_{i=1}^{80} \frac{x^i}{i} + \frac{x^{81}}{81(1-x)}, \quad \ln 2 = 2 \sum_{j=0}^{39} \frac{(1/3)^{2j+1}}{2j+1} + R, \quad 0 < R \leq \frac{2(1/3)^{81}}{81(1-1/9)}.$$

The entropy numerator is negative, so increasing its positive denominator gives an upper bound. The resulting rational comparisons give a full bracket below .904054 at $\rho = 7/25$ and below .515374 at $\rho = 1$. In particular they imply the margins .96 and .54 used below.

These endpoint checks cover every $0 < \rho \leq 7/25$. For $\rho \leq 1/8$, the entropy correction is nonpositive and $H(\rho) \leq H(1/8) < .80$. On $[1/8, 1/4]$ and $[1/4, 7/25]$, the slopes of H are respectively $7/3$ and $3/2$. The absolute derivative of the entropy correction is

$$\frac{-\rho - \ln(1-\rho)}{\rho^2 \ln 2} \leq \frac{1}{2(1-\rho) \ln 2} < \frac{25}{24},$$

using its series, $\rho \leq 7/25$ and $\ln 2 > 2/3$. Thus the bracket increases on both intervals. The accompanying script `support/network_bounds.py` checks exactly these five rational inequalities; it does not test the probabilistic or asymptotic arguments.

5.3 From truncation to all representations

Theorem 5.3 (Certificate-network contraction). *There is an absolute $c > 0$ such that for every sufficiently large dyadic n and every nontrivial irreducible of dimension D not annihilated by a pair layer,*

$$\|T_n(\lambda)\|_{\text{op}} \leq D^{-c}.$$

Consequently a fixed absolute number of sweeps has total-variation distance tending to zero.

Proof. Write $k = n - \lambda_1$ and $\mu = \bar{\lambda}$. For $k \leq .14n$ and $k \leq r \leq n - k$, (3) gives $m = \binom{r}{k} f^\mu$ and $D \leq \binom{n}{k} f^\mu$. If k/n is bounded below, take $r = 2k$. Proposition 5.2 applies uniformly over this density interval. Then $\log_2 m \geq r - o(r) + \log_2 f^\mu$. The dense cutoff with slack .01 is at most .97 r and loses exponentially small mass, so Lemma 3.1 gives a fixed exponential saving when $\log f^\mu = O(r)$. If $\log f^\mu$ is larger than a sufficiently large fixed multiple of r , use instead the cutoff $L = \frac{1}{2} \log_2 m$. Markov's inequality and (16) give $p \leq 2^{O(r) - bL}$, which is a fixed inverse power of m . These alternatives give a fixed inverse power of D .

If k/n is small, fix $0 < \gamma < \delta/(1 + \delta)$ and take $r = \lfloor k(n/k)^\gamma \rfloor$. Then $r(r/n)^\delta = O(k)$, whereas

$$\log m \geq c_\gamma k \log(n/k) + \log f^\mu.$$

Choose the fixed density cutoff sufficiently small. At $L = \frac{1}{2} \log_2 m$, Markov's inequality gives discarded mass $\exp(O(k))m^{-b/2}$, an inverse power of m . The retained bound is $m^{-1/4}$. Again $\log m$ is a fixed positive fraction of $\log D$.

For the remaining shapes use all n cards, whose multiplicity is D . Lemma 2.1 removes first columns longer than $n/2$. Every other remaining shape has first row at most .86 n . Let a be the larger of its first row and column, transposing for the dimension estimate if necessary. If $.14n \leq a \leq .86n$, the first-row hook inflation over $a!$ is $\exp(O((n/a) \log n)) = \exp(o(n))$. To verify this, for columns $j \leq a/2$ sum $\lambda'_j - 1$ and divide by $a/2$; for later columns use $\lambda'_j \leq n/j \leq 2n/a$ and the harmonic sum of $1/(a - j + 1)$. The remaining hook product is the hook product of the tail, whose tableau dimension is at least one. Hence

$$D \geq \binom{n}{a} \exp(-o(n)), \quad \log_2 D \geq (.58 - o(1))n,$$

since $H_2(.14) > .58$. If $a < .14n$, the mean hook length is at most a : indeed the sum of hooks is $\frac{1}{2}(\sum_i \lambda_i^2 + \sum_j (\lambda'_j)^2) \leq an$. Arithmetic-geometric mean and Stirling give a stronger bound. The full-density cutoff below .54 n , with small slack, therefore beats the multiplicity exponentially. If $\log D$ is larger than a sufficiently large fixed multiple of n , switch to $L = \frac{1}{2} \log_2 D$ and (16) as before. This proves the stated power bound for K_n and, after taking square roots, for T_n .

For a sufficiently large fixed integer r , Plancherel and $\|T_n(\lambda)^r\|_{\text{HS}}^2 \leq D \|T_n(\lambda)\|_{\text{op}}^{2r}$ bound the squared relative L^2 distance by $\sum_{\lambda \neq (n), (1^n)} D^{2-2rc}$. Choose r so the exponent is at most -1 and apply Lemma 2.5. Alternatively, in the range $k \leq .14n$, the more precise hook estimate (4) bounds the same sum directly level by level. Lemma 2.3 gives the matching lower order in physical time. $\square$

6 Exponential smoothing for every partial deck

We now extract two reusable consequences of an exponential density cost per tracked card: a pointwise bound after a fixed number of compositions, and a contraction factor involving the dimension of the diagram below the first row. The density input already follows from Theorem 1.1, since $(l/N)^\eta \leq 1$; renaming coordinates covers every order and both positive

orientations. Here we give an independent proof by encoding cycle closures through the switch bits they force. The proof applies uniformly from the empty tuple to the full deck and makes no representation estimate until the final lower-diagram transfer.

Let $N = 2^d$, with $d \geq 0$, and let T_N be one sweep of independent fair switches in any fixed order of the d distinct cube coordinates. Permutations map input positions to output positions, and products act from right to left. The inverse-permutation law is denoted by T_N^* . Thus

$$H_N \in \{T_N^* T_N, T_N T_N^*\}$$

is a forward sweep followed by its reversed coordinate order, with fresh coins in both halves; the choice of the first order distinguishes the two displayed orientations. On

$$\mathcal{X}_{N,l} = \{(x_1, \dots, x_l) : x_i \in \{0, 1\}^d, x_i \neq x_j (i \neq j)\}, \quad 0 \leq l \leq N,$$

write $H_N(x, y)$ for the transition probability, and $u_{N,l}$ for uniform measure on this set. Put $(N)_l = N!/(N-l)!$, with $(N)_0 = 1$. The empty tuple has transition probability one.

Theorem 6.1 (Strong density bound in both orientations). *There are absolute constants $p \in (1, 2]$ and $C < \infty$ such that, for every $N = 2^d$, every coordinate order, every $0 \leq l \leq N$, either choice of H_N , and every $x \in \mathcal{X}_{N,l}$,*

$$\frac{1}{(N)_l} \sum_{y \in \mathcal{X}_{N,l}} ((N)_l H_N(x, y))^p \leq e^{Cl}. \quad (20)$$

After increasing C , the same constants give

$$N^{-l} \sum_{y \in \{0,1\}^{dl}} (N^l H_N(x, y))^p \leq e^{Cl}, \quad (21)$$

where the kernel is extended by zero on tuples with repeated positions. Both conclusions hold with rows replaced by columns.

We prove the injection-density bound first. The power normalization will follow by an explicit falling-factorial comparison. The central counting estimate permits fixed permutations before and after the fresh layers; this is what allows it to be used after conditioning on the other parts of the network.

6.1 Encoding cycles by omitted switch bits

We need a counting observation on cycles. Let m be a cube size, D a fixed set of at most k positions in it, and π the permutation obtained using any fixed pre-permutation, fair switch layers in L distinct coordinates, and any fixed post-permutation. If Y is the number of cycles of π contained entirely in D , there are absolute $c, \gamma > 0$ and L_0 such that

$$\Pr(Y \geq t) \leq 2^{-cLt} \quad (L \geq L_0, t \text{ integer} \geq \max(1, k2^{-\gamma L})). \quad (22)$$

The contact estimate of Lemma 3.2 applies after relabeling inputs by the fixed pre-permutation. If the L coordinates do not fill the whole cube, apply it separately in each L -coordinate block: $\sum_i w_i \log_2 w_i \leq w \log_2 w$ shows that any w selected paths still share at most $w \log_2 w/2$ switches in total.

We prove (22) with $\gamma = 1/1000$ and $L_0 = 256$. The event is empty if $t > k$. Otherwise, when $Y \geq t$, at least $t/2$ of its cycles have length at most $2k/t$. Select $s = \max(1, \lfloor t/2 \rfloor)$ of these cycles; in particular $s \geq t/3$. We will encode their routes while omitting the fresh switch bits forced by closing a cycle.

First we find starts and an exploration order that omit many bits. Choose each start uniformly in its cycle. Independently give D a uniform random priority order, and visit the selected cycles in the order of their starts, completing one cycle before the next. On a cycle of length w , the closing path is uniform among its w paths. Its expected contacts with other paths of that cycle are at most $\log_2 w$ by the contact estimate. At each of the L layers, a contact outside that cycle can reveal at most one bit; conditional on the starts, the contacted selected cycle precedes it with probability $1/2$. Therefore the number S of fresh switch bits on the closing paths satisfies

$$0 \leq S \leq Ls, \quad \mathbb{E}S \geq Ls/2 - s \log_2(2k/t) \geq 0.4Ls.$$

The last inequality uses $k/t \leq 2^{\gamma L}$ and $L \geq 256$. It follows that $\Pr\{S \geq Ls/4\} \geq 1/5$. Thus at least one fifth of the priority orders admit some choices of starts saving $Ls/4$ bits. This is an existence statement for each coin configuration, before counting configurations under fixed priorities.

Fix a priority order. A code consists of the unordered set of s starts, the cycle lengths in traversal order, and a finite bit string. The decoder keeps a table of already known switch bits. It traces nonclosing edges using the fixed pre- and post-permutations, reading a bit only at a previously unknown switch. On a closing edge, its input and required endpoint are known. A butterfly updating each coordinate at most once has at most one route between those endpoints; the decoder fills the unknown bits along that route with the forced values, rejecting inconsistencies. After completing the cycles, it reads all remaining unknown bits in a fixed order. Hence a valid code determines at most one full coin configuration. If $S \geq Ls/4$, its bit string has length at most $mL/2 - Ls/4$. Counting all finite strings up to that length, including unequal lengths, gives at most

$$\binom{k}{s} (2k/t)^s 2^{mL/2 - Ls/4 + 1}$$

configurations for these fixed priorities.

Every configuration in the tail event has such a code for at least one fifth of all priority orders. Averaging the preceding count over priorities and dividing by the $2^{mL/2}$ equiprobable coin configurations bounds the tail probability by

$$5 2^{1+s\{\log_2(6e)+2\gamma L-L/4\}} \leq 2^{-Lt/64} \quad (L \geq 256).$$

Here $\log_2 \binom{k}{s} + s \log_2(2k/t) \leq s\{\log_2(6e) + 2\gamma L\}$ uses $s \geq t/3$. This proves (22).

6.2 There-and-back recursion on cube nodes

Now realize H_N as there-and-back switches (using the pass order or its reverse first, according to which product is used), all with independent coins. It is a binary tree of cube nodes: each size $2m$ node uses switches first across one axis, then gives two children (panels of m positions, indexed by columns of the switch pairs) on which independent child there-and-back permutations act, then again switches on its original axis. Size one leaves do nothing. Call the first layers of switches over the tree its input layers, and second layers its output layers.

Within a node, write $R(x, y)$ for the density of output placement at y for k distinguished cards starting at x , relative to uniform on such placements. Write $J(x, y)$ for the density in an ideal experiment at the node, replacing the two child permutations by independent uniforms. We use at root or any node, for $p = 1 + \theta > 1$, the inequality

$$\mathbb{E}_{\text{unif}} R(x, y)^p G(y) \leq \mathbb{E}_{\text{true}} \left[G(y) \prod_v J_v(X_v, Y_v)^\theta \right] \quad (23)$$

for any nonnegative G . Left expectation is in y ; right uses the actual switch experiment with this input x , taking J_v at this and descendant nonleaf nodes with the placements of the distinguished cards passing through them (local inputs X_v and outputs Y_v). Densities for empty placements are 1. To see (23), in the ideal node experiment take x_i, y_i the inputs into and outputs of children. Given final y (when $J > 0$), the conditional expectation of the product of child true densities $R_i(x_i, y_i)$ equals $R(x, y)/J(x, y)$, by the density change to the true experiment from ideal. And R vanishes when J does. Jensen thus bounds the left side by

$$\mathbb{E}_{\text{ideal}} G(y) J(x, y)^\theta \prod_i R_i(x_i, y_i)^p.$$

Condition on this node's outer switch layers (input and output), so y_i in this formula are independent uniforms, and y is a function of them and the outer choices. Apply (23) inductively at each child (against nonnegative functions of their outputs); this gives exactly the desired inequality. The size one case is immediate.

For x, y in the size $2m$ node, let the multigraph on the k cards have two matchings, given by sharing an input column or sharing an output column, of sizes a, b respectively. It consists of paths (possibly single vertices) and even cycles (possibly of length two); let c_* be the number of cycles. Call coloring the cards by the two children allowed if opposite along matching edges. With r the number in the first child, we have

$$J(x, y) = (2m)_k 2^{a+b-2k} \sum_{\text{allowed colors}} \frac{1}{(m)_r (m)_{k-r}} = 2^{c_*} \mathbb{E} Z(r), \quad Z(r) = \frac{(2m)_k}{2^k (m)_r (m)_{k-r}}. \quad (24)$$

Indeed realizing a given coloring from input costs 2^{a-k} ; given correct child output columns the final switches cost 2^{b-k} ; the child-permutation probabilities between columns are as in the sum, with columns distinct per color by the constraints. There are $2^{k-a-b+c_*}$ allowed choices, taken equally in the last expectation.

Every allowed coloring is feasible: it puts at most one card of each color in each of the m input columns and each output column. Each path or cycle component has two colorings. Even components contribute equally to the two colors; each odd path component contributes one independent sign to their difference.

The following bound holds uniformly:

$$\mathbb{E} Z(r) \leq \exp(Ck \log(2m)/m). \quad (25)$$

To prove it, put $z = 2r - k$. Each odd-vertex path component contributes one independent uniform sign to z ; let q be their number. Clearly $q \leq k$. Making all odd components favor the same color is feasible, so $k + q \leq 2m$ and hence $q \leq w := 2m - k$ as well. At the balanced count $r_0 = \lceil k/2 \rceil$, writing $k = 2a$ or $2a + 1$ gives

$$Z(r_0) = \prod_{i=0}^{a-1} \frac{2m - 2i - 1}{2m - 2i} \leq 1.$$

The cases $k = 0$, $q = 0$, or $w = 0$ are consequently immediate.

Suppose first that $1 \leq k \leq \epsilon m$, for a small absolute ϵ . The factorial ratio

$$\frac{Z(r)}{Z(r_0)} = \frac{(m-r)!(m-k+r)!}{\lfloor w/2 \rfloor! \lceil w/2 \rceil!}$$

is unchanged by replacing z by $-z$. Moving i steps from the balanced count multiplies successive factors of the form $1 + O((i+1)/m)$. Thus it is at most $e^{C(z^2+1)/m}$. Since z is a sum of at most k fair signs, $\Pr\{z^2 \geq v\} \leq 2e^{-v/(2k)}$, and for $m > 2Ck$,

$$\mathbb{E} e^{Cz^2/m} \leq 1 + \frac{4Ck}{m - 2Ck}.$$

Choosing ϵ small makes $\mathbb{E}Z(r) \leq e^{C'k/m}$, which implies (25) in this range.

For $k > \epsilon m$ and $w > 0$, Stirling's formula gives the uniform bound

$$\frac{Z(r)}{Z(r_0)} = \frac{((w-z)/2)!((w+z)/2)!}{\lfloor w/2 \rfloor! \lceil w/2 \rceil!} \leq e^{O(\log(2m)) + wI(z/w)},$$

where $I(x) = ((1-x)\log(1-x) + (1+x)\log(1+x))/2$ is extended continuously at the endpoints. The sign sum has point probabilities at most $e^{-qI(z/q)}$. Convexity and $I(0) = 0$ give

$$I(z/w) \leq (q/w)I(z/q) \quad (0 < q \leq w).$$

The two exponential rate factors therefore cancel. Summing over at most $2m+1$ possible values of z leaves $e^{O(\log(2m))}$, which is at most $e^{Ck \log(2m)/m}$ since $k > \epsilon m$. This completes the uniform imbalance estimate, including full occupancy.

6.3 Summing cycle moments over scales

Apply (23) at root with $G = 1$ and l distinguished cards. At each height $j \geq 1$ (nodes of size $2^j = 2m$), distinguished counts k_v at the nodes sum to l , so (25) costs only e^{Cl} over the tree for fixed θ . It remains to obtain, in the true H_N runs, for some small absolute $\theta > 0$

$$\mathbb{E} 2^\theta \sum_v c_v \leq e^{Cl} \quad (26)$$

where c_v is the cycle count c_* of (24) for the placements there.

Indeed, the logarithm of the noncycle factor at height j is at most $Clj/2^j$, and $\sum_{j \geq 1} j2^{-j} < \infty$. It is the cycle factor, rather than this deterministic imbalance factor, that requires the remaining multiscale argument.

Condition on all input layers, fixing the assignments and inputs throughout the tree. At a node, given α, β the two child permutations on columns, c_v is the number of cycles of $\beta^{-1}\alpha$ entirely contained in the set D_v of input columns with both entering positions marked (distinguished). To see this, in a cycle of the two matchings one uses both inputs of participating columns, one going through each child. Following this circuit, going through the output matching from the α -card at one such input column i , and then through the input matching, brings us to the α -card for column $\beta^{-1}\alpha(i)$. Conversely a cycle of the comparison contained in D_v traces such a circuit, irrespective of the outermost output switches there. In particular c_v with inputs fixed depends just on smaller height output layers.

For $h = 1, 2, 4, \dots$ group node heights j in bands $h \leq j < 2h$. For such j consider conditioning on all inputs as above and also all output layers at heights $< h/2$. For large h , α in the above comparison has still its last $L = h/2$ switch layers (distinct axes in this child) fresh. Indeed these output layers of a height $j-1$ child use output coins at heights $j-L, \dots, j-1$; each layer runs over all the m columns in that child. Even conditioning further on everything except these layers, (22) applies to the comparison and D_v , the latter of size at most k_v . Consequently for sufficiently small $\theta > 0$,

$$\mathbb{E} \left[2^{2\theta h c_v} \mid \text{inputs, outputs}_{< h/2} \right] \leq \exp(Ck_v h 2^{-\gamma' h}) \quad (27)$$

for some absolute $\gamma' > 0$; in the conditioning we mean layer coins. Indeed for positive counts at or above $k_v 2^{-\gamma' L}$, multiplication by $2^{2\theta h t}$ still gives geometric decay in ht from (22), and for counts below threshold one bounds the moment by $2^{2\theta h k_v 2^{-\gamma' L}}$. For $k_v \geq 1$ the additive exponentially small term fits as well by taking γ' small; $k_v = 0$ is automatic. For bounded h , (27) holds by enlarging the constant since $c_v \leq k_v$.

Conditional on the indicated layers, nodes on a fixed level have independent c_v , using only their disjoint subtrees. So (27) also bounds the $2\theta h$ moment in this notation for the level sum, with k_v replaced by l . Hölder over the at most h levels in the band gives

$$\mathbb{E} \left[2^{2\theta \sum_{j \text{ in band}} \sum_{v \text{ at } j} c_v} \mid \text{inputs, outputs}_{<h/2} \right] \leq \exp(Clh2^{-\gamma'h}).$$

The next smaller band of the same parity of $\log_2 h$ has heights $h/4 \leq j < h/2$. Its cycles depend only on output layers strictly below $h/2$, so its exponential factor is measurable under the conditioning for the band starting at h . The tower property therefore permits integrating each parity from the largest band downwards. It bounds that parity's 2θ moment by

$$\exp \left\{ Cl \sum_{h=1,2,4,\dots} h2^{-\gamma'h} \right\} \leq e^{C'l}.$$

Cauchy–Schwarz combines the two parities at exponent θ and proves (26). Independence was used within a height, while the tower property handles the dependence across heights.

Completion of Theorem 6.1. Inserting (24) and (25) into (23), then applying (26), gives (20) with $p = 1 + \theta$. Decrease θ if necessary so that $p \leq 2$. The proof applies to either order at the root and throughout its descendants, so it proves both orientations with the same absolute constants.

To change normalization, put $s_l = (N)_l / N^l$. Then

$$N^{-l} \sum_y (N^l H_N(x, y))^p = s_l^{-(p-1)} \frac{1}{(N)_l} \sum_y ((N)_l H_N(x, y))^p.$$

The decreasing list $\log(1 - i/N)$, $0 \leq i < N$, has mean $N^{-1} \log(N!/N^N) \geq -1$, since $N! \geq (N/e)^N$. The mean of its first l terms is no smaller, so $s_l \geq e^{-l}$. This proves (21) after an increase of C . Both positive products are self-adjoint, hence $H_N(x, y) = H_N(y, x)$; the column statements follow. The cases $l = 0$ and $N = 1$ are immediate. $\square$

6.4 A fixed number of compositions gives a pointwise bound

Corollary 6.2 (Strong pointwise smoothing). *There are an absolute integer $u \geq 1$ and an absolute $C < \infty$ such that, for every $N = 2^d$, every coordinate order, every $0 \leq l \leq N$, every $x, y \in \mathcal{X}_{N,l}$, and either $H_N = T_N^* T_N$ or $H_N = T_N T_N^*$,*

$$H_N^u(x, y) \leq \frac{e^{Cl}}{(N)_l}. \quad (28)$$

In particular, with unnormalized regular trace on S_N ,

$$\text{Tr}_{\text{reg}} |T_N|^{2u} \leq e^{CN}. \quad (29)$$

Proof. All function norms in this proof use the probability measure $u_{N,l}$. Write $A = e^{C_0 l}$ for a common bound on the L^p norms of the transition densities supplied by Theorem 6.1. The kernel formula and Minkowski's inequality give $\|H_N\|_{1 \rightarrow p} \leq A$. Self-adjointness gives $\|H_N\|_{p' \rightarrow \infty} \leq A$, where $p' = p/(p-1)$. As a Markov operator, H_N has ∞ -to- ∞ norm one. Riesz–Thorin interpolation consequently gives, for every $b \geq 1$,

$$\|H_N\|_{b \rightarrow pb} \leq A^{1/b}.$$

Let r be the least nonnegative integer with $p^r \geq p'$. Composing the bounds for $b = 1, p, \dots, p^{r-1}$ yields

$$\|H_N^r\|_{1 \rightarrow p^r} \leq A \sum_{j=0}^{r-1} p^{-j}.$$

Since the underlying measure is a probability measure, $\|f\|_{p'} \leq \|f\|_{p^r}$. One further application of $H_N : p' \rightarrow \infty$ therefore proves $\|H_N^{r+1}\|_{1 \rightarrow \infty} \leq e^{Cl}$. Applying this to $(N)_l \mathbf{1}_{\{x\}}$, whose L^1 norm is one, gives (28) with $u = r + 1$.

For $l = N$, the injection module is the regular permutation module on S_N . Every diagonal entry of $(T_N^* T_N)^u$ is at most $e^{CN}/N!$, so its unnormalized trace is at most e^{CN} . Since $|T_N|^{2u} = (T_N^* T_N)^u$, this is (29). $\square$

6.5 The lower-diagram dimension factor

The density theorem also distinguishes permutations of the tracked labels. We now make that distinction explicit. This is useful when a bound involving only the number of boxes below the first row does not yet control the dimension of the lower diagram.

Proposition 6.3 (Strong tail-dimension bound). *There are absolute constants $a > 0$ and $C < \infty$ such that, for every dyadic N , every coordinate order, and every partition $\lambda = (N - k, \beta) \vdash N$, with $\beta = (\lambda_2, \lambda_3, \dots) \vdash k$,*

$$\|T_N(\lambda)\|_{\text{op}}^2 \leq e^{Ck} D_\beta^{-a}. \quad (30)$$

Here $D_\beta = \dim V_\beta$, including $D_\emptyset = 1$. One may take $a = 1/p' = (p - 1)/p$ for the exponent in Theorem 6.1.

Proof. The case $k = 0$ is immediate. On ordered k -tuples, permutations of the k coordinate labels act on the right and commute with all position permutations. Choose an ordering x_A of each unordered k -set A . The fiber over A is then the regular representation of S_k . Since H_N commutes with this right action, its block from one fiber to another acts by convolution. In the Fourier decomposition of one fiber, the β isotype is $V_\beta \otimes V_\beta^*$, with the right action on the second factor. The block acts as $M_{AB} \otimes I$; it acts on the first, multiplicity factor and may mix the copies of the right type. Up to inverse or transpose conventions that do not affect the norm, its multiplicity matrix is

$$M_{AB} = \sum_{\pi \in S_k} H_N(x_A, x_B \pi) \rho_\beta(\pi).$$

For unit vectors $v, w \in V_\beta$, Schur orthogonality [8, Proposition 3.14(ii)] and unitarity give

$$\frac{1}{k!} \sum_{\pi \in S_k} |\langle v, \rho_\beta(\pi) w \rangle|^2 = D_\beta^{-1}, \quad |\langle v, \rho_\beta(\pi) w \rangle| \leq 1.$$

Because $p' \geq 2$, the normalized $L^{p'}$ norm of this matrix coefficient is at most $D_\beta^{-1/p'}$. Hölder therefore gives

$$\|M_{AB}\|_{\text{op}} \leq k! N^{-k} D_\beta^{-1/p'} \left\{ \frac{1}{k!} \sum_{\pi \in S_k} (N^k H_N(x_A, x_B \pi))^p \right\}^{1/p}. \quad (31)$$

To sum this estimate, put $w_0 = k! N^{-k}$ and denote the braces to the power $1/p$ by F_{AB} . There are $\binom{N}{k}$ fibers, so $\sum_B w_0 = (N)_k / N^k \leq 1$. The density estimate gives

$$\sum_B w_0 F_{AB}^p = N^{-k} \sum_{y \in \mathcal{X}_{N,k}} (N^k H_N(x_A, y))^p \leq e^{Ck}.$$

Weighted Hölder shows that every row sum of the scalar matrix $(\|M_{AB}\|_{\text{op}})$ is at most $e^{Ck}D_\beta^{-1/p'}$. The same bound holds for column sums: self-adjointness and relabeling equivariance give

$$H_N(x_A, x_B\pi) = H_N(x_B, x_A\pi^{-1}).$$

The Schur test now bounds the full operator on the right- β isotypic space by $e^{Ck}D_\beta^{-1/p'}$. Explicitly, if $b_{AB} = \|M_{AB}\|_{\text{op}}$ and both scalar sums are at most L , then

$$\sum_A \left\| \sum_B M_{AB} v_B \right\|^2 \leq \sum_A \left(\sum_B b_{AB} \right) \sum_B b_{AB} \|v_B\|^2 \leq L^2 \sum_B \|v_B\|^2.$$

As a left S_N module, this right- β isotypic space consists of D_β copies of

$$\text{Ind}_{S_{N-k} \times S_k}^{S_N} (\mathbf{1} \otimes V_\beta).$$

The skew diagram λ/β is a horizontal strip of size $N - k$: its interlacing conditions are $\lambda_i \geq \beta_i \geq \lambda_{i+1}$, and here $\beta_i = \lambda_{i+1}$. The horizontal Pieri rule therefore places V_λ in this induced module. The preceding operator bound applies to its λ block. Taking $H_N = T_N^* T_N$ and using $\|H_N(\lambda)\|_{\text{op}} = \|T_N(\lambda)\|_{\text{op}}^2$ proves the assertion. The other orientation gives the same singular-value conclusion. $\square$

6.6 An alternative cycle-moment proof by entropy windows

We give a second proof of the cycle-moment estimate (26), retaining the node comparison and color imbalance estimate already established. Instead of separating dyadic bands into two parities, we control their expectation under every tilted law of the output coins. The inverse-height weights in the entropy chain rule give each layer a bounded total coefficient. We retain the fixed input layers and the node cycle counts c_v from the proof of Theorem 6.1.

We first give a certificate version of the cycle estimate. Consider a butterfly of q distinct-coordinate layers, with arbitrary fixed permutations before and after it, and a fixed set D of at most k positions. Let Y count the cycles of the resulting permutation contained in D , and let $r \geq 1$ be an integer. Put $\varepsilon = 1/1000$. If at least r cycles lie in D , with $r > k2^{-\varepsilon q}$, at least $r/2$ of them have length at most $2k/r$. A uniformly chosen closing path of one of these cycles has at most $\log_2(2k/r) \leq 1 + \varepsilon q$ internal contacts on average, by the contact bound used in Section 6.1.

Choose one root uniformly in each short cycle, and retain each short cycle independently with probability $1/16$. Conditional on a particular cycle being retained, its closing path has at most $q/16$ expected contacts with paths in other retained cycles: at each layer there is only one other path at its switch, and its cycle is retained with probability at most $1/16$. Markov's inequality shows that the probability of more than $q/4$ such external contacts is at most $1/4$. The probability of more than $q/4$ internal contacts is at most $4/q + 4\varepsilon$. For all sufficiently large absolute q , the expected number of retained roots passing both tests is therefore larger than

$$\frac{1}{16} \frac{r}{2} \left(1 - \frac{1}{4} - \frac{4}{q} - 4\varepsilon \right) > \frac{r}{64}.$$

Consequently some deterministic choice contains $R = \lceil r/64 \rceil$ cycles whose closing paths share at most $q/2$ switches with all other paths in those selected cycles. Deleting further selected cycles only improves this property.

A certificate specifies the unordered root set and the length of each selected cycle; traverse the cycles in increasing root order. There are at most

$$\binom{k}{R} (2k/r)^R \leq 2^{R\{C_0 + 2\varepsilon q\}}$$

certificates. Fix one certificate and expose its routes in that order. Before a closing route is queried, its initial point and target are known. The unique geometrical route between them is therefore known too. Whether it has at most $q/2$ already exposed switches is measurable at this time. If that test succeeds, at least $q/2$ fresh fair bits must take prescribed values for the cycle to close. Its conditional cost is at most $2^{-q/2}$. Iterating over the R cycles and then summing certificates proves

$$\Pr\{Y \geq r\} \leq e^{-cqr} \quad \text{if } r > k2^{-\varepsilon q} \text{ and } q \geq q_0, \quad (32)$$

for absolute $c > 0$ and q_0 . This reasoning conditions only on already exposed bits; it does not condition on a subsequently discovered cycle.

Let $b_t = \sum_{v: \text{height}(v)=t} c_v$ be the cycle count at height t in the there-and-back tree, and put $q = \lfloor t/2 \rfloor$. Condition on all input coins and on output coins at heights below $t - q$. First suppose that $q \geq q_0$. At a height- t node, write the two child permutations as α, β , as in the cycle identification above. Freeze β and everything in α except its last q output layers. These are fresh distinct-coordinate layers, so (32) applies. Integrating its tail, and decreasing a positive absolute a_1 if necessary, gives

$$\log \mathbb{E} \left[e^{a_1 t c_v} \mid \text{inputs, output heights} < t - q \right] \leq C k_v e^{-a_2 t}.$$

The factor t multiplying the exponentially small threshold is absorbed by decreasing $a_2 > 0$. Small heights are covered by $c_v \leq k_v$ and increasing C . Nodes at one height use disjoint remaining subtrees and $\sum_v k_v = l$. Thus

$$\log \mathbb{E} \left[e^{a_1 t b_t} \mid \text{inputs, output heights} < t - q \right] \leq C l e^{-a_2 t}. \quad (33)$$

To make the change of law precise, fix the input coins and let P be the product law of the output coins. Order those coins by increasing height. For any probability Q on this finite coin space, put

$$\begin{aligned} H_s &= \mathbb{E}_Q D(Q(\text{height-}s \text{ coins} \mid \text{lower heights}) \parallel P(\text{height-}s \text{ coins})), \\ H &= D(Q \parallel P) = \sum_s H_s. \end{aligned}$$

Here $D(\cdot \parallel \cdot)$ denotes relative entropy. Applying the entropy inequality conditionally to the window in (33) gives, for $t \geq 2$,

$$\mathbb{E}_Q b_t \leq \frac{C l e^{-a_2 t} + \sum_{s=t-\lfloor t/2 \rfloor}^{t-1} H_s}{a_1 t}.$$

Here b_t is measurable in output heights strictly below t . The tilted law used in the conditional inequality is the marginal law of the window given its lower-height history, with all future coins integrated out. The entropy chain rule says that its conditional relative entropy, averaged over that history under Q , is exactly the displayed sum of H_s . The height-one term is at most l directly. A fixed layer s belongs only to windows with $s+1 \leq t \leq 2s$, and $\sum_{t=s+1}^{2s} t^{-1} \leq \log 2$. Summation therefore yields

$$\mathbb{E}_Q \sum_t b_t \leq C_1 l + C_2 D(Q \parallel P). \quad (34)$$

For $0 < \delta \leq 1/C_2$, the finite-space entropy variational formula now gives

$$\log \mathbb{E}_P e^{\delta \sum_t b_t} = \sup_Q \left\{ \delta \mathbb{E}_Q \sum_t b_t - D(Q \parallel P) \right\} \leq \delta C_1 l.$$

The estimate is uniform in the fixed inputs, so it also holds after averaging them. Inserting it into the exact ideal-node recursion (23), with $\theta \log 2 \leq \delta$, proves the density theorem by this alternative route; the smoothing and lower-diagram transfers then apply without change.

7 Uniform contact cancellation on an exact injection level

A tracked path that meets no other tracked path contributes nothing to the alternating sum over retained coordinates. We use this cancellation to obtain a sparse-level contraction estimate, with constants uniform in the number of tracked cards. Combining it with the lower-diagram bound of Proposition 6.3 will give dimension-power decay when the tracked fraction is very small.

The main issue is controlling the norm of the surviving kernel. We estimate its square by a return experiment with common and independent paths. Conditional on the return randomness, a weighted forest count bounds the probability that every tracked path meets another. This avoids an independence assumption on the encounter edges.

Let $n = 2^d$ with $d \geq 1$, and let T_n be the average operator of a single directed sweep. Write $X = [n]_{\neq}^k$ for the ordered injective k -tuples and $Y = [n]^k$ for all ordered k -tuples. Both spaces below have counting measure. Let $J : \ell^2(X) \rightarrow \ell^2(Y)$ be extension by zero; it is an isometry. A function $f : X \rightarrow \mathbb{C}$ is called harmonic if, for each slot, its sum over that slot is zero when the other slots are fixed. The sum uses the positions not occupied by the fixed slots. Equivalently, Jf has zero sum over each unrestricted coordinate of Y , with the other coordinates fixed. Denote this subspace by $\mathcal{H}_k$.

Proposition 7.1 (Uniform sparse contact estimate). *There is an absolute constant $C_* > 0$ such that for every dyadic $n = 2^d \geq 2$, every $1 \leq k < n$, and every partition $\lambda = (n-k, \beta) \vdash n$,*

$$\|T_n(\lambda)\|_{\text{op}}^2 \leq \min \left\{ 1, \left(\frac{C_* dk}{n} \right)^{k/2} \right\}. \quad (35)$$

For $k = 1$ this operator is zero. In particular, with an absolute constant C independent of n, k, λ ,

$$\|T_n(\lambda)\|_{\text{op}}^2 \leq C^k (1+d)^{Ck} (k/n)^{k/2}. \quad (36)$$

The same statements hold with the coordinate order reversed.

Proof. We first prove an operator bound on $\mathcal{H}_k$. The representation transfer at the end will explain why it applies to V_λ .

Alternating coordinate kernels. For $A \subseteq [k]$, $x \in X$, and $y \in Y$, define

$$F_A(x, y) = n^{|A|} \Pr\{\pi x_i = y_i \text{ for every } i \in A\}, \quad P_A(x, y) = n^{-k} F_A(x, y),$$

where π is a sweep permutation. The empty constraint has probability one. The matrix P_A has rows indexed by X , columns indexed by Y , and row sums one. Its row law has the coordinates in A follow one common sweep, while the other coordinates are independent uniform positions. Let K_X be the usual sweep transition kernel on X , acting on functions by averaging their values at the output. Then $K_X = P_{[k]}J$. If $A \neq [k]$ and $f \in \mathcal{H}_k$, summing over any omitted output coordinate gives $P_A Jf = 0$. Hence

$$K_X f = \sum_{A \subseteq [k]} (-1)^{k-|A|} P_A Jf \quad (f \in \mathcal{H}_k). \quad (37)$$

A path through a directed sweep is determined by its two endpoints: at each stage the coordinates already updated equal the corresponding output coordinates, and those not yet updated equal the input coordinates. This defines a path even when the proposed paths cannot be realized simultaneously. For $x \in X, y \in Y$, make a graph on $[k]$ by joining two labels when their prescribed paths enter the same switch at some stage. Switches are indexed

by both stage and pair of positions. Let $E(x, y)$ be the event that this graph has no isolated vertex.

Suppose label i is isolated. For $i \notin A$, the prescribed path of i uses d switches disjoint from every other prescribed path. Its probability is $2^{-d} = 1/n$, independently of the prescriptions for A . If those prescriptions conflict, both probabilities are zero. Thus $F_{A \cup \{i\}}(x, y) = F_A(x, y)$ in every case, and the alternating sum in (37) cancels pointwise. Define the nonnegative rectangular matrices

$$M_A(x, y) = P_A(x, y) \mathbf{1}_{E(x, y)}.$$

We have proved the exact identity

$$K_X f = \sum_{A \subseteq [k]} (-1)^{k-|A|} M_A J f, \quad \|K_X|_{\mathcal{H}_k}\|_{\text{op}} \leq \sum_{A \subseteq [k]} \|M_A\|_{\ell^2(Y) \rightarrow \ell^2(X)}.$$

For $k = 1$ the event E is empty and the operator on $\mathcal{H}_1$ is zero. Assume henceforth that $k \geq 2$.

The square of a retained kernel. Fix A and a row $x \in X$. After sampling y according to $P_A(x, \cdot)$, run a fresh reverse sweep for the labels in A , and let each other label run its own independent reverse path. Such a private path uses independent fair stay-or-swap choices in the successive directions. Its final position is uniform. Because the forward-supported tuple y_A is injective, this defines a stochastic kernel on all return tuples $z \in Y$:

$$Q_A(y, z) = n^{-|A^c|} \Pr\{\pi^{-1} y_i = z_i \text{ for every } i \in A\}.$$

Here π is a fresh forward sweep; its inverse has the reverse-sweep law. For $x' \in X$, the equality of the prescribed-image events gives $Q_A(y, x') = P_A(x', y)$. Thus the restriction of Q_A to injective return tuples is the adjoint kernel; the extension to all z restores its row sum to one.

Let $E_{\text{rev}}(y, z)$ say that the prescribed reverse paths have no isolated vertex. Reversing paths preserves all switch encounters, so $E(x', y) = E_{\text{rev}}(y, x')$. Dropping the first-leg encounter condition and then the injectivity constraint on x' gives

$$\sum_{x' \in X} (M_A M_A^*)(x, x') \leq \sum_{y \in Y} P_A(x, y) \sum_{z \in Y} Q_A(y, z) \mathbf{1}_{E_{\text{rev}}(y, z)}. \quad (38)$$

Terms with $P_A(x, y) = 0$ can be omitted. The right side is the probability that the mixed return paths have no isolated vertex. We now bound it uniformly in both A and x .

Product bounds for the midpoint occupation. Write $a = |A|$. There is one common network carrying the a labels in A , and one private network for each of the $k - a$ other labels. A vertex in the next construction records a network and an initial position in that network. Distinct networks give distinct vertices, even when the physical position agrees.

In the common network the occupied midpoint set is the image of the fixed a -set $\{x_i : i \in A\}$ under a sweep. If its current occupancy marginals are p_v , then for every position set D ,

$$\Pr\{D \text{ is occupied}\} \leq \prod_{v \in D} p_v. \quad (39)$$

This holds at the deterministic initial set. A fair switch replaces the two endpoint marginals by their mean. For D containing exactly one endpoint, average the two old bounds; for D containing both, the probability is unchanged while the product of those two marginals can only increase; the other sets are unaffected. This proves preservation under each switch. At the end of a sweep each card is uniform, so every common-network marginal is a/n .

Each private network has one uniform occupied vertex, independently of the common network and of all other private networks. Assign to a typed vertex v the weight

$$p_v = \begin{cases} a/n, & v \text{ belongs to the common network,} \\ 1/n, & v \text{ belongs to a private network.} \end{cases}$$

If $a = 0$, omit the common network. The random set $\mathcal{O}$ of occupied typed vertices has exactly k elements and, for every fixed typed set D ,

$$\Pr\{D \subseteq \mathcal{O}\} \leq \prod_{v \in D} p_v, \quad \sum_v p_v = k. \quad (40)$$

A set requiring two different vertices of one private network has probability zero, which is consistent with this bound.

A graph determined by the return randomness. Condition only on the fresh randomness used for the return trip. Sample the entire common switch wiring. For each private label, sample its d independent stay-or-swap bits and use those same bits for every possible starting position of that private network. This is a coupling of the possible starts: for each fixed start it has exactly the private-path law already defined. It changes no entry of Q_A . A fixed private bit at a stage applies either the identity or the flip of that coordinate to all positions. Consequently, in every network, the map from a starting position to the position entering any specified stage is a permutation.

Join two distinct typed vertices when their return paths enter the same physical switch at some stage. For a fixed vertex and a fixed target network there are at most $2d$ neighbors: at a given stage the switch has two positions, and each has exactly one preimage under the target network's position map. The weighted degree therefore satisfies

$$\sum_{w: w \sim v} p_w \leq \Delta, \quad \Delta = 2dk/n. \quad (41)$$

The midpoint occupation is independent of the fresh return randomness, so (40) remains valid under this conditioning. The encounter graph of the return paths is exactly the graph induced by $\mathcal{O}$ in this typed graph.

Counting forests with uniform constants. Every graph on k vertices without isolated vertices contains a spanning forest whose components all have at least two vertices. Root each component and order the children at each vertex. If the forest has j components, then $1 \leq j \leq \lfloor k/2 \rfloor$. For fixed k, j , there are at most 4^k plane rooted forest shapes with ordered components: adding one common root embeds this class in the plane rooted trees with $k+1$ vertices, counted by the k th Catalan number, which is at most 4^k .

For a fixed forest shape $\mathcal{F}$, give an embedding ϕ into the typed graph the weight $\prod_{u \in V(\mathcal{F})} p_{\phi(u)}$. Initially require injectivity and require that every forest edge maps to a graph edge. We may increase the sum by dropping injectivity, retaining all factors with their multiplicities in this weight. Sum the leaves first: for a fixed image of its parent, each leaf contributes at most Δ by (41). Delete that leaf and continue. At the end, each of the j roots contributes $\sum_v p_v = k$. The full sum is at most $k^j \Delta^{k-j}$.

The component orderings can be divided out before dropping injectivity. Indeed a fixed embedded forest with j distinct components gives $j!$ distinct shape-embedding pairs by ordering those components, even if some component shapes coincide. Together with (40), the union bound over witnessing forests consequently gives

$$\Pr\{\mathcal{O} \text{ has no isolated vertex} \mid \text{return randomness}\} \leq 4^k \sum_{1 \leq j \leq k/2} \frac{k^j}{j!} \Delta^{k-j}. \quad (42)$$

This argument only uses a witnessing forest on the occupied set; it does not assume that the encounter edges are independent.

If $\Delta \leq 1$, each $k - j \geq k/2$, and

$$4^k \sum_{1 \leq j \leq k/2} \frac{k^j}{j!} \Delta^{k-j} \leq 4^k \Delta^{k/2} \sum_{j=0}^{\infty} \frac{k^j}{j!} = (4e)^k \Delta^{k/2}.$$

This is uniform in the conditioned return graph and hence also holds after averaging its randomness. By (38), every row sum of the symmetric nonnegative matrix $M_A M_A^*$ has that upper bound. Schur's test, or its maximum-row-sum bound, gives

$$\|M_A\|_{\text{op}}^2 \leq (4e)^k \Delta^{k/2}.$$

There are 2^k choices of A , so the triangle inequality in (37) yields

$$\|K_X|_{\mathcal{H}_k}\|_{\text{op}}^2 \leq (16e)^k (2dk/n)^{k/2}. \quad (43)$$

For $\Delta > 1$, the contraction bound $\|K_X\|_{\text{op}} \leq 1$ gives the same conclusion after enlarging the absolute constant. For example, $C_* = 512e^2$ suffices in (35).

Passing to the exact representation level. The permutation representation on X is induced from the trivial representation of the stabilizer S_{n-k} . By Young branching and Frobenius reciprocity, V_λ , with $\lambda_1 = n - k$, occurs in this representation: remove the k boxes below its first row in a valid corner-removal order. It does not occur on $(k-1)$ -tuples, because occurrence there would require a first row of length at least $n - k + 1$.

For any specified slot, the functions lifted from the other $k-1$ slots form an invariant subspace containing no copy of V_λ . Its orthogonal complement consists exactly of functions whose sum in that slot is zero. Thus every V_λ -isotypic component on X lies in $\mathcal{H}_k$. The averaging operator K_X on functions has the Fourier action of T_n^* under the convention that a permutation acts by pullback through its inverse. Its norm on a copy of V_λ is therefore $\|T_n(\lambda)\|_{\text{op}}$. Applying (43) proves the proposition. The proof uses only that each coordinate is updated once and works unchanged in the reversed coordinate order. $\square$

Remark 7.2 (Counting roots among physical positions). For comparison with counting roots among physical positions, the right side of (42) equals

$$4^k (k/n)^k \sum_{1 \leq j \leq k/2} \frac{n^j}{j!} (2d)^{k-j}.$$

For $j \leq n/2$, the bound

$$\log \frac{n^j}{(n)_j} = \sum_{r=0}^{j-1} -\log(1 - r/n) \leq \sum_{r=0}^{j-1} \frac{r}{n-r} \leq j$$

shows that $n^j/j! \leq e^j \binom{n}{j}$. Thus we also obtain the root-count bound

$$\Pr\{\mathcal{O} \text{ has no isolated vertex} \mid \text{return randomness}\} \leq (k/n)^k \sum_{1 \leq j \leq k/2} (4e)^k \binom{n}{j} (2d)^{k-j}. \quad (44)$$

The weighted version identifies all types and occupation factors explicitly; both forms retain the same number $k - j$ of contact costs.

Corollary 7.3 (Exponential sparse-level saving). *There are absolute constants $C, d_0 > 0$ such that for every dyadic $n = 2^d$ and $\lambda = (n - k, \beta) \vdash n$ satisfying $1 \leq k < n$ and $\log(n/k) \geq d^{3/4}$,*

$$\|T_n(\lambda)\|_{\text{op}}^2 \leq e^{Ck} (k/n)^{k/4}.$$

For $d \geq d_0$, the factor e^{Ck} can be omitted.

Proof. Put $s = \log(n/k)$. For a nonzero operator, (35) bounds the logarithm of its squared norm by $-ks/2 + (k/2)\log(C_*d)$. Since $\log(C_*d) = o(d^{3/4})$, it is at most $-ks/4$ for all sufficiently large d in the stated range. The remaining finitely many d are covered by increasing C , using $\|T_n\|_{\text{op}} \leq 1$ and $s \leq d \log 2$. A zero operator is covered directly. $\square$

Corollary 7.4. *There are absolute $c > 0$ and d_0 such that, for every $N = 2^d$ with $d \geq d_0$ and every partition $\lambda = (N - k, \beta) \vdash N$ satisfying $1 \leq k < N$ and $\log(N/k) \geq d^{3/4}$,*

$$\|T_N(\lambda)\|_{\text{op}}^2 \leq D_\lambda^{-c}.$$

Proof. Put $s = \log(N/k)$ and $L = \log D_\beta$. For sufficiently large d , (36) has logarithm at most $-ks/4$, since its positive error is $O(k \log(1+d))$. Proposition 6.3 gives the second logarithmic bound $C_1 k - aL$. If $L \leq (2C_1/a)k$, then

$$\log D_\lambda \leq k(1+s) + L \leq C_2 ks,$$

so the first bound suffices. If $L > (2C_1/a)k$, the second bound is at most $-aL/2$. Taking the better of the two bounds then gives

$$\log \|T_N(\lambda)\|_{\text{op}}^2 \leq -\max\{ks/4, aL/2\} \leq -c_1(ks + L).$$

The tableau inequality $D_\lambda \leq \binom{N}{k} D_\beta \leq (eN/k)^k D_\beta$ again proves the result. Zero operator norm is covered directly. $\square$

8 Cycle traces and contraction by the transposition Casimir

The estimates in this section keep track of the number of boxes below the first row of a representation. The relevant scale is

$$h_n(k) = k \log(en/k), \quad 1 \leq k \leq n.$$

We will prove a density bound for every injection and a contraction bound with this same scale, then combine them to control a fixed trace moment. This gives a complete additional route from the switching network to the full permutation law.

Let $n = 2^d$, and let μ_n be the law of one directed coordinate sweep. In any unitary representation its average is $T_n = \Pi_d \cdots \Pi_1$, where Π_i averages the independent fair switches in direction i . Put $K_n = T_n^* T_n$ and write ν_n for its permutation law. Chronologically, ν_n runs a sweep and then its reflection, with fresh independent switches. For $n = 2m$, the network consists of an outer pair layer, two independent networks with law ν_m , and another outer pair layer. We write $K_n(\lambda)$ for the operator in the irreducible representation V_λ , and $D_\lambda = f^\lambda = \dim V_\lambda$. The same group-algebra operator also acts on tensor spaces and on injections; its indicated carrier will always specify which action is meant.

Write $X_{n,k} = [n]_{\neq}^k$ for the ordered distinct k -tuples of positions, and $u_{n,k}$ for their uniform probability measure. There are $(n)_k = n(n-1) \cdots (n-k+1)$ such tuples. For $x, y \in X_{n,k}$ define

$$K_{n,k}(x, y) = \Pr_{\pi \sim \nu_n} (\pi x = y), \quad R_x(y) = (n)_k K_{n,k}(x, y),$$

where πx means applying π to every entry of x . Thus R_x is a density relative to $u_{n,k}$. In particular its L^p norms use a probability measure, whereas all matrix traces and Schatten norms below are unnormalized.

Theorem 8.1 (Cycle density and Casimir contraction). *There are absolute constants $a, C > 0$ such that for every dyadic n , every $1 \leq k \leq n$, and every $x \in X_{n,k}$,*

$$\|R_x\|_{L^{65/64}(u_{n,k})} \leq e^{Ch_n(k)}.$$

For every nontrivial partition $\lambda \vdash n$, put $k = n - \lambda_1$. Then

$$\|K_n(\lambda)\|_{\text{op}} \leq e^{-ah_n(k)}, \quad D_\lambda \text{Tr } K_n(\lambda)^{65} \leq e^{Ch_n(k)}.$$

*Consequently there is an absolute integer l for which $\|\mu_n^{*l} - U_{S_n}\|_{\text{TV}} \rightarrow 0$ as $n \rightarrow \infty$ through powers of two. Here U_{S_n} is the uniform permutation law and total variation includes the factor $1/2$. The same bound holds from every deterministic initial permutation. When $n = 2^d$, these l sweeps take ld physical shuffles.*

The density proof counts cycles through positive tensor traces. For the operator bound we use the transposition Casimir, the sum of the positive operators $I - U((i j))$. Its scalar on V_λ is comparable to nk . Two estimates for a random split of this sum control, respectively, the loss of total Casimir mass and the imbalance between the two children. They allow an induction with a fixed multiplicative margin. Finally, Hausdorff–Young turns the injection-density bound into the 65th moment, and Schatten Hölder applies that moment to powers of the directed sweep.

8.1 Cycle traces and tuple spreading

Our first objective is the $L^{65/64}$ density bound. A selected cycle at a network node contributes a factor two to the number of possible routings. We therefore need an exponential moment for the sum of these cycle counts. Positive tensor traces first bound the full cycle count; a conditional estimate across levels then controls the selected cycles.

If π is a permutation let $\kappa(\pi)$ denote the number of cycles, counting fixed points. Define for real $x \geq 1$

$$Z_b(x) = \mathbb{E}_{\pi \sim \nu_b} x^{\kappa(\pi)}$$

where b is a power of 2. We first show that there are absolute $C < \infty$ and $\gamma > 0$ such that, for $b = 2^w$,

$$\log Z_b(1 + 2^{\lfloor w/8 \rfloor}) \leq Cb^{1-\gamma}. \quad (45)$$

For integer $q \geq 1$, $Z_b(q)$ is the trace of K_b on the tensor space $(\mathbb{C}^q)^{\otimes b}$ with permutations acting by permuting tensor legs. In fact the trace of a permutation counts its fixed words on an alphabet of size q . The following identity is the common tensor calculation underlying this proof and the harmonic cycle estimates.

Lemma 8.2 (Tensor-swap recursion). *For a dyadic m and an integer $q \geq 1$, let D be the action of K_m on $(\mathbb{C}^q)^{\otimes m}$. For $S \subseteq [m]$, let $A_S = \text{Tr}_{S^c} D$, where the partial trace is unnormalized. Then*

$$Z_{2m}(q) = 2^{-m} \sum_{S \subseteq [m]} \text{Tr } A_S^2, \quad Z_{2m}(q) \leq Z_m(q)Z_m(1 + (q-1)/2).$$

The scalar function $Z_m(x)$ in the second formula is defined for every real $x \geq 1$, including a nonintegral argument.

Proof. In this representation the recursion at size $2m$ gives the operator

$$P(D \otimes D)P, \quad D = K_m \text{ on } (\mathbb{C}^q)^{\otimes m},$$

where $P = \prod_{i=1}^m ((I + F_i)/2)$ with F_i interchanging leg i between the two copies (number the pair indices by $i = 1, \dots, m$, using the same ordering in both copies). Here $P^2 = P$ and D is positive semidefinite. Taking a cyclic trace and expanding P gives

$$Z_{2m}(q) = 2^{-m} \sum_{S \subseteq [m]} \text{Tr} [(\text{Tr}_{S^c} D)^2], \quad (46)$$

where Tr_{S^c} is ordinary, unnormalized partial trace over legs in the complement. Indeed for the product of swaps on S , first trace out the non-swapped legs and then use $\text{Tr}(F(A \otimes A)) = \text{Tr}(A^2)$ with F the whole swap of the remaining copies.

The operator $A_S = \text{Tr}_{S^c} D$ is positive semidefinite, with trace $Z_m(q)$. For an individual permutation operator in D , its partial trace is $q^{\kappa_0(\pi, S)}$ times a permutation operator on the legs in S , where κ_0 counts cycles disjoint from S . This follows by writing matrix entries in the word basis: an input color index is equated to the output index at the next site along the cycle, and input is set equal to output on legs traced out and summed. Thus each deleted whole cycle has a free common index, and on other cycles the deleted legs simply shorten the cycle to its undeleted legs. It follows that $\|A_S\| \leq \mathbb{E}_{\pi \sim \nu_m} q^{\kappa_0(\pi, S)}$, where $\|\cdot\|$ on operators denotes operator norm. Use $\text{Tr} A_S^2 \leq \|A_S\| \text{Tr} A_S$. With S a uniform subset, independent of π , we obtain

$$\begin{aligned} Z_{2m}(q) &\leq Z_m(q) \mathbb{E}_{\pi \sim \nu_m} \prod_{\text{cycles } C_1 \text{ of } \pi} (1 + (q-1)2^{-|C_1|}) \\ &\leq Z_m(q) Z_m(1 + (q-1)/2). \end{aligned} \quad (47)$$

□

A slowly growing alphabet. We now derive the stated bound for Z_b from the recursion. This proof retains the decay obtained by following the alphabet parameter until it reaches its smallest value. Let $h_w(r) = 2^{-w} \log Z_{2^w}(1 + 2^r)$ for integers $r \geq 0$. For $w \geq 1, r \geq 1$, (47) gives

$$h_w(r) \leq \frac{1}{2} h_{w-1}(r) + \frac{1}{2} h_{w-1}(r-1).$$

At the boundary $r = 0$, concavity gives $Z_m(3/2) \leq Z_m(2)^\alpha$, where $\alpha = \log_2(3/2)$. Consequently

$$h_w(0) \leq \theta h_{w-1}(0), \quad \theta = (1 + \alpha)/2 < 1, \quad h_0(r) \leq (r+1) \log 2.$$

To iterate these inequalities, start at r and, while positive, stay in place or decrement by one with equal probabilities. After reaching zero, remain there and multiply by θ at each subsequent step. The recurrence bounds $h_w(r)$ by $(r+1) \log 2$ times the expected product of these factors over w steps.

Suppose $r \leq w/8$. If B counts successes in $\lfloor w/2 \rfloor$ independent fair trials, the probability that the process has not reached zero by that time is at most

$$\Pr(B < r) \leq 2^r \mathbb{E} 2^{-B} = 2^r (3/4)^{\lfloor w/2 \rfloor} \leq C e^{-cw},$$

since $2^{1/8} \sqrt{3/4} < 1$. On the complementary event the product of factors is at most $\theta^{w/2}$. Thus

$$h_w(r) \leq (r+1) \log 2 \{C e^{-cw} + \theta^{w/2}\}.$$

Taking $r = \lfloor w/8 \rfloor$ proves (45), with an absolute $\gamma > 0$ decreased as necessary to absorb the linear factor in w .

Conditional cycle counts. The trace bound concerns an unconditioned network of one size. To use it on the many levels of a larger network, we leave a specified interval of switch levels fresh and require a bound uniform in all other switches. We next use (45) for switches

sampled in the recursion defining ν_n . Nodes of size 2^j will be called level j ($1 \leq j \leq d$), with $n/2^j$ nodes at that level indexed by fixed prefixes of length $d - j$. Regard the input and output outer switches of every node as its switch variables at that level; the raw switch variables are independent. For a realization, draw at a node the edges taken by all its cards between its input pair indices and output pair indices (two parts); the graph is bipartite 2-regular, allowing double edges. Its two full matchings, together giving the edges with multiplicity, come from the two child bijections A, B on the $m = 2^{j-1}$ pair indices: the outer switches do not change these indices and there is one card per input pair going to each child. The number of cycle components, including cycles of two edges, is $\kappa(A^{-1}B)$ (traverse using B -edges forward, A -edges backward). Let C_j be the sum of these numbers over level j . It depends only on switches at levels strictly below j , regardless of card identities coming into the nodes.

Fix $1 \leq k \leq n$, and put $W_j = \min(k/2, C_j)$. With

$$\delta = 1/64$$

we claim the absolute bound

$$\log \mathbb{E} 2^{\delta \sum_{j=1}^d W_j} \leq Ck \log(en/k). \quad (48)$$

For $j \geq 64$ let

$$w = \lfloor j/2 \rfloor, \quad b = 2^w, \quad I_j = \{j - w, \dots, j - 1\}.$$

Condition on all level switch variables outside I_j . We will bound the exponential moment at the higher exponent δj . Take $q = 1 + 2^{\lfloor w/8 \rfloor}$, so $q \geq 2^{\delta j}$ ($\lfloor w/8 \rfloor = \lfloor j/16 \rfloor \geq j/64$ for $j \geq 64$). Within each child of size $m = 2^{j-1}$ of a node at level j , the interval I_j supplies w outer layers of input switches and the corresponding output layers. We can additionally condition on all the output switch variables there. The child input switches in I_j act as independent w -bit sweeps in m/b blocks (first w bits of the child position index used, its other bits fixed within a block); the blocks are aligned across the two children. Thus for a node, $A^{-1}B$ now has the form

$$U_A^{-1} G U_B$$

where G is a fixed permutation of the m slots and U_A, U_B are independent permutations with law the product of μ_b over those blocks. Indeed each child map first applies the indicated input switches, then its deeper fixed submaps and its conditioned output switches. In the q -color tensor representation on m slots, let T_{bl} be the mean operator of the input sweeps in blocks, and $U(G)$ the representation of G . Conditionally,

$$\mathbb{E} q^{\kappa(A^{-1}B)} = \text{Tr}(T_{\text{bl}}^* U(G) T_{\text{bl}}) \leq \text{Tr}(T_{\text{bl}} T_{\text{bl}}^*) = Z_b(q)^{m/b}.$$

Here we bounded the absolute trace against a positive semidefinite matrix using unitarity, and used the tensor product over blocks and cyclicity. Different level- j nodes use independent remaining randomness. The bound is uniform so we can drop the extra output conditioning. We get, with $\mathcal{F}_{\bar{I}_j}$ denoting the switches outside I_j ,

$$\log \mathbb{E}[2^{\delta j W_j} \mid \mathcal{F}_{\bar{I}_j}] \leq \min((\log 2) \delta j k/2, Cnb^{-\gamma}). \quad (49)$$

Indeed the second bound applies to C_j by multiplying over the $n/2^j$ nodes and using (45) with $(n/2^j)(m/b)b^{1-\gamma} \leq nb^{-\gamma}$.

Combining overlapping level intervals. Estimate (49) is conditional, so it does not require different cycle counts to be independent. We will use it first on collections of disjoint

intervals, and then average those collections by Hölder. To combine intervals without paying a Hölder exponent of more than j on level j , observe that for a fixed level l ,

$$\sum_{j \geq 64: l \in I_j} \frac{1}{j} \leq \sum_{j=l+1}^{2^l} \frac{1}{j} \leq 1.$$

Using only $64 \leq j \leq d$, we can find weights $\theta_S \geq 0$, summing to 1, on subcollections $\mathcal{S}$ of these indices with pairwise disjoint intervals, such that $\sum_{S \ni j} \theta_S = 1/j$ for each such j . For clarity, assign to each interval, in order of increasing left endpoints (ties ordered arbitrarily), a measurable subset of $[0, 1]$ of length $1/j$, disjoint from the sets of earlier overlapping intervals. This is possible because those earlier overlapping intervals all contain the current left endpoint, so their sets' total length leaves enough room. Partition by membership in the assigned sets, taking lengths as weights. Weighted Hölder now gives

$$\mathbb{E} 2^{\delta \sum_{64 \leq j \leq d} W_j} \leq \prod_{\mathcal{S}} \left(\mathbb{E} 2^{\delta \sum_{j \in \mathcal{S}} j W_j} \right)^{\theta_S}$$

(positive weights suffice in the product, and the exponents inside the expectations on the right give the one on the left when averaged with those weights). Within a collection, use (49) on the largest j first and continue downwards. The terms from smaller j' in the collection depend only on levels $\leq j' - 1$, hence outside I_j since $I_{j'}$ ends at $j' - 1$ and the intervals are disjoint with $j' - 1 < j - 1$. Taking logs, the resulting bounds (49) occur with factors $1/j$. Handling $j < 64$ deterministically, we obtain an upper bound for the left hand side of (48) by

$$Ck + \sum_{j \geq 64} \min(Ck, Cn 2^{-\gamma \lfloor j/2 \rfloor} / j) \leq C'k \log(en/k),$$

where the sum bound can use all $j \geq 64$: after $j_0 = \lceil (2/\gamma) \log_2(n/k) \rceil$, $n 2^{-\gamma \lfloor j/2 \rfloor} \leq 2^{\gamma k} 2^{-(\gamma/2)(j-j_0)}$. Thus the tail costs at most Ck , and there are only $O(\log(en/k))$ terms before it. This proves (48).

From cycle weights to an injection density. The cycle moment is now available under the actual switch law. The next calculation relates it to the auxiliary uniform choices of legal child colors. Keeping these two probability laws separate is necessary when we estimate a power of the density. Denote by $[n]_{\neq}^k$ the ordered distinct k -tuples of positions, of cardinality $(n)_k = n(n-1) \cdots (n-k+1)$; positions can again be indexed by bit words, and πx denotes acting by the permutation π on every coordinate of x . The following row bound is what we need:

$$\begin{aligned} K_{n,k}(x, y) &= \Pr_{\pi \sim \nu_n} (\pi x = y), & R_x(y) &= (n)_k K_{n,k}(x, y) \\ \implies \|R_x\|_{1+\delta} &\leq \exp(Ck \log(en/k)). \end{aligned} \tag{50}$$

for every such x , with norm on uniform $y \in [n]_{\neq}^k$.

Here is a partial route-counting identity. Fix the input and output tuples at a node. Draw the selected edges (the specified cards) between its input and output pair indices. Their components with edges are paths or cycles, say there are p_0 paths and s cycles, with double edges forming a cycle as well. There are $k + p_0$ vertices incident to edges (here k refers to the number of cards specified at this node). A proper edge coloring with two colors corresponds to assigning child halves, with two choices on each component by alternation. For each such coloring, the probability of the required outer switches, input and output together, is $2^{-(k+p_0)}$: exactly one switch is prescribed per incident vertex, and the prescriptions are consistent since the tuples are distinct and the coloring is proper. In each child we now prescribe input and output tuples on the reduced indices, again distinct, for the tags of its color (keep their given

order). With an empty tuple the kernel is interpreted as 1. Multiplying by n^k for a node of size $n = 2m$, the identity at that node is

$$n^k K_{n,k}(x, y) = 2^s \mathbb{E}_{\text{color}} \left[\prod_{a=0}^1 m^{k_a} K_{m,k_a}(x'_a, y'_a) \right]$$

where the proper coloring is uniform and the child data (sizes k_a) are induced by it. At size 1 the normalized kernel is 1. Recursing defines a base coloring experiment for fixed x, y , with independent uniform coloring choices at each level's nodes conditional on the already induced data there, inducing child data all the way down. Let S be the sum of selected-graph cycle counts s over its nodes. Thus $n^k K_{n,k} = \mathbb{E}_{\text{base}} 2^S$.

More precisely, for any outcome of this coloring tree, the probability under the actual full switches of taking x to y with those specified child assignments along the way is $n^{-k} 2^S$ times its probability in the base experiment. Indeed at each internal node v , with local tuple size and component counts k_v, p_v, s_v (cards, paths with edges, cycles), the probability of the needed outer switches is $2^{-(k_v+p_v)}$, versus $2^{-(p_v+s_v)}$ for the designated coloring under the base experiment given its incoming data. For this fixed tree the switch requirements are prescriptions of raw coins, disjoint across the nodes, and together necessary and sufficient: tuples remain injective in each child (proper coloring) and the induced endpoints at size 1 require nothing more, with all unselected cards free of endpoint restrictions. The ratio thus multiplies to $\prod_v 2^{s_v-k_v} = 2^S n^{-k}$, since the k_v 's sum to the original k on each level. Conversely a full switch realization from x determines a unique selected endpoint and coloring tree for it (actual child routes).

On a realization of the full switches starting with tuple x , call the corresponding total cycle count S_x , using its actual endpoint and child routes. Any cycle in a selected partial graph is a full cycle component of that node's graph from the realization, since its vertices already have degree 2 on the cycle. Also each uses at least two selected cards, with disjoint cards across cycles of a level. It follows that $S_x \leq \sum_j W_j$ (now with the original full tuple size k) on every realization. By convexity and the refined identity, with $u = (n)_k/n^k \leq 1$,

$$\mathbb{E}_{y \text{ uniform}} R_x(y)^{1+\delta} \leq \frac{u^{1+\delta}}{(n)_k} \sum_y \mathbb{E}_{\text{base}(x,y)} 2^{(1+\delta)S} = u^\delta \mathbb{E}_{\text{actual}} 2^{\delta S_x}.$$

Now (48) proves (50).

8.2 A contraction measured by the transposition Casimir

The density estimate is complete. We now prove the operator contraction on the same scale $h_n(k)$. The induction splits the positions into two halves, so its main task is to compare the parent Casimir scalar with the two child scalars seen by a vector fixed by the outer switches.

We use standard representation theory of the symmetric group over $\mathbb{C}$: irreducibles are indexed by partitions λ of n , of dimensions f^λ (numbers of standard Young tableaux). We use Young's branching rule for restriction down the symmetric groups (multiplicity given by paths of successively removing single boxes while keeping diagrams of partitions), and the standard transposition-sum content formula (by the Young–Jucys–Murphy content theorem [31, Eq. (2.1), Proposition 5.3 and Theorem 5.8]), that $\sum_{i < j} (i j)$ acts in λ by the sum of column-minus-row indices of its boxes. Indices of rows and columns here start at 1. The permutation module on ordered distinct k -tuples is the coset module with stabilizer S_{n-k} , so multiplicity of λ is the dimension of its stabilizer-fixed space by Frobenius reciprocity. In particular for $k = n - \lambda_1$, λ occurs in this module and (if $k \geq 1$) not in the corresponding

$k - 1$ tuple module. Its multiplicity at this k is $f^{\bar{\lambda}}$, where $\bar{\lambda}$ consists of all rows below the first: paths in the branching rule to the trivial shape $(n - k)$ must remove just that tail. Also

$$f^{\lambda} \leq \binom{n}{k} f^{\bar{\lambda}} \quad (51)$$

by choosing the entries of a standard tableau below the top row.

Let U denote the unitary action of S_n on the representation currently under consideration. Write $D_{ij} = I - U((ij))$ on a unitary representation (indices i, j are positions). They are positive semidefinite. Put $\mathcal{C} = \sum_{i < j} D_{ij}$, which in λ acts by

$$c = c_{\lambda} = \frac{1}{2}(n^2 - \sum_i \lambda_i^2) + \sum_i (i - 1)\lambda_i, \quad nk/2 \leq c \leq nk, \quad k = n - \lambda_1. \quad (52)$$

For the lower bound use $\sum \lambda_i^2 \leq n\lambda_1$. For the upper, the sum of squares below row 1 is at least k , and the sum of row-minus-one indices over tail boxes is at most $k(k + 1)/2$ (ordering tail boxes by row, the a -th has row index at most $a + 1$). Define

$$\Phi_n(c) = \frac{c}{n}(2 + \log(n^2/c)), \quad \Phi_n(0) = 0.$$

We prove there are absolute $B > 1, \eta > 0$ such that on every nontrivial irreducible λ for $n = 2^d$,

$$\|K_n\|_{\lambda} \leq B^{-1} \exp(-\eta \Phi_n(c_{\lambda})). \quad (53)$$

The norm notation means the operator norm in the indicated irreducible. In particular the bound implies $\|K_n\|_{\lambda} \leq \exp(-ak \log(en/k))$ there for an absolute $a > 0$: $k/2 \leq c/n \leq k$ gives $\Phi_n(c) \geq (k/2)(2 + \log(n/k))$.

We will do an induction for (53) after some random-split estimates. In outline, K_n is the child product-law operator compressed by Π_1 , so we study Rayleigh values in unit vectors invariant under the bit-1 switch group. Under restriction to the two halves' product subgroup, the induction hypothesis will contribute two child Φ -values (evaluated at the partial Casimir scalars on the irreducibles of the factors), in place of the $\Phi_n(c)$ in (53). The invariant vector lets us analyze the distribution of the partial Casimir values (spectral weights in the vector) by randomly flipping which member of each pair lies in each half. We control the sum of these two scalars relative to $c/2$, and their difference, for the Φ -comparison; we also use the high probability of two nontrivial child factors in the spectral weighting for large k so that the factor B^{-1} in (53) gives slack.

8.2.1 Two estimates for a random split

Partition the positions into the $m = n/2$ switch pairs of bit 1, using subscripts $p0, p1$ in a pair p for its bit-1 values 0 and 1, respectively. Draw independent fair signs s_p . Write $\xi_{p0} = s_p$, $\xi_{p1} = -s_p$, and let L, R be the positions with $\xi = +1, -1$ respectively. The two partial Casimirs $\mathcal{C}_L, \mathcal{C}_R$ are the sums of D_{ij} for pairs of positions both in the indicated side. They commute for a fixed split. Use the operators

$$\Delta = \mathcal{C} - 2(\mathcal{C}_L + \mathcal{C}_R), \quad \mathcal{R} = \mathcal{C}_L - \mathcal{C}_R.$$

On λ , both Δ and $\mathcal{R}$ are between $-cI$ and cI by positivity of the terms of $\mathcal{C}$. The sum defect Δ measures how much Casimir mass is lost in the split, while $\mathcal{R}$ measures the difference between the two sides. They require different estimates: an exponential moment of Δ itself and an exponential moment of $\mathcal{R}^2$.

Lemma 8.3 (Exponential moments for a transversal split). *There are absolute constants $t_0, t_1 > 0$ and $C_0 < \infty$ such that, for every dyadic $n \geq 2$ and every nontrivial $\lambda \vdash n$, with $k = n - \lambda_1$ and $c = c_\lambda$,*

$$\left\| \mathbb{E}_s e^{t\Delta/n} \right\|_\lambda \leq C_0 \quad \text{at } t = t_0 \log(en/k), \quad \left\| \mathbb{E}_s e^{t_1 \mathcal{R}^2/(nc)} \right\|_\lambda \leq C_0, \quad (54)$$

The expectations are over the independent fair signs defining the transversal split, and both norms are operator norms on V_λ .

Proof. The imbalance moment. We start with the second estimate. Write $F_i = \sum_{j \neq i} D_{ij}$. Then $\mathcal{R} = \frac{1}{2} \sum_i \xi_i F_i = \sum_p s_p W_p$ with $W_p = (F_{p0} - F_{p1})/2$. On λ ,

$$K_0 = \max_p \|W_p\| \leq n, \quad M^2 = \left\| \sum_p W_p^2 \right\| \leq 2nc,$$

using $0 \leq F_i \leq 2(n-1)I$, $F_i^2 \leq 2(n-1)F_i$, $\sum_i F_i = 2cI$, and $\sum_p W_p^2 \leq \frac{1}{2} \sum_i F_i^2$ (by $(F_{p0} + F_{p1})^2 \geq 0$ for each pair). For such self-adjoint matrices the following moment bound suffices, for integers $l \geq 1$:

$$\left\| \mathbb{E}_s (\sum_p s_p W_p)^{2l} \right\| \leq (C(M\sqrt{l} + K_0 l))^{2l}. \quad (55)$$

Here are details without a dimension factor. Expand the noncommuting product. In the expectation of the scalar signs, for sign indices $p_1, \dots, p_{2l}$ we have

$$\mathbb{E}_s \prod_{u=1}^{2l} s_{p_u} = \sum_{\mathcal{B}} \prod_{B_0 \in \mathcal{B}} \left(\mathbf{1}_{\{p_u \text{ all equal for } u \in B_0\}} \chi_{|B_0|} \right), \quad \chi_b = \left. \frac{d^b}{dz^b} \log \cosh z \right|_{z=0}$$

over set partitions $\mathcal{B}$ of the $2l$ places. Indeed the log of the generating expectation for $\exp(\sum_u z_u s_{p_u})$ is $\sum_p \log \cosh(\sum_{u: p_u=p} z_u)$ as power series near 0. Exponentiating and taking the multilinear coefficient gives the expansion. Singletons have zero coefficient, and $|\chi_b| \leq b! C^b$ by analyticity near 0.

In the term for a fixed partition with h blocks of sizes at least 2, there are within-block index equalities in the sum of operator words $W_{p_1} \cdots W_{p_{2l}}$, but indices across blocks can be summed without distinctness restrictions. The norm of this contracted sum (before the factors χ_b) is bounded by $M^{2h} K_0^{2l-2h}$. To see this, apply from right to left, keeping an extra index register $\mathbb{C}^m$ (with orthonormal basis indexed by pairs p) per open block in tensor product with the representation. On first encountering a block, use the column map $x \mapsto \sum_p W_p x \otimes e_p$ of norm M , inserting its register; on final occurrence use its adjoint ($W_p = W_p^*$); on intermediate occurrences use $\sum_p W_p \otimes |e_p\rangle\langle e_p|$ with that register, of norm at most K_0 . At each step tensor with identities on other open registers and permute registers as needed. This composition realizes precisely the sum over assignments of indices to blocks, as the first encounter branches over the new index, intermediate encounters use that same index, and the final one sums it out.

Summing absolute bounds using ordered compositions $b_1 + \cdots + b_h = 2l$ with $b_i \geq 2$, partitions are accounted for with factor $(2l)!/(h! \prod_i b_i!)$. The $b_i!$ cancel in the bound from $\prod_i (b_i! C^{b_i})$, there are at most 2^{2l} compositions in total, and $(2l)!/h! \leq (2l)^{2l-h}$. Now $l^{2l-h} M^{2h} K_0^{2l-2h} = (M\sqrt{l})^{2h} (K_0 l)^{2l-2h}$ with $h \leq l$, giving (55).

For $1 \leq l \leq k$, (55) and (52) imply $\|\mathbb{E}_s(\mathcal{R}^2/(nc))^l\|_\lambda \leq (Cl)^l (M/\sqrt{nc} \leq \sqrt{2}, K_0/\sqrt{nc} \leq \sqrt{2/k})$. For $l > k$, use $\|\mathcal{R}^2/(nc)\| \leq c/n \leq k$ on λ instead, so the same bound holds. The exponential series, using $l! \geq (l/e)^l$, now proves the $\mathcal{R}$ bound in (54) at a sufficiently small $t_1 > 0$.

The sum defect on injections. The imbalance bound is now proved. For the sum defect we will first replace signs by an entrywise majorant, then bound that majorant through a Gaussian tensor calculation. This larger carrier avoids a factor D_λ in the estimate. We prove the first estimate in (54) on the whole ordered distinct k -tuple module $\mathcal{V} = \ell^2(X_{n,k})$, which contains V_λ . Embed $\mathcal{V}$ in $(\mathbb{C}^n)^{\otimes k}$ with the standard ordered tuple basis, so $U(g)|x\rangle = |gx\rangle$, and write $D_\neq$ for its orthogonal projection. Let $\mathcal{V}_{\text{sym}} \subseteq \mathcal{V}$ consist of vectors invariant under permutations of the tensor slots. For a real vector u on positions put $p_u = |u\rangle\langle u|/n$ and $\mathcal{L}^u = \sum_{a=1}^k p_u^{(a)}$. On $\mathcal{V}$, the identity $\sum_i \xi_i = 0$ gives

$$\Delta = - \sum_{i < j} \xi_i \xi_j D_{ij} = n D_\neq \mathcal{L}^\xi D_\neq + B^\xi, \quad (B^\xi f)(x) = \sum_{a < b} \xi_{x_a} \xi_{x_b} (f(x) + f(x^{a \leftrightarrow b})), \quad (56)$$

where the tuple on the right interchanges two slots. In fact in the transposition sum one separates replacements of a single occupied position by an unoccupied one from interchanges of two occupied positions. Both have off-diagonal coefficient $\xi_i \xi_j$ for positions i, j involved. The diagonal at x , for its occupied set X , equals $-\sum_{i \in X, j \notin X} \xi_i \xi_j - \sum_{i < j: i, j \in X} \xi_i \xi_j = k + \sum_{i < j: i, j \in X} \xi_i \xi_j$.

Removing the opposite signs inside a switch pair. All entries of powers of the last expression in (56) on distinct tuples are given by polynomials in the position variables with nonnegative coefficients before substituting the signs. Replace ξ by v with $v_{p0} = v_{p1} = s_p$ in this expression (the resulting operator $\tilde{\Delta} = n D_\neq \mathcal{L}^v D_\neq + B^v$ need not equal the transposition formula with that replacement). For any such monomial with the original signs, the absolute value of its average is bounded by its average with the replacement: both averages vanish on odd pair-index parity at any p , and the replacement gives $+1$ on all even parities. For $t \geq 0$, it follows by the series and triangle inequality that entrywise

$$\left| (\mathbb{E}_s e^{t\Delta/n})_{xy} \right| \leq (\mathbb{E}_s e^{t\tilde{\Delta}/n})_{xy}.$$

The latter matrix is real symmetric entrywise nonnegative and commutes with permutations of the tensor slots (in fact $\tilde{\Delta}$ commutes with them for every set of signs). Its operator norm, which bounds the former norm, is attained on slot-symmetric distinct tensors. Indeed a symmetric entrywise nonnegative matrix has a nonnegative eigenvector for its largest eigenvalue realizing the spectral radius (or maximize using entrywise absolute values in the Rayleigh bound), and here it can be averaged over slot permutations without vanishing. Entrywise domination in absolute value by nonnegative matrices bounds operator norm by applying them to entrywise absolute vectors.

On the slot-symmetric distinct subspace use the orthonormal basis indexed by k -sets X , the normalized sums of orderings. In this basis $\tilde{\Delta}$ has diagonal $V_s(X) = (\sum_{i \in X} v_i)^2$, coming from k on the diagonal of $n D_\neq \mathcal{L}^v D_\neq$ and $2 \sum_{i < j: i, j \in X} v_i v_j$ from B^v on symmetric functions. Its off-diagonal is $v_i v_j$ between X and $(X \setminus \{i\}) \cup \{j\}$ ($i \in X, j \notin X$), zero otherwise. Let J be the unweighted adjacency matrix of this single-exchange graph. Conjugating $\tilde{\Delta}$ by the diagonal signs $v_X = \prod_{i \in X} v_i$ gives $J + V_s$ with V_s diagonal. Thus in computing $(e^{t\tilde{\Delta}/n})_{XY}$ by this conjugation, the sign factor is $v_X v_Y = \prod_{p \in O} s_p$, where $O = O(X, Y)$ consists of pair indices of odd combined occupancy in X, Y (counting multiplicity). Write $h = |O|$, always even.

A Gaussian bound along each path. We need to average the endpoint sign $v_X v_Y$ together with the nonnegative potential V_s . An expansion over paths keeps this sign visible and turns the potential into a Gaussian covariance. In the exponential kernel of $(J + V_s)/n$ over duration t , expand in graph paths using J/n , with a multiplier on each timed path $X(u)$ given by

$$\exp \left(\frac{1}{n} \int_0^t V_s(X(u)) du \right).$$

Specifically, using symmetry of the kernel, its X, Y entry is

$$\sum_{r=0}^{\infty} n^{-r} \sum_{X=X_0 \sim X_1 \sim \dots \sim X_r=Y} \int_{0 < u_1 < \dots < u_r < t} \exp \left(n^{-1} \sum_{i=0}^r (u_{i+1} - u_i) V_s(X_i) \right) du_1 \cdots du_r,$$

where $\sim$ denotes adjacency in J , $u_0 = 0$, $u_{r+1} = t$, $X(u) = X_i$ between successive times u_i, u_{i+1} , and $r = 0$ means no jump ($X = Y$). This is the ordinary iterated exponential expansion (Duhamel iteration with the exponential of the diagonal term retained between jumps); it converges absolutely by boundedness in finite dimension and the simplex volumes $t^r/r!$. The nonnegative baseline weights without the multipliers sum to $(e^{tJ/n})_{XY}$. On a fixed timed path let $b_p(u) \in \{0, 1, 2\}$ denote occupancy at pair p , so $\sum_p b_p = k$. Define a centered real Gaussian z on the m pair indices with covariance

$$H = \frac{2}{n} \int_0^t b(u) b(u)^\top du.$$

The path multiplier is $\mathbb{E}_z \exp(\sum_p z_p s_p)$. Thus after including $v_X v_Y$, averaging the signs on the path gives $\mathbb{E}_z \prod_{p \in O} \sinh z_p \prod_{p \notin O} \cosh z_p$, bounded in absolute value by $\mathbb{E}_z e^{\|z\|_2^2/2} \prod_{p \in O} |z_p|$, using $\cosh y \leq e^{y^2/2}$, $|\sinh y| \leq |y|e^{y^2/2}$ (by the series). We have $\text{Tr } H \leq 4kt/n$, $H_{pp} \leq 8t/n$. From now on take $t = t_0 \log(en/k)$ for an absolute $t_0 > 0$ sufficiently small, starting with $t_0 \leq 1/8$; then $\text{Tr } H \leq 1/2$ since $(k/n) \log(en/k) \leq 1$. Tilting the Gaussian by $e^{\|z\|_2^2/2}$ gives mass $\det(I - H)^{-1/2} \leq e^{\text{Tr } H}$ and covariance $H(I - H)^{-1} \leq 2H$ after normalization, as seen in an eigenbasis (allowing zero eigenvalues, and using $-\log(1 - y) \leq 2y$ for $0 \leq y \leq 1/2$). By scalar Gaussian absolute moments and Hölder on the h factors, the sign-averaged path multiplier with the endpoint sign included has absolute value at most

$$C(C_1 \sqrt{th/n})^h$$

with the power taken as 1 if $h = 0$. The constants C, C_1 here can be taken absolute for all $0 < t_0 \leq 1/8$, using e.g. $\mathbb{E}|Z|^h = (h-1)!! \leq h^{h/2}$ for a standard normal Z and positive even h . Uniformly summing the path weights, we have on this symmetric subspace the entrywise nonnegative averaged matrix $\mathbb{E}_s e^{t\tilde{\Delta}/n}$ bounded entrywise above by

$$C(e^{tJ/n})_{XY} (C_1 \sqrt{th/n})^h. \quad (57)$$

Absorbing the endpoint factor into a tensor operator. The path estimate (57) still depends on h , the number of pair indices with odd endpoint occupancy. A common Gaussian coordinate will supply exactly the moments needed to absorb this factor. After that comparison, only the norm of one explicit positive tensor average remains. We absorb this extra weight into a tensor operator. With $\mathbf{1}$ the all-ones vector on positions, on the same symmetric distinct basis $(J + kI)/n$ is the restriction of $D_{\neq} \mathcal{L}^1 D_{\neq}$. All slot sums here commute with slot permutations. We can drop a scalar factor $e^{-tk/n}$, and in the remaining exponential power series use entrywise nonnegativity in the ordered basis to drop the distinctness requirement on intermediates (then compare on normalized sums of orderings). Thus $e^{tJ/n}$ is bounded entrywise by the symmetric-distinct compression of

$$e^{t\mathcal{L}^1} = \bigotimes_{a=1}^k (I + \beta p_{\mathbf{1}}), \quad \beta = e^t - 1.$$

To dominate the weight depending on $O(X, Y)$ as well, take a duplicated Gaussian vector g on positions with

$$g_{p0} = g_{p1} = g'_p + y_*,$$

where g'_p are independent standard real Gaussians and y_* is an independent centered real Gaussian of variance C_*t/n . Then (57) is bounded entrywise by C times the symmetric distinct compression of

$$G_t := \mathbb{E}_g \bigotimes_{a=1}^k (I + \beta p_g). \quad (58)$$

Indeed expand each tensor product, checking on ordered endpoints x, y with sets X, Y . Identity slots (inactive in a contributing term) require the same position at both ends. Thus in the active slots $E \subseteq [k]$, the occurrences of pair indices among x_a, y_a for $a \in E$ have the same parities (odd-index set $O(X, Y)$). In (58) we get the additional factor $\mathbb{E}_g \prod_{a \in E} g_{x_a} g_{y_a}$ compared to the tensor expansion with p_1 . The Gaussian monomial expectations are nonnegative; expanding g , assign for each odd pair-index one occurrence to y_* and the remaining occurrences everywhere to the independent g'_p , all the latter degrees then even. Those even moments of standard normals are at least 1, and for positive even h , $\mathbb{E} y_*^h = (C_*t/n)^{h/2} (h-1)!! \geq (C_1 \sqrt{th/n})^h$ by taking C_* absolutely large enough; for instance $(h-1)!! = h!/(2^{h/2}(h/2)!) \geq (h/e^2)^{h/2}$. This also works with moment 1 at $h = 0$. Other assignments contribute nonnegative expectations. Summing over orderings with the basis normalizations for the symmetric-distinct compression preserves the entrywise bound.

Let $Q : \mathcal{V}_{\text{sym}} \rightarrow (\mathbb{C}^n)^{\otimes k}$ be the isometric inclusion, whose k -set basis vectors are normalized sums of their orderings. The comparisons just proved give

$$\begin{aligned} \|\mathbb{E}_s e^{t\Delta/n}\|_{V_\lambda} &\leq \|\mathbb{E}_s e^{t\Delta/n}\|_{\mathcal{V}} \\ &\leq \|\mathbb{E}_s e^{t\tilde{\Delta}/n}\|_{\mathcal{V}_{\text{sym}}} \|\cdot\|_{\text{op}} \\ &\leq C \|Q^* G_t Q\|_{\text{op}} \leq C \|G_t\|_{\text{op}}. \end{aligned} \quad (59)$$

The middle inequalities use entrywise domination in the indicated orthonormal bases and the nonnegative eigenvector argument above. The last inequality is the norm bound for an orthogonal compression. It remains to prove $\|G_t\|_{\text{op}} \leq 2$ on the full tensor product.

Bounding the Gaussian tensor average. On the one-slot space, p_g is supported on the pair-sum subspace with orthonormal basis $(e_{p0} + e_{p1})/\sqrt{2}$. There it becomes $|\tilde{g}\rangle\langle\tilde{g}|/m$, where $\tilde{g}_p = g'_p + y_*$ has covariance $I + (C_*t/n)\mathbf{1}\mathbf{1}^\top$ in m dimensions. Splitting each slot into this subspace and its orthogonal complement makes G_t block diagonal. Complementary slots contribute identity, so a block with $r \leq k$ pair-sum slots requires a bound only on $(\mathbb{C}^m)^{\otimes r}$.

Diagonalize the covariance by a real orthogonal change of basis. The new coordinates are independent centered Gaussians with variances at most $\Gamma = 1 + C_*t/2$. In this basis, every averaged monomial in the tensor expansion is zero or a nonnegative product of even moments. Increasing all these variances to Γ therefore gives an entrywise upper bound, and hence an operator-norm upper bound.

The resulting comparison matrix for independent coordinates of variance Γ is entrywise nonnegative and commutes with slot permutations. A nonnegative norm-maximizing eigenvector can again be averaged over these permutations without vanishing. It therefore suffices to bound the matrix on $\text{Sym}^r(\mathbb{C}^m)$.

We have reduced the problem to symmetric tensors in m independent Gaussian coordinates. The next calculation bounds each coefficient of the tensor product. Its dependence on the degree is what permits a geometric sum even when t grows like $\log(en/k)$. Here are details of the Gaussian norm estimate there, now using $\tilde{g}$ as the vector for the i.i.d. coordinate comparison. Use creation and annihilation operators $a_i^\dagger, a_i$ on the algebraic direct sum of symmetric tensor powers (only acting between finite degrees here), where on normalized occupation vectors with n_i copies of basis vector i they raise or lower n_i with coefficients $\sqrt{n_i + 1}$ or $\sqrt{n_i}$, respectively (zero for lowering at 0). For the real vector $\tilde{g}$ write $a(\tilde{g}) = \sum_i \tilde{g}_i a_i$.

Restricted to symmetric degree r , for $0 \leq j \leq r$ the sum over all size- j subsets of slots of the product of $|\tilde{g}\rangle\langle\tilde{g}|/m$ in active slots is

$$\frac{(a(\tilde{g})^\dagger)^j a(\tilde{g})^j}{j! m^j}.$$

One can see this in an orthonormal basis with first vector along $\tilde{g}$, counting the choices of j copies among those in its first mode: on an occupation vector for this basis with first-mode occupancy n'_1 , both sides multiply by $\binom{n'_1}{j}(\|\tilde{g}\|_2^2/m)^j$. Here creation along a vector, from degree u , equivalently inserts that vector then projects to symmetric tensors with the factor $\sqrt{u+1}$, with lowering the adjoint, so creation/annihilation along a unit direction obey the same formulas in a basis containing that unit vector. (At the zero vector the identity for the subset sum is immediate.) Take expectation with the i.i.d. Gaussian coordinates. The scalar Gaussian pairing identity [12] gives an order- $2j$ scalar moment as a sum over all pairings of the $2j$ places, with factor $\Gamma \mathbf{1}_{\{i=i'\}}$ per pair of coordinate indices i, i' (also seen by differentiating the Gaussian moment generating function). Applied to the numerator in the operator formula, it sums over pairings of the scalar coefficients of creators and annihilators, leaving normal order of the operators. Put $A = \sum_i a_i^2$ and let $\mathcal{N} = \sum_i a_i^\dagger a_i$ be tensor degree. For ℓ creator-creator pairs there are also ℓ annihilator-annihilator pairs and a bijective pairing across the remaining $j - 2\ell$ of each. Each fixed pairing in the numerator gives factor Γ^j times the contracted operator

$$(A^\dagger)^\ell \mathcal{N}^{j-2\ell} A^\ell,$$

using falling powers $x^{q'} = x(x-1)\cdots(x-q'+1)$, including $x^0 = 1$. Indeed creators commute among themselves, as do annihilators among themselves. The cross terms in the middle sum a normally ordered removal and replacement of an ordered selection of $q' = j - 2\ell$ particles: on occupations (n_i) this counts $(\sum_i n_i)^{q'}$, since a given ordered list of modes with multiplicities d_i contributes $\prod_i n_i^{d_i}$. There are $j!/((j-2\ell)!2^\ell\ell!)$ choices of the internal pairs on each side and $(j-2\ell)!$ bijections of the remaining places. Thus after division by $j!$ the count coefficient is $j!/((j-2\ell)!2^{2\ell}(\ell!)^2) \leq \binom{j}{2\ell}$.

On degree u , $\|A\|^2 \leq u(u+m)$, writing the norm for the action out of that degree. This follows using $[a_i, a_j^\dagger] = \mathbf{1}_{\{i=j\}}I$, hence $[A, A^\dagger] = 4\mathcal{N} + 2mI$. For $u \geq 2$ the squared norm out of u is the norm of $AA^\dagger$ on $u-2$, which equals the squared norm out of $u-2$ plus $4(u-2) + 2m$; the sequence starts with 0 on degrees 0,1. This gives the bound by induction. For $1 \leq j \leq r$, out of degree r the squared norm of A^ℓ is thus at most $(r(r+m))^\ell$ for $2\ell \leq j$, and the middle falling power of $\mathcal{N}$ in the contracted operator is evaluated on degree $r-2\ell$, bounded there by $r^{j-2\ell}$. Consequently the norm of the averaged size- j sum on the symmetric subspace in this range is at most

$$(\Gamma/m)^j \sum_{0 \leq \ell \leq j/2} \binom{j}{2\ell} r^{j-2\ell} (r(r+m))^\ell \leq \left(\frac{\Gamma}{m} (r + \sqrt{r(r+m)}) \right)^j$$

(and use the bound 1 at $j=0$ for the identity).

For sufficiently small absolute t_0 this sums in (58), since

$$(e^t - 1)(1 + C_* t/2) \frac{k + \sqrt{k(k+m)}}{m} \leq \frac{1}{2}.$$

In fact the last ratio is at most $C\sqrt{x}$ for $x = k/n \in (0, 1]$. Since $e^t - 1 \leq te^t$ and $t = t_0 \log(e/x)$, the expression is bounded by an absolute multiple of

$$t_0 \log(e/x) x^{1/2-t_0} + C_* t_0^2 \log(e/x)^2 x^{1/2-t_0}.$$

For $0 < t_0 \leq 1/8$, both powers of the logarithm times $x^{1/2-t_0} \leq x^{3/8}$ are uniformly bounded. With C_* already fixed by the endpoint comparison, we may therefore decrease t_0 until the displayed geometric ratio is at most $1/2$.

The sum with coefficients β^j , $0 \leq j \leq r$, then has norm at most two in every sector. Hence $\|G_t\|_{\text{op}} \leq 2$. Equation (59) now bounds $\|\mathbb{E}_s e^{t\Delta/n}\|_{V_\lambda}$ by an absolute constant, completing (54). $\square$

8.2.2 Induction of the operator bound

The two split moments now control the change in the potential Φ_n . We first derive that scalar comparison, then use the extra factor B^{-1} from two nontrivial children to close the induction. For the split with fixed halves (the children in the recursion), denote by H_1 the outer switch group. In $\lambda \neq (n)$, to bound the norm of K_n it suffices to bound its Rayleigh values on unit vectors v invariant under H_1 , since K_n is positive semidefinite and compressed by the projection Π_1 onto these invariants (if there is no such vector the norm is zero). For such a vector, the Rayleigh value is that for the operator of the middle product law from the two independent children, since $\Pi_1 v = v$. Under restriction to $S_m \times S_m$, decompose into an orthogonal sum of product irreducibles (using the usual tensor-product form of irreducibles of a direct product of finite groups), with the joint partial Casimir values c_0, c_1 from the two factors. We use c_0, c_1 as classical random variables with the joint spectral probabilities in v , i.e. squared norms of the joint spectral projections applied to v . Their law for this fixed split is also their law for each split in the sign randomization used in (54): any such transversal split is the image of the fixed one by an element of H_1 , which fixes v , and the operators for the splits correspond by conjugation. Write

$$D_0 = c - 2(c_0 + c_1), \quad D_1 = c_0 - c_1, \quad \mathcal{E} = \Phi_n(c) - \Phi_m(c_0) - \Phi_m(c_1).$$

In this scalar law we have $\mathbb{E} e^{tD_0/n} \leq C_0$ and $\mathbb{E} e^{t_1 D_1^2/(nc)} \leq C_0$ by (54) evaluated against v . Also $|D_0|, |D_1| \leq c$. Put $u = c/n$ and $x_i = 4c_i/c$ for $i = 0, 1$, with mean $\bar{x}$ of the two x_i . Then

$$\mathcal{E} = u[(1 - \bar{x})(2 + \log(n/u)) + \frac{1}{2} \sum_{i=0}^1 x_i \log x_i] \leq \frac{D_0}{n}(1 + \log(n/u)) + \frac{D_0^2 + 4D_1^2}{nc},$$

where $0 \log 0 = 0$, $x \log x \leq (x-1) + (x-1)^2$ for $x \geq 0$, and $x_0 - 1 = -D_0/c + 2D_1/c$, $x_1 - 1 = -D_0/c - 2D_1/c$. Using $|D_0| \leq c$ and $\log(n/u) \geq 0$, the D_0 terms on the right are at most $(D_0)_+(2 + \log(n/u))/n$, writing $z_+ = \max(z, 0)$. Here $k/2 \leq u \leq k$, so $2 + \log(n/u) \leq C \log(en/k)$. For an absolute $a_0 > 0$ sufficiently small we have $2a_0(2 + \log(n/u)) \leq t$, $8a_0 \leq t_1$. Cauchy–Schwarz under the scalar law therefore gives

$$\mathbb{E} e^{a_0 \mathcal{E}_+} \leq (\mathbb{E} e^{t(D_0)_+/n})^{1/2} (\mathbb{E} e^{t_1 D_1^2/(nc)})^{1/2} \leq C$$

using (54) as above and $e^{t(D_0)_+/n} \leq 1 + e^{tD_0/n}$. In particular

$$\mathbb{E} e^{\eta \mathcal{E}_+} \leq 1 + C' \eta \quad (0 \leq \eta \leq a_0) \quad (60)$$

by decreasing the exponent via Hölder ($C^{\eta/a_0} \leq 1 + (C-1)\eta/a_0$ for $C \geq 1$). Also

$$\Pr(c_0 = 0 \text{ or } c_1 = 0) \leq \Pr(D_0 \geq c/2) + \Pr(|D_1| \geq c/4) \leq C \exp(-c'k) \quad (61)$$

for an absolute $c' > 0$. Indeed if one $c_i = 0$, either their sum is at most $c/4$ or $|c_0 - c_1| \geq c/4$. The two probability terms are bounded by $C_0 e^{-tc/(2n)}$ and $C_0 e^{-t_1 c/(16n)}$ using (54), with (52). Take an absolute integer K large enough that for $k \geq K$ the probability of a zero c_i in (61) is at most $1/4$.

Assuming (53) on the nontrivial child irreducibles, the Rayleigh value above is bounded by

$$\mathbb{E} B^{-z} \exp \left[-\eta(\Phi_m(c_0) + \Phi_m(c_1)) \right], \quad z = \mathbf{1}_{\{c_0 > 0\}} + \mathbf{1}_{\{c_1 > 0\}},$$

by taking the operator norm product on each product summand (there the middle operator is the tensor product of the two child K_m 's in the respective irreducibles; trivial factors have norm 1 and $c_i = 0$). For $k \geq K$ and $1 < B \leq 2$,

$$\mathbb{E} B^{1-z} \leq 1 - \frac{1}{8}(B - 1)$$

by (61), since $z = 2$ has probability at least $3/4$ with deficit $1 - 1/B$, and otherwise the excess over 1 is at most $B - 1$. Using (60) and $B^{1-z} \leq 2$ gives

$$\mathbb{E}[B^{1-z} e^{\eta \mathcal{E}}] \leq 1 - (B - 1)/8 + 2C'\eta \leq 1$$

if $\eta > 0$ is sufficiently small depending on B (and $\eta \leq a_0$). This gives the required bound on the Rayleigh value and is the induction step for (53) when $k \geq K$.

The remaining fixed levels. The induction just proved applies when $k \geq K$. For the finitely many smaller levels, a simpler form of the isolated-path cancellation suffices: we may allow constants depending on k and bound a Hilbert–Schmidt norm directly. Proposition 7.1 obtains constants uniform in k by estimating a squared kernel and using Schur's test. We give the fixed- k argument here so the base of this induction uses only harmonic zero sums and the elementary forest count below.

Lemma 8.4 (Fixed first-row deficit). *For every fixed integer $k \geq 2$ there are constants C_k, n_k such that, for every dyadic $n = 2^d \geq n_k$ and every $\lambda \vdash n$ with $\lambda_1 = n - k$,*

$$\|K_n(\lambda)\|_{\text{op}} \leq C_k (d/n)^{k/2}.$$

For $k = 1$, $K_n(\lambda) = 0$ at every admissible size.

Proof. For $k = 1$, the directed sweep sends a single position uniformly over all positions (each new bit is fair conditionally as its layer is reached), so the nontrivial irreducible with $n - \lambda_1 = 1$ is annihilated on the one-position module. For fixed $k \geq 2$ and large n , take a function f in the λ subspace of the ordered distinct k -tuple module with $\lambda_1 = n - k$. It is orthogonal to functions omitting any slot, by the branching facts (each such entire function space is of types available with $k - 1$ slots). Testing against indicators specifying the other slots, we see its extension by zero to the unrestricted tuple space has zero sum in each slot with the others fixed (also automatically so if the others are not distinct).

From a given distinct x to a candidate y in a directed sweep, each proposed path is determined by those two endpoints: updated bits must already equal the corresponding bits of y . For tags a, b , let $h = h_{ab}$ be the largest differing bit index of x_a, x_b . At layer h their paths use a shared switch exactly if their y -bits before h agree, in which case they require opposite y -bits at h for validity. They cannot share at an earlier layer (look at bit h), or at a later layer on a valid route (their prefixes through h must then differ). A collision making routing impossible occurs if some such pair instead agrees in y through h ; conversely if this never occurs all paths are on distinct positions at every boundary (compare the unupdated suffix of x before processing h , or the updated prefix of y once h has been processed, for each pair). In that case the needed switch choices are consistent: at a shared switch the two tags distinct before it are sent to distinct outputs of the switch. Each used switch coin is prescribed, with a shared switch reducing the number of prescriptions by one from one-per-tag-per-layer. It

follows that the kernel for the directed sweep on these tuples, extended by zero to invalid y , is $n^{-k}F_{[k]}$, using the functions

$$F_S = \prod_{a < b: a, b \in S} \left(1 - \mathbf{1}_{\{y_{a, < h_{ab}} = y_{b, < h_{ab}}\}} (-1)^{y_{a, h_{ab}} + y_{b, h_{ab}}}\right).$$

Here the $< h$ subscript extracts the indicated prefix, possibly empty. Each factor on a valid route is 2 for a shared switch, 1 otherwise. Against the extended f summed on the unrestricted tuple space, we can replace $F_{[k]}$ by $F' = \sum_{S \subseteq [k]} (-1)^{k-|S|} F_S$ by the zero sums (F_S uses no y -slots outside S). It is bounded by a constant depending on k , and cancels to zero if any vertex is isolated in the graph on tags with potential interaction edges $y_{a, < h_{ab}} = y_{b, < h_{ab}}$: adding an isolated vertex does not change F_S .

With uniform distinct x and independent unrestricted uniform y , the probability of no isolates is at most $C_k(d/n)^{\lceil k/2 \rceil}$ for large n . Indeed then the graph contains a spanning forest with no isolates, thus at least $\lceil k/2 \rceil$ edges. For unrestricted independent uniform coordinates also in x , the constraints along any such forest (requiring $x_a \neq x_b$ along its edges and the indicated prefix matches) cost $(d/n)^{\# \text{edges}}$ by successively integrating leaves. Given the other tags' values at a leaf-removal step, each possible last differing index $h \in \{1, \dots, d\}$ relative to the neighbor of the leaf has probability $2^{h-1}/n$ in x and the prefix match in y costs $2^{-(h-1)}$, totaling d/n . Conditioning x to be globally distinct costs at most factor 2 for large n at fixed k , and there are only boundedly many forests at fixed k in the union bound.

Thus with these tuple measures (distinct for x , unrestricted for y), $\mathbb{E}_{x,y} |F'(x,y)|^2 \leq C_k(d/n)^{\lceil k/2 \rceil}$. The kernel applied to f at x is $\mathbb{E}_y F'(x,y) \tilde{f}(y)$ with $\tilde{f}$ the zero extension. By Cauchy-Schwarz in y , then taking the norm using uniform distinct x , the output norm is at most $C_k(d/n)^{k/4} \|f\|_2$; here the L^2 -norms use probability measures and the unrestricted norm of f is no larger than $\|f\|_2$. This kernel on distinct-tuple functions is T_n^* for the action permuting the tuple basis. Since λ occurs and the averaged permutations preserve its subspace, $\|T_n^*\|_\lambda = \|T_n\|_\lambda$ gives

$$\|K_n\|_\lambda \leq C_k(d/n)^{k/2}$$

for each fixed $k \geq 2$ at large n . □

Choosing the constants and completing the induction. Both the large-level induction and the fixed-level bound have been proved. The choices below make them hold with one pair of absolute constants on every dyadic size. First fix the integer K from (61). Since $c/n \leq k$ and $u(2 + \log(n/u))$ is increasing for $0 < u \leq n$,

$$\Phi_n(c) \leq k(2 + \log(n/k)).$$

For each $1 \leq k < K$, Lemma 8.4 gives a bound $o(n^{-k/8})$. There is therefore one absolute n_0 such that these bounds imply (53) whenever $n > n_0$, simultaneously for all $1 < B \leq 2$ and $0 < \eta \leq 1/8$. Indeed throughout those ranges,

$$B^{-1} e^{-\eta \Phi_n(c)} \geq \frac{1}{2} e^{-k/4} k^{k/8} n^{-k/8}.$$

Only finitely many sizes and shapes remain. On each nontrivial shape at $n \leq n_0$, the norm of K_n is strictly less than one by Lemma 2.2: equality would give a vector fixed by every coordinate layer and hence by the hypercube edge transpositions, which generate S_n . Choose $1 < B \leq 2$ close enough to one that B^{-1} strictly exceeds all these finitely many norms. Then choose $\eta > 0$ small enough to satisfy (53) on this finite set, $\eta \leq \min(1/8, a_0)$, and the large-level induction condition $2C'\eta \leq (B-1)/8$ from (60).

The direct fixed-level estimates now cover $k < K$ above n_0 , and the induction step covers $k \geq K$. This proves (53) on every dyadic size with one pair of absolute constants B, η .

8.3 Hausdorff–Young amplification to the full permutation

We have separately proved a uniform injection-density bound and an operator contraction. To combine them, first convert the density bound into the 65th trace moment in a single irreducible. Then use the operator contraction for the remaining sweep factors. The passage to powers of T_n uses Schatten Hölder, so it does not require T_n to be normal.

We view probability laws on S_n also by their densities relative to normalized uniform measure. For a density or function f , use Fourier matrix $\hat{f}(\lambda) = \mathbb{E}_g \text{uniform}(f(g)U_\lambda(g))$, with U_λ a unitary irreducible of dimension $d_\lambda = f^\lambda$. By finite group orthogonality/Plancherel, $\|f\|_2^2 = \sum_\lambda d_\lambda \|\hat{f}(\lambda)\|_{\text{HS}}^2$ with the (unnormalized) Hilbert–Schmidt norm; this holds with the displayed convention since one is expanding against the complex conjugates of matrix coefficients (scaled by $\sqrt{d_\lambda}$ for an orthonormal basis). We need the single-irrep Hausdorff–Young bound [10, Lemma 5.1(1)], using Schatten norms S^s (the ℓ^s -norms of the singular values, with S^∞ the operator norm):

$$d_\lambda^{1/s} \|\hat{f}(\lambda)\|_{S^s} \leq \|f\|_p, \quad 1 < p \leq 2, \quad s = p/(p-1). \quad (62)$$

For completeness, at $p = 1, s = \infty$ the bound without dimension factor is unitarity, and at $p = s = 2$ it follows from Plancherel. To interpolate, pair by trace against a matrix A of S^p -norm 1 and take $\|f\|_p = 1$, ignoring the trivial zero case. In $0 \leq \text{Re } z \leq 1$, replace f by f_z using its phases times $|f|^{p(1-z/2)}$ at nonzero entries (keep zeros); similarly in a singular value factorization of A keep both unitary factors and replace nonzero singular values β by $\beta^{p(1-z/2)}$ to give A_z . Apply the three-lines theorem to $d_\lambda^{z/2} \text{Tr}(A_z \hat{f}_z(\lambda))$. On the first boundary use $L^1 \rightarrow S^\infty$ and the trace norm of A_z , on the second $L^2 \rightarrow S^2$ and Hilbert–Schmidt; the norms of the inputs in these bounds are 1 and the analytic family is bounded on the strip. Taking $z = 2(1 - 1/p)$ and Schatten duality proves (62).

The injection multiplicity and the 65th moment. The density estimate controls the Fourier matrix after averaging over a tuple stabilizer. Averaging these stabilizer projections in turn recovers K_n at the cost of the ratio D_λ/f^λ . This ratio is at most $\binom{n}{k}$, even when the lower-diagram dimension f^λ is large. It is this multiplicity comparison that allows a level-dependent density bound to control the dimension-weighted trace moment. Take $p = 1 + \delta$ from (50), so $s = 65$. For a fixed nontrivial λ , use $k = n - \lambda_1$. For ordered distinct x , average the density of ν_n over right multiplication by its tuple stabilizer H_x . The resulting function is $g \mapsto R_x(gx)$, with L^p -norm bounded by (50); its Fourier matrix in λ is $K_n P_x$, where P_x is the average of U_λ over H_x . Averaging these projections in x uniformly gives $(f^\lambda/d_\lambda)I$, by conjugation symmetry, Schur’s lemma and the fixed-space dimension above. By (62) and the triangle inequality in S^s we have in λ

$$d_\lambda^{1/s} (f^\lambda/d_\lambda) \|K_n\|_{S^s} = d_\lambda^{1/s} \|\mathbb{E}_x K_n P_x\|_{S^s} \leq \exp(Ck \log(en/k)).$$

Taking the s -th power and using (51) gives

$$d_\lambda \text{Tr}_\lambda(K_n^s) \leq \left(\frac{d_\lambda}{f^\lambda} \right)^s \exp(Csk \log(en/k)) \leq \exp(C_2 k \log(en/k)). \quad (63)$$

Here K_n is positive semidefinite and C_2 is absolute (using $\binom{n}{k} \leq (en/k)^k$).

Completion for directed sweeps. After l directed sweeps with integer $l > s$, the nontrivial Fourier matrices of the position permutation law/density are T_n^l . Subtracting 1 from the density keeps these and gives zero in the trivial representation. Schatten Hölder for matrix products gives $\|T_n^s\|_{\text{HS}} \leq \|T_n\|_{S^{2s}}^s$, so in nontrivial λ

$$\|T_n^l\|_{\text{HS}}^2 \leq \|K_n\|_\lambda^{l-s} \text{Tr}_\lambda(K_n^s).$$

Choose a fixed absolute integer $l > s$ sufficiently large, with $(l - s)a - C_2 \geq 4$ using a from the consequence of (53). By (53), (63) and Plancherel, the squared L^2 distance of the density to 1 is bounded by

$$\sum_{k=1}^{n-1} 2^k \exp(-4k \log(en/k)) = o(1).$$

We used that the number of shapes with a given k is bounded by the number of partitions of k , hence by 2^k , and increased l so the exponent bound applies before counting shapes. The sum tends to zero, for example splitting at $\sqrt{n}$ and using $\log(en/k) \geq 1$ in the large range. Total variation is at most half the L^2 distance by Cauchy–Schwarz. The estimate is independent of the starting permutation, since changing that permutation translates the law. As one sweep consists of d physical shuffles, this proves Theorem 8.1.

9 Harmonic restriction and cycle moments

The density estimates already proved describe the joint routes of tracked cards. We now retain more of the representation carried by those cards. For a diagram whose first row has length $n - k$, its first occurrence on ordered tuples has exactly k slots. Summing out any one slot therefore annihilates that representation. This harmonic cancellation controls how much of its level can disappear when the network splits into two halves.

The first goal is a contraction on the scale $k \log(en/k)$. A separate cycle-moment estimate will then control the dimension of the diagram below the first row. We give two transfers from that moment, one by clipping kernel entries and one by averaging matrix coefficients. Both retain the lower-diagram dimension, whereas the final tuple-trace argument can sum all representations without separating that dimension.

Let T_n be one sweep on $n = 2^d$ positions, and put $K_n = T_n^* T_n$. Reversing the coordinate order conjugates this law to the law of $T_n T_n^*$ by the permutation that reverses the binary coordinates. Thus all statements below also hold for that orientation, with the same constants and with the starting tuple relabeled by this permutation. At the one-position base, the empty product gives $T_1 = K_1 = I$. If $n = 2m$, then

$$K_n = P(K_m \otimes K_m)P, \tag{64}$$

where P independently averages the m switches between corresponding positions of the two halves. Thus K_n is positive, self-adjoint, and a contraction in every unitary representation. Throughout this section traces count dimension, and logarithms are natural unless a base is shown. The regular trace of an operator A means $\text{tr}_{\text{reg}} A = \sum_{\lambda \vdash n} D_\lambda \text{tr}_{V_\lambda} A$. On a tuple space it means the ordinary trace of its matrix in the tuple basis; rescaling all basis vectors from counting to probability norm does not change it. For $\lambda = (n - k, \beta) \vdash n$, write $D_\lambda = \dim V_\lambda$, $f_\beta = \dim V_\beta$, and

$$\Lambda_n(k) = k \log(en/k) \quad (k > 0), \quad \Lambda_n(0) = 0, \quad W_n(\lambda) = \|K_n|_{V_\lambda}\|_{\text{op}} = \|T_n|_{V_\lambda}\|_{\text{op}}^2.$$

Theorem 9.1. *There are absolute constants $c, C, \zeta > 0$, a number $\delta \in (0, 1)$, and an integer p_0 with the following properties. For every dyadic n and $\lambda = (n - k, \beta) \vdash n$ with $k \geq 1$,*

$$W_n(\lambda) \leq e^{-c\Lambda_n(k)}, \tag{65}$$

$$W_n(\lambda) \leq e^{C\Lambda_n(k)} f_\beta^{-\zeta}. \tag{66}$$

Independently, for every $1 \leq k \leq n$, let $X_{n,k}$ be the ordered injective k -tuples of positions, with uniform probability measure, and let $\mathcal{K}_{n,k}(x, y) = (n)_k \Pr_{K_n}(x \mapsto y)$ be its kernel density.

Then

$$\sup_x \mathbb{E}_y \mathcal{K}_{n,k}(x, y)^{1+\delta} \leq e^{C\Lambda_n(k)}, \quad (67)$$

$$\mathrm{tr}_{X_{n,k}} K_n^{p_0} \leq e^{C\Lambda_n(k)}. \quad (68)$$

Consequently, after changing absolute constants,

$$\|T_n|_{V_\lambda}\|_{\mathrm{op}} \leq \exp[-c\{\Lambda_n(k) + \log f_\beta\}], \quad (69)$$

$$W_n(\lambda) \leq e^{-c_1\Lambda_n(k)} D_\lambda^{-c_2} \quad (70)$$

for absolute $c_1, c_2 > 0$.

9.1 Harmonic tuple restriction

We first describe the probability law on the two child representations that arises from one maximizing vector. The law is a spectral weighting, not an independent random choice of Young diagrams. The removal estimates below hold uniformly over that weighting.

The position action of S_n on $X_{n,k}$ commutes with the action of S_k on the tuple labels. Pieri's rule [26, 27] gives the multiplicity-free injection decomposition [7, Section 3.2, equation (3.1)]: one copy of $V_\mu \otimes V_\xi$ precisely when μ/ξ is a horizontal strip of size $n - k$. In particular the position type $(n - k, \beta)$ has label type β at this smallest tuple size. It does not occur on $X_{n,k-1}$. Consequently its functions are harmonic: summing over any one tuple entry, with all the others fixed, gives zero.

Work in the position- λ isotypic space of $\ell^2(X_{n,k})$, which is $V_\lambda \otimes V_\beta$ under the commuting position and slot-permutation actions. Use counting norm here and take a unit eigenvector v attaining a positive value of $W_n(\lambda)$. Equation (64) implies $Pv = v$. Decompose v by the set of labels in each half and then by the two position types. The squared component norms define a probability law. Write a, b for the numbers of labels in the halves, $a + b = k$, and r, s for their first-row deficits. Branching and the tuple realization give

$$0 \leq r \leq a, \quad 0 \leq s \leq b, \quad r, s \geq k - m, \quad W_n(\lambda) \leq \mathbb{E}[W_m(\mu)W_m(\nu)]. \quad (71)$$

The last lower bounds follow from $\mu, \nu \subseteq \lambda$. The decompositions by label sets commute with the half-position groups, so they do not create additional types.

Lemma 9.2 (Removal and splitting). *For the component law above, a is a mixture of $t + \mathrm{Bin}(k - 2t, 1/2)$. In particular*

$$\Pr(|a - k/2| \geq z) \leq 2e^{-2z^2/k}, \quad \mathbb{E}e^{u(a-b)^2/k} \leq 1 + Cu \quad (0 \leq u \leq 1/4). \quad (72)$$

For every fixed $\eta > 0$ and integer $l \geq 1$,

$$\mathbb{E} \left[\mathbf{1}_{\{m-a-r \geq \eta m\}} \binom{a-r}{l} \right] \leq (C_\eta \sqrt{k/n})^l, \quad (73)$$

and the same statement holds in the other half. Thus, with $\alpha = 1/8$ and

$$L_1 = (a - r) \mathbf{1}_{\{a \leq 3m/4, r < \alpha m\}}, \quad L_2 = (b - s) \mathbf{1}_{\{b \leq 3m/4, s < \alpha m\}},$$

we have $\mathbb{E}(L_i^{L_i}) \leq (C\sqrt{k/n})^l$ and

$$\mathbb{E}e^{u \log(en/k) L_i} \leq 1 + Cu \quad (0 \leq u \leq u_0) \quad (74)$$

for an absolute $u_0 > 0$. A second useful form, with $\alpha = 1/16$, is

$$\Pr((a - r) \mathbf{1}_{\{r < \alpha m\}} \geq x) \leq Ce^{-cx \log(en/k)} \quad (x \in \mathbb{Z}_{\geq 1}). \quad (75)$$

Proof. Randomly orient the crossing pairs after sampling a configuration with probability $|v|^2$. A doubly occupied pair contributes one label to each side, and each singly occupied pair contributes an independent fair choice. This proves the mixture and the concentration bounds.

For the removal bounds it suffices to take $1 \leq l \leq k$; the binomial coefficients vanish when $l > k$. Fix the first half S_0 . For $J \subseteq [k]$, let $D_J^{S_0}$ sum the coordinates with labels in J over S_0 , leaving all the other labeled coordinates in place. Its target has counting measure on the ordered injective $(k - |J|)$ -tuples. We use the zero extension $\tilde{v}$ of v to write these sums without collision exclusions. For a transversal S of the matched pairs define D_J^S in the same way, and put

$$Q_l = \mathbb{E}_S \sum_{|J|=l} \|D_J^S v\|_2^2.$$

Here $S = gS_0$ for a uniformly chosen element g of the matching switch group. Since $Pv = v$, every such g fixes v and $D_J^{gS_0} v = U_{k-l}(g) D_J^{S_0} v$, where $U_{k-l}(g)$ permutes the retained positions. Thus

$$Q_l = \sum_{|J|=l} \|D_J^{S_0} v\|_2^2.$$

We will bound this one quantity below using the fixed-half representation decomposition and above using the random transversal. Write $D_i = D_{\{i\}}^{S_0}$. In a sector with a labels and position type $\mu = (m - r, \theta)$,

$$\sum_i D_i^* D_i \succeq (a - r)(m - a - r + 1)I. \quad (76)$$

Indeed the operator on the left is the arrangement adjacency operator plus aI ; the position-label content identity [1, Lemma 3.1 and Proposition 3.3] gives $aI + \Omega_{\text{pos}} - \Omega_{\text{lab}} - \binom{m-a}{2}I$, where Ω is the transposition class sum. In the following calculation, $c(\rho) = \sum_{(i,j) \in \rho} (j - i)$ denotes its content scalar, rather than the complementary positive Casimir scalar of Section 8. If the label type is ξ , Pieri gives $\xi \supseteq \theta$. Put $h = a - r$. The content formula and successive addition of its h extra boxes give

$$c(\mu) = \binom{m-r}{2} + c(\theta) - r, \quad c(\xi) \leq c(\theta) + hr + \binom{h}{2}.$$

Substitution proves (76). Iterating over l distinct labels and dividing by the $l!$ possible orders gives the lower bound

$$Q_l \geq (\eta m)^l \mathbb{E} \left[\mathbf{1}_{\{m-a-r \geq \eta m\}} \binom{a-r}{l} \right].$$

For clarity, in a sector whose left-label set is A , a deletion contributes only when $J \subseteq A$. Afterward the left-label set is $A \setminus J$, so the target still distinguishes the original sector when J is fixed. Deletion also intertwines the two half-position groups, so different position types remain orthogonal. After j deletions the surviving type still has deficit r , while the occupancy is $a - j$. Iterating (76) therefore supplies $(a - r)_l (\eta m)^l$ on the indicated event. The $l!$ orders of each deleted label set give the displayed binomial coefficient.

For the upper bound, give the two positions of each pair opposite fair signs $\epsilon(x)$, so $S = \{x : \epsilon(x) = 1\}$. For fixed retained tuple y and deleted labels J , harmonicity gives the exact formula

$$D_J^S v(y) = 2^{-l} \sum_z \tilde{v}(y, z) \prod_{i \in J} \epsilon(z_i).$$

Indeed expanding the product of half-indicators leaves this term only: every term missing a sign has a zero sum in that coordinate. On squaring and averaging, two deleted assignments

can correlate only if their sets of pair indices with odd occupancy agree. Suppose that set has $l - 2t$ members. A partner assignment is specified by at most $\binom{m}{t}$ choices of doubly occupied pair sites, two endpoint choices at each of the $l - 2t$ odd sites, and $l!$ assignments of the chosen positions to the deleted labels. Hence the class has size at most $l!2^{l-2t}\binom{m}{t}$. Ignoring conflicts with the retained tuple only increases this bound.

Cauchy–Schwarz in each such class bounds its squared signed sum by the class size times the sum of $|\tilde{v}(y, z)|^2$. Now sum over y and J . In a fixed full k -tuple there are at most $\lfloor k/2 \rfloor$ doubly occupied pair sites. Choose t of them for the deleted double pairs and then choose the remaining $l - 2t$ deleted labels. Thus there are at most $\binom{\lfloor k/2 \rfloor}{t} \binom{k}{l-2t}$ choices, with overcounting harmless. Since $\|v\|_2 = 1$, this proves

$$Q_l \leq 4^{-l} \sum_{0 \leq t \leq l/2} l!2^{l-2t} \binom{m}{t} \binom{\lfloor k/2 \rfloor}{t} \binom{k}{l-2t} \leq (C\sqrt{mk})^l. \quad (77)$$

The last step uses $k \leq 2m$ and $l/(t!^2(l-2t)!) \leq 3^l$: each summand is at most $C^l m^t k^{l-t} \leq C^l (mk)^{l/2}$, and the number of summands is absorbed into the absolute constant. Comparing the two bounds on Q_l proves (73). For either cutoff loss $L = L_i$, put $H = \log(en/k)$. Expanding $e^{uHL} = (1 + (e^{uH} - 1))^L$ shows that its expectation is at most one plus a geometric series with ratio $C\sqrt{x}((e/x)^u - 1)$, where $x = k/n$. This is $O(u)$ uniformly on $0 < x \leq 1$ for $u \leq 1/4$, proving (74).

For (75), first restrict to $a \leq .8m$ and $r < m/16$. The lower removal factor is at least $.1m$, so on $a - r \geq x$,

$$\Pr(a \leq .8m, r < m/16, a - r \geq x) \leq \frac{(C\sqrt{k/m})^j}{\binom{x}{j}} \quad (1 \leq j \leq x).$$

For small k/n take $j = x$. On the remaining density range H is bounded; take $j = \lfloor x/M \rfloor$ with a sufficiently large absolute M , using $\binom{x}{j} \geq (x/j)^j$. The bounded $x < M$ cases are absorbed in the constant. If $a > .8m$ and $r < m/16$, then $k > .6n$ is impossible by $r \geq k - m$. For $k \leq .6n$ the event requires a deviation of at least $.2m$, while also $k > .8m$. Its probability is Ce^{-ck} by (72); here H is bounded and $x \leq k$. $\square$

9.2 Three ways to close the level induction

We have controlled both the imbalance of the two tuple occupancies and the loss from an occupancy to a child representation level. The next step is to turn those two controls into the first bound in Theorem 9.1. Three potentials do this with different sources of strictness: a fixed multiplicative margin, a large square-root correction, or a unit square-root correction coupled to a fixed-level estimate. We first complete the constant-margin proof. The two square-root alternatives then show how to replace that margin; the last also supplies the fixed-level estimate (85).

Capped potential comparisons. The three inductions use the same elementary estimates. For $c > 0$ set

$$E_c(x) = \begin{cases} x \log(ec/x), & 0 < x \leq c, \\ c, & x > c, \\ 0, & x = 0, \end{cases} \quad R_c(x) = \sqrt{\min(x, c)}.$$

Take $c = \alpha m$, where $0 < \alpha \leq 1/8$ is fixed, and write $H = \log(en/k)$, $D = (a - k/2)^2/k$ and $U_1 = (a - r)\mathbf{1}_{\{r < c\}}$, with U_2 defined in the other half. The identities $E_{2c}(k) = 2E_c(k/2)$ and $a + b = k$ give

$$0 \leq E_{2c}(k) - E_c(a) - E_c(b) \leq a \log(2a/k) + b \log(2b/k) \leq 4D.$$

For the first inequality on the right, the magnitude of the negative second derivative of E_c is at most $1/x$; integrating from the balanced split compares its gap with the displayed entropy gap. The last inequality is the relative-entropy bound by the corresponding chi-square divergence on two points. Concavity and $E_c(0) = 0$ also give

$$E_c(a) - E_c(r) \leq U_1 \{1 + \log_+(c/a)\}.$$

When $a \geq k/4$, the braces are at most H . When $a < k/4$, their excess over H costs at most $a \log(k/a) \leq k/e$, while $D \geq k/16$. Thus

$$E_{2c}(k) - E_c(r) - E_c(s) \leq CD + H(U_1 + U_2). \quad (78)$$

All these inequalities include zero occupancies by continuity.

The map $x \mapsto \sqrt{\min(x, c)}$ is a contraction as a function of $\sqrt{x}$. Hence $|R_c(a) - R_c(k/2)| \leq \sqrt{2D}$, and $R_c(a) - R_c(r) \leq \sqrt{U_1}$. With $s_0 = \min(k, 2c)$ this yields

$$R_{2c}(k) - R_c(r) - R_c(s) \leq -(\sqrt{2} - 1)\sqrt{s_0} + 2\sqrt{2D} + \sqrt{U_1} + \sqrt{U_2}. \quad (79)$$

These comparisons separate occupancy imbalance from the loss of representation level, so the estimates in Lemma 9.2 can be applied without conditioning on either event.

A constant multiplicative margin.

First take $x_* = 1/100$ and define

$$H_N(j) = Nh(j/N), \quad h(y) = \begin{cases} y(1 + \log(x_*/y)), & 0 < y \leq x_*, \\ x_*, & y > x_*, \\ 0, & y = 0. \end{cases}$$

The function is nondecreasing and concave, and $H_n(k) \geq c\Lambda_n(k)$. Put $\Delta = H_n(k) - H_m(r) - H_m(s) \geq 0$. There are absolute $\beta, C > 0$ such that

$$\mathbb{E}e^{\beta\Delta} \leq C. \quad (80)$$

Here is the loss calculation. For $k/n \geq 3/5$, the diagram containment bound gives $r, s \geq m/5$ and the gap is zero. Otherwise introduce $H_m(a) + H_m(b)$ and use the preceding entropy comparison. If $a > 4m/5$ but $r < x_*m$, containment implies $k < (1 + x_*)m$. Therefore $D \geq cm$, which bounds the entire capped loss, at most x_*m . For all other occupancies the same calculation, with $1 + \log_+(4x_*m/k)$ in place of H , gives

$$\Delta \leq CD + J(V_1 + V_2), \quad D = (a - k/2)^2/k, \quad J = 1 + \log_+(4x_*m/k),$$

where $V_1 = (a - r)\mathbf{1}_{\{r < x_*m, a \leq 4m/5\}}$, and similarly for V_2 . On this indicator $m - a - r \geq m/10$. The binomial moments in (73) imply

$$\mathbb{E}e^{tJV_i} \leq 1 + \sum_{l \geq 1} [(e^{tJ} - 1)C_{1/10}\sqrt{k/n}]^l.$$

The ratio is uniformly small for a sufficiently small fixed t . Hölder and (72) prove (80).

For all sufficiently large n , uniformly over $k \geq 2$, the same component law satisfies

$$\Pr(r, s \geq 1) \geq 1/3, \quad \Pr(r = s = 0) \leq 1/12. \quad (81)$$

For density at least $3/5$ this is deterministic. Otherwise, when k is large the two occupancies lie between $k/4$ and $4m/5$ except with vanishing probability, and a zero child level would

make a V_i at least $k/4$. The preceding exponential estimate excludes this. For bounded $k \geq 2$, the probability of any removal loss tends uniformly to zero by the $l = 1$ bound. The binomial mixture has two positive occupancies with probability at least $1 - 2^{1-k} \geq 1/2$, and cannot have both zero.

Choose a finite base threshold for (81). Every nontrivial sweep norm at those finitely many sizes is strictly below one: equality in a product of orthogonal projections would give a vector fixed by every cube-edge transposition, hence by S_n . Choose $0 < A \leq 1/2$ so the finite base norms are at most e^{-2A} . We induct with

$$W_n(\lambda) \leq e^{-A - \varepsilon H_n(k)} \quad (k \geq 2). \quad (82)$$

Level one is annihilated by exact one-card uniformity. Let $G = \{r \neq 1, s \neq 1\}$ and $b = \#\{r, s \text{ that are at least } 2\}$. After dividing the recursive bound by the proposed right side, the remaining factor is

$$\mathbb{E}[\mathbf{1}_G e^{-A(b-1)} e^{\varepsilon \Delta}].$$

Without the last factor this is bounded uniformly below one: the event of two positive levels loses at least $1 - e^{-A}$, whereas only two zero levels can gain $e^A - 1 \leq 2(1 - e^{-A})$. Equations (81) bound the factor by $1 - (1 - e^{-A})/6$. Moreover (80) and Hölder give $\mathbb{E}e^{\varepsilon \Delta} \leq C^{\varepsilon/\beta} = 1 + O(\varepsilon)$. Since $\mathbf{1}_G e^{-A(b-1)} \leq e^A$, reinstating $e^{\varepsilon \Delta}$ costs only $O(\varepsilon)$, uniformly. Also choose $\varepsilon H_n(k) \leq A$ at the base sizes. This proves (65).

A large square-root correction. A second invariant dispenses with the constant margin by inserting a square-root term. With $\alpha = 1/16$, set

$$G_N(j) = N\phi(j/N), \quad \phi(y) = \begin{cases} y \log(e\alpha/y), & 0 < y \leq \alpha, \\ \alpha, & y \geq \alpha, \\ 0, & y = 0, \end{cases}$$

$$F_N(j) = G_N(j) + 100\sqrt{\min(j, \alpha N)}.$$

For $Y = F_n(k) - F_m(r) - F_m(s)$ and $s_0 = \min(k, \alpha n)$,

$$Y \leq -100(\sqrt{2} - 1)\sqrt{s_0} + C_{100}\{1 + D + H(U_1 + U_2)\}, \quad (83)$$

where $H = \log(en/k)$, $D = (a - k/2)^2/k$, and the U_i are the uncut losses in (75). Indeed (78) and (79), together with $\sqrt{D} \leq 1 + D$ and $\sqrt{U_i} \leq U_i$ for integer losses, give (83). The split and loss tails imply, for small $\tau \geq 0$, a bound

$$\mathbb{E}e^{\tau C_{100}\{1+D+H(U_1+U_2)\}} \leq e^{C'\tau}.$$

For example the tail in (75) bounds $\mathbb{E}e^{cHU_i}$ uniformly for a small fixed $c > 0$; Hölder and convexity then give the displayed bound at smaller exponents. Since $s_0 \geq \alpha k$, one fixed threshold K makes $\mathbb{E}e^{\tau Y} \leq 1$ for every $k > K$ and all sufficiently small τ .

For $2 \leq k \leq K$, take n large enough that the potentials are uncapped at these levels. Put $L = (a - r) + (b - s)$, and decompose the gap without conditioning on the loss event:

$$Y = Q - 100S + R, \quad Q = a \log(2a/k) + b \log(2b/k), \quad S = \sqrt{a} + \sqrt{b} - \sqrt{k},$$

where $0 \leq R \leq (H + C_K)L$. The binomial mixture gives $\mathbb{E}Q \leq 4\mathbb{E}D \leq 1$. Both occupancies are positive with probability at least $1/2$, and then $S \geq 1/2$, so $\mathbb{E}S \geq 1/4$. The first removal moment gives $\mathbb{E}L \leq C_K e^{-H/2}$. Consequently $\mathbb{E}Y \leq -24 + o(1) \leq -12$ for all sufficiently large n , uniformly over these finitely many levels.

The same bound also justifies a uniform Taylor remainder. We have $|Y| \leq C_K + KH \mathbf{1}_{\{L > 0\}}$ and $\Pr(L > 0) \leq C_K e^{-H/2}$, and therefore

$$M_K := \sup \mathbb{E}[Y^2 e^{|Y|/(4K)}] < \infty,$$

where the supremum is over these levels, sizes, and component laws. For $0 < \tau \leq \min\{1/(4K), 12/M_K\}$, Taylor's inequality gives $\mathbb{E}e^{\tau Y} \leq 1 + \tau \mathbb{E}Y + \tau^2 M_K/2 \leq 1 - 6\tau$. The remaining finitely many sizes are covered by shrinking τ using the strict norm gap. Induction now proves the invariant $W_n(\lambda) \leq e^{-\tau F_n(k)}$.

A unit correction and fixed-level bootstrap. A third version uses coefficient one instead of 100. Use the preceding definition of G_N with $\alpha = 1/8$, and put $\Phi_N(j) = G_N(j) + \sqrt{\min(j, \alpha N)}$. The cut losses L_i in Lemma 9.2 give

$$\Phi_n(k) - \Phi_m(r) - \Phi_m(s) \leq -c_0 \sqrt{k} + C_0(a-b)^2/k + C_0 H(L_1 + L_2). \quad (84)$$

Here are the capped cases in this comparison. Put $c = \alpha m$. If $a, b \in [.4k, .6k]$ and $c/k \leq .4$, both child square roots are capped, so their saving over the root is $(2 - \sqrt{2})\sqrt{c} \geq (2 - \sqrt{2})\sqrt{k}/4 > .1\sqrt{k}$; we used $c/k \geq 1/16$. If $c/k \geq .4$, each child square root is at least $\sqrt{.4k}$ and the root is at most $\sqrt{k}$, giving the same saving. Outside this occupancy interval, $D \geq k/100$, and the square-root gap, at most $\sqrt{k}$, is bounded by $-.1\sqrt{k} + 110D$. This proves the required no-loss bound.

The entropy comparison (78) and the root loss $\sqrt{L_i} \leq L_i \leq H L_i$ cover the cut losses. The only omitted case has $a > 3m/4$ and $r < m/8$, or its counterpart. Containment gives $k < 9m/8$, hence $D > m/32$; the entire capped entropy and square-root loss is at most m , and so is bounded by $32D$. These bounds prove (84) with $c_0 = 1/10$ and an absolute C_0 . Equations (72) and (74) imply

$$\mathbb{E}e^{\tau(\Phi_n(k) - \Phi_m(r) - \Phi_m(s))} \leq e^{\tau(B - c_0 \sqrt{k})}$$

for fixed B and all sufficiently small τ . One may start this induction through the additional quantitative fact

$$\max_{\lambda: n - \lambda_1 = k} W_n(\lambda) = O_k(n^{-1/4}) \quad (k \geq 1 \text{ fixed}). \quad (85)$$

Induct on k . A positive removal loss has probability $O_k(n^{-1/2})$. Without losses, two positive child levels are smaller than k and obey the induction hypothesis. All labels in one half has probability at most 2^{1-k} , giving for $k \geq 2$ $M_k(2m) \leq 2^{1-k} M_k(m) + O_k(m^{-1/4})$. Since $2^{1-k} 2^{1/4} < 1$, this proves (85); level one vanishes. For any fixed upper level K , the estimate is eventually stronger than $e^{-\tau \Phi_n(k)}$ simultaneously for $1 \leq k \leq K$ if $\tau \leq 1/(8K)$, since $\Phi_n(k) = k \log n + O_K(1)$ there. Shrinking τ handles the remaining finite sizes. Thus it supplies one common small τ for all bounded levels, and then (84) closes induction for the larger levels. Thus all three invariants prove the first assertion of Theorem 9.1, while (85) records a separate fixed-level estimate.

9.3 Cycle moments from positive tensor traces

We next prove the density estimate. In a node of size $2m$ in the recursive network, form the bipartite graph of input and output switch cells, with one edge for each tracked card. It is a union of paths and even cycles, including double edges. Let c_v^* be its cycle count; let c_v be the count when all cards are tracked. A node of size 2^j has height j . Write C_j^*, C_j for their sums over all nodes at height j . Then $C_j^* \leq k/2$ and $C_j^* \leq C_j$. The full count at a node is the number of cycles of the relative permutation of its two children. In particular it depends only on the randomness below that node, irrespective of the tracks arriving from above.

For $q \geq 1$ put $Z_l(q) = \mathbb{E}_{g \sim K_{2^l}} q^{\#\text{cycles}(g)}$. This scalar generating function is defined for every real $q \geq 1$. For an integer q it is also the unnormalized trace of K_{2^l} on $(\mathbb{C}^q)^{\otimes 2^l}$. The tensor identity in Lemma 8.2, specialized to this integer alphabet, gives

$$Z_l(q) \leq Z_{l-1}(q)Z_{l-1}(1 + (q-1)/2) \quad (q \in \mathbb{Z}_{\geq 1}). \quad (86)$$

For completeness, divide the positive child operator by its trace and call the resulting density matrix D . Expanding the crossing average over a uniformly chosen subset S of tensor positions gives $Z_l(q) = Z_{l-1}(q)^2 \mathbb{E}_S \text{tr}(D_S^2)$. The partial trace of a permutation is its induced permutation on S , multiplied by q for each cycle avoiding S . Therefore its norm is at most that scalar. Bounding $\text{tr}(D_S^2)$ by $\|D_S\|_{\text{op}}$ and averaging S contributes at most $1 + (q-1)2^{-|C|} \leq 1 + (q-1)/2$ per cycle, proving the recursion.

Two useful explicit versions are

$$2^{-l} \log Z_l(1 + 2^h) \leq 2^h (.81)^l \quad (h = 0, 1, \dots), \quad (87)$$

$$\log Z_l(1 + w) \leq w(2^l)^\gamma, \quad \gamma = \log_2(1 + \log_2(3/2)) < 1 \quad (w = 2^h, h \in \mathbb{Z}_{\geq 0}). \quad (88)$$

For $q > 2$ the normalized induction factor is $3/4$; for $q = 2$ use $Z_l(3/2) \leq Z_l(2)^{\log_2(3/2)}$, giving factor $(1 + \log_2(3/2))/2 < .81$. The same two cases yield the sharper exponent in (88), since $3/2 \leq 2^\gamma$. The base $Z_0(q) = q$ satisfies both bounds.

Here is the conditional form needed to sum heights. Let $\mathcal{F}_h$ be the sigma-field generated by all switches at heights at most h , with the trivial sigma-field when $h < 1$. Reveal $\mathcal{F}_{j-l-1}$. Each child of a height- j node has conditional average A^*UA , with U a fixed unitary and A a tensor product of l -coordinate sweeps on 2^l -slot cells. Independence of the two children and Hilbert–Schmidt Cauchy–Schwarz show

$$\mathbb{E}[q^{C_j} \mid \mathcal{F}_{j-l-1}] \leq Z_l(q)^{n/(2 \cdot 2^l)}. \quad (89)$$

Indeed, writing F_0, F_1 for the two conditional tensor mean operators, $\text{tr}(F_0^* F_1) \leq \text{tr}((AA^*)^2) \leq \text{tr}(AA^*)$, and the trace factors over the cells. Fresh switches in different height- j nodes are conditionally independent.

Choose $\delta > 0$ with $4\delta < 1 - \gamma$. Partition heights into bands $[b, 2b - 1]$ for powers of two b , and separate alternating bands. For $b \geq 2$, take $l = b/2$ and round $2^{2\delta b}$ up to an integer alphabet, then its excess above one up to a power of two. Equations (88)–(89) give

$$\mathbb{E}[2^{2\delta b C_j} \mid \mathcal{F}_{b/2-1}] \leq \exp(Cn 2^{-\eta b}), \quad \eta = (1 - \gamma)/2 - 2\delta > 0.$$

Conditional Hölder over the at most b heights gives the same bound for $2^{2\delta} \sum C_j$ in the band. A smaller band of the same parity ends below $b/2$, so its factor is measurable for this conditioning. Integrate the highest band first and iterate; Cauchy–Schwarz combines the two parities. For $J \geq 2$ the result is

$$\mathbb{E} 2^{\delta \sum_{j \geq J} C_j} \leq \exp(Cn 2^{-\eta' J}), \quad \eta' = \eta/2. \quad (90)$$

The use of full counts is essential: replace tracked counts by full counts in every band where conditional integration will be used before performing that integration. Tracked lower counts can otherwise depend on the routes chosen above them.

Taking J a sufficiently large constant multiple of $\log_2(en/k)$, the deterministic bound on lower heights and (90) give, uniformly over initial tuples,

$$\mathbb{E} 2^{\delta \sum_j C_j^*} \leq e^{C\Lambda_n(k)}. \quad (91)$$

Equivalently one may use (87): in a band starting at b , the conditional logarithmic bound is $(n/2)2^{\lceil 2\delta b \rceil} (.81)^{\lfloor b/2 \rfloor} \leq Cn e^{-\gamma_0 b}$ for a sufficiently small fixed δ . Using tracked deterministic bounds below $\gamma_0^{-1} \log(n/k)$ and full counts above it proves the same estimate with explicit geometric decay in band height.

9.4 An entropy coding proof of the cycle moment

The tensor argument proves the required moment. The following coding argument also controls the cycle count under an arbitrary change of the switch law, charging its increase to relative entropy. This is the additional conclusion we retain from this alternative proof. In this subsection entropy is measured in bits.

At height t the switch array consists of both outer layers of every node of size 2^t ; all these raw switch bits are independent under the reference fair law. Let Q be any probability law on the full array of raw switches at all heights of the n -position reflected network. Write $\mathcal{B}_t$ for the vector of bits at height t and define the nonnegative number

$$I_t = |\mathcal{B}_t| - H_Q(\mathcal{B}_t \mid \mathcal{B}_1, \dots, \mathcal{B}_{t-1}),$$

where conditional Shannon entropy is already averaged under Q . Thus $I_t \geq 0$ and $\sum_t I_t = D(Q \parallel \text{fair})$.

Proposition 9.3 (Cycle costs under a change of law). *For any fixed input tuple of k distinct positions and any law Q on the raw switches, the actual tracked cycle counts satisfy*

$$\mathbb{E}_Q \sum_j C_j^* \leq C\Lambda_n(k) + CD(Q \parallel \text{fair}), \quad (92)$$

with an absolute constant C .

Proof. With $h = \lfloor j/2 \rfloor$, the full cycle counts obey

$$\mathbb{E}_Q C_j \leq Cne^{-cj} + \frac{C}{j} \sum_{t=j-h}^{j-1} I_t \quad (j \geq 2). \quad (93)$$

To prove this, code the window bits given smaller heights. At one height- j node, with child size $m = 2^{j-1}$, send literally all window bits except the first h input layers of one child. The relative child permutation is now D_0A , with D_0 known and A a butterfly with $hm/2$ unknown bits. If D_0A has r cycles, send r , one representative per cycle, and the ordered list of cycle lengths. These data cost at most $C[\log_2(2m) + r \log_2(em/r)]$ bits, using $\binom{m}{r}$ and $\binom{m-1}{r-1}$.

Traverse each cycle from its representative. Send each as-yet-unknown switch bit used on its A paths, except on the last path of that cycle: its required output is already known, so its unique butterfly route recovers all those bits without sending them. The specified lengths tell the decoder exactly when this omission occurs. For each fixed choice of the public orders described next, use a fixed-width field for r , an enumerative code for its representative set, and an enumerative code for the composition of m into the transmitted lengths. The decoder then reads a bit precisely when the traversal first queries an unknown switch outside a closing path. It consequently knows when the whole message has ended. The resulting code is prefix-decodable. Every switch is recovered after all positions have been traversed.

Use independent public random priority orders on vertices to choose a uniform representative within each cycle and then to order the transmitted representative set. The decoder uses that same second priority order; no extra factorial cost is needed to specify the cycle order. For any set of s butterfly paths, the number of internal contacts, counted at both ends, is at most $s \log_2 s$. Indeed, splitting a block into two predecessor blocks of sizes s_1, s_2 adds at most $2 \min(s_1, s_2)$ contacts, and

$$s_1 \log_2 s_1 + s_2 \log_2 s_2 + 2 \min(s_1, s_2) \leq (s_1 + s_2) \log_2 (s_1 + s_2).$$

The closing path of a length- s cycle therefore has, on average, at least $h - \log_2 s$ contacts outside its cycle. Each outside cycle is later with probability $1/2$, so those switches are then unknown. The expected saving is at least

$$\frac{r}{2} \{h - \log_2(m/r)\} - C[\log_2(2m) + r \log_2(em/r)].$$

This remains a valid lower bound if some terms are negative, and covers cycles of length one. The public orders are independent of the switch array. Conditional on the smaller heights and on those orders, the code remains a prefix code for the entire window: different height- j nodes use disjoint sets of its bits. Its entropy is at most its expected length. Average over the public orders and the smaller heights. The difference between the literal window length and its conditional entropy is exactly $\sum_{t=j-h}^{j-1} I_t$, by the entropy chain rule. Thus the saving just computed is charged to that window deficit, without assuming independence of the bits under Q . The inequality $Cr \log_2(em/r) \leq (h/4)r + C'me^{-c'h}$ and the $n/2^j$ nodes give (93).

At each height the tracked count is at most k and at most the full count. Moreover each I_t occurs with bounded total coefficient in the windows, since $\sum_{j>t, j-\lfloor j/2 \rfloor \leq t} 1/j = O(1)$. Sum the deterministic tracked bound below a sufficiently large multiple of $\log(en/k)$ and (93) above it. The first sum and the exponentially decaying errors cost $O(\Lambda_n(k))$; the bounded window coefficients cost $O(\sum_t I_t)$. This proves (92). $\square$

Apply this to the law tilted by $2^\varepsilon \sum C_j^*$. The variational identity

$$\log_2 \mathbb{E} 2^\varepsilon \sum C_j^* = \varepsilon \mathbb{E}_Q \sum C_j^* - D(Q \| \text{fair})$$

proves (91) when $\varepsilon C < 1$. Thus the tensor-trace argument and the coding argument each supply a complete, separate proof of the required cycle moment.

9.5 From routed cycles to densities and multiplicities

Consider specified injective endpoints for h links in a size- $2m$ node. Let e be the total number of shared input and output pairs, and let c be the cycle count in their union. The links admit exactly 2^{h-e+c} proper child colorings. Each has switch probability 2^{-2h+e} . If $R = (2m)^h p$ is the provisional transition density, the child recursion is

$$R = \sum_{\text{proper colors}} 2^{-h+e} R_0 R_1 = 2^c \text{Avg}_{\text{proper colors}} R_0 R_1. \quad (94)$$

Jensen with exponent $1 + \delta$ charges a factor $2^{\delta c}$. After unrolling to one-slot leaves,

$$(n^k p(x, y))^{1+\delta} \leq n^k \mathbb{E}_{K_n} [\mathbf{1}_{\{\pi x=y\}} 2^{\delta \sum C_j^*}].$$

The remaining coefficients are the actual switch probabilities: each link is counted once per height, and specified recursive routings are disjoint events. Summing in y and using $(n)_k/n^k \leq 1$ proves (67). Symmetry supplies the corresponding column estimate. We will also use the provisional normalization directly:

$$\sup_x \sum_y n^{-k} (n^k p(x, y))^{1+\delta} \leq e^{C\Lambda_n(k)}. \quad (95)$$

Indeed summing the preceding unrolled inequality proves this before the factor $(n)_k/n^k$ is inserted. Equivalently it follows from the density bound after enlarging C , since

$$\log \frac{n^k}{(n)_k} = \sum_{i=0}^{k-1} -\log(1 - i/n) \leq \int_0^k -\log(1 - x/n) dx \leq k.$$

The integral is finite also at $k = n$.

The row and column bounds imply bounded operators $K_n : L^1 \rightarrow L^{1+\delta}$ and $K_n : L^{(1+\delta)/\delta} \rightarrow L^\infty$, each with norm at most $M = e^{C\Lambda_n(k)}$ after enlarging C . Riesz–Thorin interpolation decreases the reciprocal exponent by $\delta/(1+\delta)$ at each application. After $p_0 = \lceil (1+\delta)/\delta \rceil$ applications, with probability-space inclusion for the last step, $K_n^{p_0} : L^1 \rightarrow L^\infty$ has norm at most M^{p_0} . Equivalently,

$$\sup_{x,y} |X_{n,k}| K_n^{p_0}(x,y) \leq e^{C\Lambda_n(k)}. \quad (96)$$

The diagonal kernel bound gives (68).

There are two ways to use the label multiplicity f_β . For the first transfer, the goal is to use the f_β equivalent slot copies of one position vector. Clip the entries $p(x,y)$ of K_n above $n^{-k}\sqrt{f_\beta}$, setting them to zero, and call the retained matrix M_0 . The provisional moment bound (95) and symmetry give

$$\|K_n - M_0\|_{\text{op}} \leq e^{C\Lambda_n(k)} f_\beta^{-\delta/2}. \quad (97)$$

Since every row sum is at most one and $|X_{n,k}| \leq n^k$, $\|M_0\|_{\text{HS}}^2 \leq \sqrt{f_\beta}$. Clipping commutes with relabeling the tuple slots. Compressing to position type λ , whose label type is β , therefore repeats each singular value at least f_β times. Its compressed norm is at most $f_\beta^{-1/4}$. If $\log f_\beta$ exceeds a sufficiently large constant times $\Lambda_n(k)$, these two estimates give $W_n(\lambda) \leq e^{-c \log f_\beta}$; otherwise (65) applies. This proves (69).

The clipping proof now gives the combined level-and-tail estimate. To obtain the explicit tail bound (66) by a different mechanism, average a matrix coefficient over all slot copies. Tensor a maximizing unit vector in V_λ with an orthonormal basis of V_β , obtaining orthonormal functions u_j , $1 \leq j \leq f_\beta$, in probability norm. Define

$$F(x,y) = \frac{1}{f_\beta} \sum_j \overline{u_j(x)} u_j(y).$$

Then $\mathbb{E}(FK) = W_n(\lambda)$ and $\mathbb{E}|F|^2 = f_\beta^{-1}$. The diagonal $G(x) = f_\beta^{-1} \sum_j |u_j(x)|^2$ is slot-invariant and has mean one. Each orbit of the slot-permutation group is the set of $k!$ orderings of one k -element set, so has probability $\binom{n}{k}^{-1}$. Nonnegativity therefore gives $G(x) \leq \binom{n}{k}$. Cauchy–Schwarz gives the same bound for $|F(x,y)|$. With $p = 1 + \delta$ and $p' = p/(p-1) > 2$, Hölder gives

$$W_n(\lambda) \leq \|K\|_p \|F\|_{p'} \leq e^{C\Lambda_n(k)} \binom{n}{k}^{1-2/p'} f_\beta^{-1/p'}. \quad (98)$$

This proves (66) with $\zeta = 1/p'$. Since $D_\lambda \leq \binom{n}{k} f_\beta$, geometric interpolation with (65) proves (70).

Finally, the tuple trace gives a completion that needs no tail-dimension interpolation. At exact level k the multiplicity in $X_{n,k}$ is f_β , while the regular multiplicity is $D_\lambda \leq \binom{n}{k} f_\beta$. For an integer $p > p_0$, positivity and the estimates (65) and (68) yield

$$\text{tr}_{\text{reg}} K_n^p - 1 \leq \sum_{k=1}^{n-1} e^{-[(p-p_0)c-C-1]\Lambda_n(k)}. \quad (99)$$

Alternatively (70) bounds the regular Fourier sum directly by $\sum_{k \geq 1} 2^k e^{-c_1 M \Lambda_n(k)}$ once $c_2 M \geq 2$. Both sums tend to zero for a sufficiently large fixed exponent. Schatten Hölder gives $\|T_n^p|_{\mathbf{1}^\perp}\|_{\text{HS}}^2 \leq \text{tr}_{\text{reg}} K_n^p - 1$ without assuming normality of T_n . Thus these arguments mix the entire permutation law after an absolute number of sweeps, each consisting of exactly d physical shuffles.

10 Compressed generators and adaptive alphabets

A fixed representation level does not determine the dimension: diagrams with the same first row can have very different shapes below it. We now separate these two contributions in another way. Compression of the transposition generator gives the level estimate. A fourth trace moment then gives a negative power of the full dimension, at a positive cost proportional to the level scale. Combining the two cancels that cost.

The fourth-moment argument realizes a diagram in an induced tensor representation. Its alphabet size controls the multiplicity paid at each split. An alphabet that was economical at the root can become too large for a small descendant diagram. We change it at that point and prove that the total cost of all such changes is summable. The proof below first establishes the estimate for one split, then specifies the changes and accounts for their cost over the entire recursion tree.

Put $F_n = T_n T_n^*$, so that $F_n = P(F_m \otimes F_m)P$ for $n = 2m$. If $\lambda = (n - r, \rho) \vdash n$, write $h_n(r) = r \log(n/r) + r$ for $r > 0$, and $h_n(0) = 0$. The coordinate names here are ordered so that P is the final matching of T_n . Thus F_n is the other positive orientation of the sweep, with the same singular values as $T_n^* T_n$. All traces are unnormalized; $0 \log 0 = 0$ throughout.

Theorem 10.1. *There are absolute positive constants c, c', C such that for every dyadic n and every partition $\lambda = (n - r, \rho) \vdash n$,*

$$\|F_n|_{V_\lambda}\|_{\text{op}} \leq e^{-ch_n(r)}, \quad (100)$$

$$\text{Tr}_{V_\lambda} F_n^4 \leq \exp\{-c' \log D_\lambda + Ch_n(r)\}. \quad (101)$$

Consequently $\|T_n|_{V_\lambda}\|_{\text{op}}$ is at most $\exp[-c_0\{\log D_\lambda + h_n(r)\}]$ for an absolute $c_0 > 0$.

10.1 Compression of the transposition generator

Let

$$L_n = \frac{1}{n} \sum_{i < j} (I - (ij)), \quad T = L_n - L_0 - L_1, \quad J = L_0 - L_1,$$

where L_0 and L_1 act on the first and second halves, respectively, and each uses normalization $1/m$ when $n = 2m$. Transpositions in these formulas denote their unitary representation matrices. The transposition content formula [31] makes L_n scalar on V_λ , with value

$$\ell = r - \frac{\binom{r}{2} + c(\rho)}{n}, \quad r/2 \leq \ell \leq r. \quad (102)$$

Here $c(\rho)$ is the sum of the box contents of ρ . The three operators in this display commute with each other before compression, although they need not commute with P .

Lemma 10.2. *Let $1 \leq r \leq n - 1$ and let P act on the ordered injective r -tuple module by simultaneously switching matched positions. Put $p_* = r/n$. For absolute $t_0, t_1, v_0 > 0$,*

$$\|Pe^{tT}P\| \leq e^{Cp_*|t|}, \quad |t| \leq t_0, \quad (103)$$

$$\|Pe^{\pm t_1 \log(1/p_*)T}P\| \leq 1 + o(1), \quad p_* \longrightarrow 0, \quad (104)$$

$$\|Pe^{vJ^2/r}P\| \leq 1 + O(v), \quad 0 \leq v \leq v_0. \quad (105)$$

The estimates are uniform in n, r .

Proof. Embed injections in the tensor power of the one-site space and let D be the mask deleting repeated sites. Write $s_j = 1$ or -1 according as the j th coordinate lies in the first or

second half. The one-coordinate part of T is a projection onto the signed uniform vector. Its correction on distinct tuples is $n^{-1} \sum_{j < l} s_j s_l (I + (jl))$. The one-coordinate part of J is the sign diagonal minus the difference of the two half-uniform rank-one projections; its correction is $-(2m)^{-1} \sum_{j < l} (s_j + s_l)(I + (jl))$.

Use the symmetric and antisymmetric basis within matched site pairs. A pair site is odd when an odd number of its tensor slots use the antisymmetric vector. The endpoints selected by P have no odd pair sites; two antisymmetric slots at the same pair site have even parity. The injection mask kills basis indices with more than two tensor slots at one pair site and preserves the odd-site pattern. At a pair site occupied by slots j, l , it is the projection $(I - s_j s_l)/2$ in the original position basis. In the symmetric and antisymmetric basis the second term flips both slot signs, so this projection has absolute row sum one. The product over disjoint doubly occupied sites has the same bound. Hence mask insertions between steps have absolute row mass at most one.

Here is the transition count in this basis. Write $e_{p,+}, e_{p,-}$ for the two vectors at pair site p , and let S_a flip the sign of slot a . Let τ_{ab} exchange the two tensor slots, so the correction to T is $n^{-1} \sum_{a < b} S_a S_b (I + \tau_{ab})$. Only states with at most two slots per pair site survive the mask. If exactly k pair sites are odd, at most $2k$ slots lie at odd sites. The mask preserves the entire set of odd sites, as well as the slot occupancies, so the following bounds can be checked before each mask insertion.

The one-slot rank-one part of T sends a minus slot from a site p to a minus slot at any site q , with coefficient $1/m$, and kills plus slots. When $p \neq q$, it toggles the parity at precisely those two sites. Even-even moves have total absolute mass at most $rm/m = r$. Odd-odd moves have at most $2k$ source slots and k target sites, giving $2k^2/m$. For moves preserving k , an odd source and even target cost at most $2k$, whereas an even source and odd target cost at most rk/m . The same-site moves have total mass at most r/m . Thus this last term is $O(p_*)$, even when many occupied sites have even parity.

The two-slot correction toggles the parities of the two sites holding a, b ; the slot exchange changes neither their occupancies nor these parity changes. Its two summands have total coefficient at most $2/n$ per slot pair. Pairs at two even sites give mass $O(r^2/n) = O(r)$; pairs at two odd sites give $O(k^2/n)$; and pairs at one site of each parity give $O(kr/n) = O(k)$. A pair of slots at the same site leaves its parity unchanged. There are at most $r/2$ such pairs, so their mass is $O(r/n) = O(p_*)$. Combining the two parts, and using $r/m \leq 2$, gives the absolute transition bounds

$$\begin{array}{c|ccc} \text{change in odd-site count} & +2 & -2 & 0 \\ \hline T & Cr & Ck^2/m & C(k + p_*) \end{array}$$

for both rows and columns by self-adjointness. In particular the fixed-count cost at $k = 0$ is $O(p_*)$, not $O(r)$. Weight level $k = 2z$ by $z!(rm)^{-z/2}$. The off-diagonal masses then become at most $C\sqrt{p_*}(z + 1)$ upward and $C\sqrt{p_*}z$ downward. Every return path with $2u$ off-diagonal steps has exponential-series weight bounded by

$$e^{Cp_*|t|} \frac{(C\sqrt{p_*}|t|(u + 1))^{2u}}{(2u)!} e^{C|t|u}.$$

There are at most 4^u jump-direction lists with $2u$ up or down steps; their intervening stays sum to the displayed exponential factor. The weights telescope to one on every path returning to level zero. Since $(2u)! \geq (2u/e)^{2u}$, the series excluding its zero-step term is bounded by a geometric series with ratio $C'p_*t^2e^{C|t|}$. For fixed sufficiently small $|t|$ it is $O(p_*t^2)$, which is absorbed in $e^{Cp_*|t|}$. The matrices are self-adjoint, so the same absolute row bound applies to columns and Schur's test proves (103). At $|t| = t_1 \log(1/p_*)$, this ratio tends to zero if t_1 is a sufficiently small positive constant, while $p_*|t| \rightarrow 0$. This proves (104) uniformly in n, r .

For J , the one-slot sign diagonal flips one sign at its current site. The difference of the half-uniform projections sends $e_{p,+}$ to $m^{-1} \sum_q e_{q,-}$ and $e_{p,-}$ to $m^{-1} \sum_q e_{q,+}$. In the first case only the target site's parity changes, and in the second only the source site's parity changes. Thus all these moves change k by exactly one. Their upward mass is $O(r)$; downward moves either use one of the at most $2k$ slots at odd sites or choose one of the k odd target sites, for total $O(k + rk/m) = O(k)$. The correction $-n^{-1} \sum_{a < b} (S_a + S_b)(I + \tau_{ab})$ also toggles just one site's parity. Its full mass is $O(r^2/n)$, and its downward mass is $O(kr/n)$ because the toggled site must be odd. These are respectively $O(r)$ and $O(k)$, including coincident pair sites. Mask insertions preserve both conclusions. Consequently J has upward and downward masses at most Cr and Ck . A return of length $2j$ therefore has total mass at most $(Crj)^j$. Expanding $e^{vJ^2/r}$ and using $j^j/j! \leq e^j$ gives (105). The mask can be inserted between every step because its absolute row mass does not exceed one. Thus these estimates apply to injections, not only to unrestricted tensors. $\square$

Proof of (100). We first obtain a uniform estimate without the logarithmic factor. Suppose at size m that $F_m \preceq e^{-a_m L_m}$ on every irreducible, with $0 < a_m \leq t_0$. On the parent λ -space, centrality and (103) imply

$$\|P(F_m \otimes F_m)P\| \leq e^{-a_m \ell} \|Pe^{a_m T}P\| \leq e^{-a_m \ell + Ca_m r/n} \leq e^{-a_m(1-2C/n)\ell}.$$

At one fixed sufficiently large dyadic size a positive a_m is available from the strict finite-size norm gap. Starting above $4C$, the product of the factors $1 - 2C/n$ over doubled sizes is positive, since the sum of their deficits is finite. Thus for an absolute $a_0 > 0$ we have $\|F_n|_{V_\lambda}\| \leq e^{-a_0 \ell}$; finitely many smaller sizes are absorbed by reducing a_0 .

This estimate will pay an $O(r)$ error. To obtain the additional $\log(n/r)$ factor, we strengthen it by inducting on

$$\|F_n|_{V_\lambda}\| \leq \exp\{-aH_n(\ell) + D_0 \max(0, 2r - 1)\}, \quad H_n(x) = x \log(n/x) + x. \quad (106)$$

Take a unit vector in the range of P in a minimal r -tuple realization of V_λ . Decompose it by the set of labels in each half and by the two child position types. Squared component norms give the spectral probability law used below. If the occupancies are $j_0 + j_1 = r$ and the child first-row deficits are r_0, r_1 , then $0 \leq r_i \leq j_i$, hence $r_0 + r_1 \leq r$. Let ℓ_0, ℓ_1 be the corresponding child generator values and put $z_i = \ell_i/\ell$. Since $\ell_i \leq r_i \leq r \leq 2\ell$, the variables z_i lie in $[0, 2]$. The decrement is

$$Z = H_n(\ell) - \sum_i H_m(\ell_i) = \left(\ell - \sum_i \ell_i \right) \log(n/\ell) + \ell \sum_i \{z_i \log(2z_i) - z_i + 1/2\}.$$

Since $z_i \in [0, 2]$ and $z \log(2z) - z + 1/2 \leq C(z - 1/2)^2$ there,

$$Z \leq (\log(n/\ell) + C)|T| + CJ^2/r. \quad (107)$$

For a unit vector in the range of P , spectral expectation in the commuting child algebra and Lemma 10.2 give

$$\mathbb{E}e^{2aZ} \leq 3/2 \quad (r/n \leq p_0) \quad (108)$$

for sufficiently small absolute a, p_0 . For the absolute value in (107), use $(e^{|u|/2} - 1)^2 \leq C(\cosh u - 1)$ and both signs in (104); then use Cauchy-Schwarz and (105).

For $r \geq 2$, the sum of the child penalties in (106) drops by at least one, except when (r_0, r_1) is $(r, 0)$ or $(0, r)$. This persistence requires $j_0 = r$ or $j_1 = r$, respectively. On the range of P , this persistence event has weight at most 2^{1-r} . Indeed it requires all tuple coordinates

in one half. If two occupy a matched pair it is impossible; otherwise independent switches give that probability exactly. The ratio in the induction is consequently at most

$$\sqrt{(3/2)2^{1-r}} + e^{-D_0} \sqrt{3/2} \leq 1$$

when D_0 is large. Level zero is trivial and level one is annihilated because a single card is uniform after one sweep. In the range $r/n > p_0$, the ratio $H_n(\ell)/r$ is bounded by a constant depending only on p_0 , so a sufficiently large D_0 makes the right side of (106) at least one. This starts and closes that induction, with bounded sizes covered by the same finite choice of constants.

It remains to remove the positive penalty. For a fixed sufficiently small $\theta > 0$, geometrically interpolate the bound $e^{-a_0\ell}$ with (106). Because $2D_0r \leq 4D_0\ell$, choose θ so that $4\theta D_0 \leq (1-\theta)a_0$. The penalty is then canceled and we obtain $\|F_n|_{V_\lambda}\| \leq e^{-\theta a H_n(\ell)}$. Finally $r/2 \leq \ell \leq r$ gives $H_n(\ell) \geq \frac{1}{2}h_n(r)$ and proves the level estimate. $\square$

10.2 The induced tensor spaces and a single split

The level estimate is now complete. The remaining goal is the fourth-moment bound. We will prove it on larger representations containing V_λ , because a trace of a positive power can only increase under such an enlargement.

More precisely, a list of categories with sizes b_j and representations V_{η_j} defines

$$\mathcal{M} = \text{Ind}_{\prod_j S_{b_j}}^{S_n} \bigotimes_j V_{\eta_j}, \quad \sum_j b_j = n.$$

Its log trace is $\sigma = \log \text{Tr}_{\mathcal{M}} F_n^4$, with $\sigma = -\infty$ if the trace is zero. Empty categories may be omitted. An induced module will have categories of total size n . A row category consists of a diagram η of size b , an alphabet of dimension $q \geq \ell(\eta)$, where $\ell(\eta)$ is its number of rows, and an orientation, ordinary or sign-twisted. A regular category consists of a spatial diagram κ of size b and b distinct named labels. We may designate one ordinary one-row category of alphabet dimension one as the *background*; it may be empty. All other categories are *payload*, including ordinary one-row categories of alphabet dimension one. In the application the background is the original first row of λ . Every descendant of a payload category remains payload after restriction or a change of alphabet. There is at most one background category in each node, so its child counts are determined by the payload counts and the two half sizes. Isomorphic categories remain distinct when their labels or origins differ.

For a row category j , η denotes the alphabet type before the sign twist. Its spatial factor V_{η_j} in $\mathcal{M}$ is V_η in ordinary orientation and $V_\eta \otimes \text{sgn}_{S_b} \cong V_{\eta'}$ in sign-twisted orientation, where η' is the conjugate partition. In both orientations, $q \geq \ell(\eta)$ and the multiplicity $G_q(\eta)$ below refer to this pre-twist diagram.

Realize row categories in signed tensor powers. Each category has its own alphabet; let $\mathcal{H}$ be the space spanned by words specifying one alphabet symbol at each of the n position slots, with the prescribed number of slots in each category. An adjacent swap contributes -1 exactly when both symbols are odd; these adjacent swaps generate the graded permutation action. Use odd symbols precisely for sign-twisted categories. This gives the indicated representation on the stabilizer of a category allocation, and induction sums over allocations. In a regular category, each named label appears once. The commuting group is $U(q)$ for each row alphabet and S_b acting on the names of each regular category. Let E be the product of their isotypic projections onto the row types η and regular types κ . On a fixed category allocation, Schur–Weyl duality and the regular bimodule decomposition give the spatial tensor product and its multiplicity space. Summing over allocations therefore gives

$$g = \prod_{\text{row}} G_q(\eta) \prod_{\text{regular}} D_\kappa \tag{109}$$

copies of the desired spatial induced module:

$$E\mathcal{H} \cong \mathbb{C}^g \otimes \mathcal{M}.$$

Here G_q is the dimension of the polynomial $U(q)$ representation. Spatial permutations act trivially on $\mathbb{C}^g$, so E commutes with the network operators and a spatial trace on $E\mathcal{H}$ is g times the trace on $\mathcal{M}$.

A child projection W specifies counts $b_0 + b_1 = b$ and types μ, ν in the two halves of each category. For a row category, use the two child $U(q)$ isotypic projections, which commute with the diagonal $U(q)$ action and hence with E . For a regular category first fix its subset of b_0 names in the first half and project the two name groups onto μ, ν ; then sum these orthogonal projections over all $\binom{b}{b_0}$ subsets. This sum preserves each name subset and is invariant under global relabeling, so it also commutes with E . The projections W sum to the identity and commute with $Q = F_m \otimes F_m$. Only types with positive Littlewood–Richardson coefficient for the parent type contribute to WE ; we call these choices compatible. All row types in this description are taken before their inherited sign twist. Write g_i for the multiplicity (109) evaluated on the induced module in half i . Put

$$\mathcal{L} = n \log n - \sum_{\text{row}} \sum_i \eta_i \log \eta_i - \sum_{\text{regular}} b \log b.$$

Lemma 10.3. *For compatible parent and child types,*

$$\|WPE\|^2 \leq \prod_{\text{row}} G_q(\mu) G_q(\nu) e^{-(\mathcal{L} - \mathcal{L}_0 - \mathcal{L}_1)/2}.$$

Proof. Choose block density matrices A, B on the one-site alphabet. On a row block their spectra are (μ_i/m) and (ν_i/m) ; on a regular block they are scalar $b_0/(mb)$ and $b_1/(mb)$. Their traces, summed over all blocks, are one. If V applies $A^{1/2}$ in the first half and $B^{1/2}$ in the second, then

$$PVP \leq P \left(((A+B)/2)^{1/2} \right)^{\otimes n} P. \quad (110)$$

This is the needed joint-concavity instance for the operator geometric mean; see also [9, Appendix A.2.1, Lemma 4]. We include its block-matrix proof. On one matched pair, average the two positive block matrices with diagonal entries $A \otimes I, I \otimes B$ and $B \otimes I, I \otimes A$. Their positive off-diagonal blocks are the respective geometric means. The maximal positive off-diagonal block with diagonal entries X, Y is $X^{1/2}(X^{-1/2}YX^{-1/2})^{1/2}X^{1/2}$, by the Schur complement and monotonicity of square roots. Averaging and then tensoring proves (110); continuity handles zero eigenvalues. The signed flips cause no change because the matrices respect category blocks.

For one row category, average the eigenbasis of A over $U(q)$. On child type μ , Schur’s lemma makes the averaged operator $(A^{1/2})^{\otimes b_0}$ scalar on its multiplicity space. Its scalar is the Schur character s_μ divided by $G_q(\mu)$, and the highest-weight monomial gives the lower bound

$$\frac{s_\mu(\sqrt{\mu_1/m}, \dots, \sqrt{\mu_q/m})}{G_q(\mu)} \geq \frac{\prod_i (\mu_i/m)^{\mu_i/2}}{G_q(\mu)}.$$

Zero exponents contribute one. Apply the same calculation to B and ν , independently in every row block and in the two halves. A regular block contributes the scalar $(b_0/(mb))^{b_0/2}(b_1/(mb))^{b_1/2}$. Multiplication therefore shows that the average of V dominates W times

$$\frac{\exp[-(\mathcal{L}_0 + \mathcal{L}_1 + \sum_{\text{regular}} b \log b)/2]}{\prod_{\text{row}} G_q(\mu) G_q(\nu)}.$$

For the parent bound put $C = (A + B)/2$. If $x_{j,i}$ are its eigenvalues in nonincreasing order in each row block, the norm of $(C^{1/2})^{\otimes b}$ on parent type η is the highest-weight product $\prod_i x_{j,i}^{\eta_i/2}$. The regular blocks of C are each $1/n$ times the identity on their b names. All eigenvalues of C sum to one. The entropy maximization

$$\prod_{\text{row}, i} x_{j,i}^{\eta_{j,i}/2} n^{-s/2} \leq \prod_{\text{row}, i} (\eta_{j,i}/n)^{\eta_{j,i}/2} n^{-s/2}$$

follows by maximizing subject to the remaining row trace $1 - s/n$, where s is the regular mass. Thus on E the right side of (110) has norm at most $\exp[-(\mathcal{L} + \sum_{\text{regular}} b \log b)/2]$. This upper bound is uniform in the eigenbases just averaged. Compress the averaged inequality by E and use $EPWPE = (WPE)^*(WPE)$. Dividing the parent bound by the child scalar proves the lemma. Sign twists change only the spatial action, so the same multiplicity-space calculation applies in either orientation. $\square$

Take $p = 4$, and let $Y = WQ \geq 0$. The needed Schatten inequality is

$$\text{Tr}(EPYPE)^p \leq \|WPE\|^{2p-4} \text{Tr}(PY^{p/2}P)^2. \quad (111)$$

Indeed apply three-lines interpolation to $Y^{pz/4}WPE$, extended by zero on the kernel of Y , between the operator norm at $\Re z = 0$ and the fourth Schatten norm at $\Re z = 1$. At $z = 2/p$ the target norm is the $2p$ norm. Raising the interpolation inequality to $2p$ gives (111): at the fourth-norm endpoint first obtain $\text{Tr}(EPY^{p/2}PE)^2$, then drop the orthogonal compression by E using Hilbert–Schmidt contraction. Thus no commutation of P with Y is required.

Lemma 10.4. *Let s be the total number of named labels and $\sigma_i = \log \text{Tr} F_m^4$ on the child induced modules. If $m \geq 2$,*

$$\text{Tr}(PY^2P)^2 \leq g_0 g_1 e^{\sigma_0 + \sigma_1} 2^{\gamma s/2} \prod_{\text{regular}} \left(2^{-b} \binom{b}{b_0} \right), \quad \gamma = \frac{\log_2 3}{2} < 1. \quad (112)$$

For $m = 1$ the inflation $2^{\gamma s/2}$ may be replaced by $2^{s/2}$.

Proof. Collapse each matched pair of tensor slots to its unordered contents. On such an orbit, P has rank at most one and its nonzero vector has constant absolute coefficient. For orbit states C, D , let $j(C, D)$ be the number of cycles in the union of the two partial matchings on named labels, counting a common edge twice as a cycle. The fraction of orientation pairs agreeing on the labels' half assignments is $2^{-s+j(C,D)}$. Indeed, if the two matchings have a, a' edges, their separate half assignments number 2^{s-a} and $2^{s-a'}$. The union has $2^{s-a-a'+j(C,D)}$ assignments, since each cycle has one redundant opposite-half condition. Dividing gives the fraction. Cauchy–Schwarz therefore bounds the squared orbit entry by

$$2^{-s+j(C,D)} \sum_{u,v} |Y^2(u, v)|^2. \quad (113)$$

Here the sum is over word pairs in the two orbits; entries of Y^2 vanish unless each named label stays in its assigned half. For such a word pair u, v , let $A_*, B_* \subseteq [m]$ be the first-half sites occupied by names in u, v , respectively, and let C_*, D_* be the corresponding second-half site sets. The partial bijections $f : A_* \rightarrow B_*$ and $g : C_* \rightarrow D_*$ follow each named label from u to v . A subset $U \subset A_* \cap C_*$ satisfies $f(U) = g(U)$ exactly when it is a union of closed components of the resulting partial permutation. These components are the cycles counted by $j(C, D)$, so the number of such subsets is $2^{j(C,D)}$.

Let τ range uniformly over the last matching switches in the second child, and apply it to the second-half sites of the output word v . This replaces g by τg , leaving f fixed. The

squared kernel weight is unchanged: W commutes with these spatial permutations, and the positive power of Q has range fixed by them. For each U , the probability of $f(U) = \tau g(U)$ is either zero or $|\mathcal{O}(g(U))|^{-1}$, where $\mathcal{O}$ is the matching group orbit. Thus averaging inside the sum over word pairs bounds the subset count by

$$\sum_{X \subset g(A_* \cap C_*)} |\mathcal{O}(X)|^{-1}.$$

A switch pair with two available slots contributes $1 + 2(1/2) + 1 = 3$; one available slot contributes $1 + 1/2 \leq \sqrt{3}$. Since $|A_* \cap C_*| \leq \min(s_0, s_1) \leq s/2$, where s_i is the number of names in half i , the product is at most $3^{s/4} = 2^{\gamma s/2}$. Sum (113) over words. For a fixed choice of name subsets the unswitched trace is $g_0 g_1 e^{\sigma_0 + \sigma_1}$. The aggregated projection is their orthogonal direct sum, giving

$$\text{Tr } Y^4 = \text{Tr}(WQ^4) = \left(\prod_{\text{regular}} \binom{b}{b_0} \right) g_0 g_1 e^{\sigma_0 + \sigma_1}. \quad (114)$$

Together with the factor 2^{-s} this proves the result. The named subsets enter through this trace identity exactly once. At $m = 1$, simply use $j(C, D) \leq s/2$. $\square$

We can now pass from the count-space trace to the desired spatial trace. Let M be the number of compatible child projections W . For each payload category of size b , the count split and two child partitions have at most $\exp(Cb^{2/3})$ choices, by the partition bound. The single background has no type choice and its counts are determined by the payload counts. Therefore $\log M \leq C \sum_{\text{payload}} b^{2/3}$. Named subsets are already aggregated in W and are not choices counted by M .

Resolve $Q = \sum_W WQ$ on $E\mathcal{H}$. The Schatten triangle inequality, followed by (111) with $p = 4$, the angle bound, and (112), gives, for $m \geq 2$,

$$\begin{aligned} ge^\sigma &= \text{Tr}(EPQPE)^4 \\ &\leq M^4 \max_W \left\{ e^{\sigma_0 + \sigma_1 - (\mathcal{L} - \mathcal{L}_0 - \mathcal{L}_1)} \left[\prod_{\text{row}} G_q(\mu) G_q(\nu) \right]^2 \right. \\ &\quad \left. \times g_0 g_1 2^{\gamma s/2} \prod_{\text{regular}} 2^{-b} \binom{b}{b_0} \right\}. \end{aligned} \quad (115)$$

The angle contributes two copies of each child row carrier, and the child trace contributes a third. Dividing by g and taking logarithms therefore yields the explicit bound

$$\begin{aligned} \sigma &\leq \max_W \left\{ \sigma_0 + \sigma_1 + 4 \log M - (\mathcal{L} - \mathcal{L}_0 - \mathcal{L}_1) \right. \\ &\quad + \sum_{\text{row}} [3 \log G_q(\mu) + 3 \log G_q(\nu) - \log G_q(\eta)] \\ &\quad \left. + \sum_{\text{regular}} [\log D_\mu + \log D_\nu - \log D_\kappa + (\gamma/2 - 1)b \log 2 + \log \binom{b}{b_0}] \right\}. \end{aligned}$$

For $m = 1$ replace γ by one, at an extra cost $O(s)$. This calculation motivates the potentials that we now telescope.

For a diagram of size b , put $I(\eta) = \sum_i \eta_i \log(b/\eta_i)$ and set $\delta = 1/32$. Give categories the potentials

$$\phi(\eta) = \delta I(\eta) \quad (\text{row}), \quad \phi(\kappa) = \log D_\kappa - \frac{1 - \delta}{2} b \log b \quad (\text{regular}). \quad (116)$$

Let Φ be their sum. For every category with child counts $b_0 + b_1 = b$, define

$$v_b = b\{\log 2 - H_2(b_0/b)\}, \quad H_2(x) = -x \log x - (1-x) \log(1-x),$$

with $v_0 = 0$. Thus H_2 uses natural logarithms in this section, and v_b is the loss from an imbalanced count split. The preceding lemmas imply the single-split accounting inequality

$$\sigma \leq \max \left\{ \sigma_0 + \sigma_1 - (\Phi - \Phi_0 - \Phi_1) - \sum_{\text{payload row}} v_b + C_1 \sum_{\text{payload}} b^{2/3} + C_1 \sum_{\substack{\text{payload row} \\ q \geq 2}} q^2 \log(e(1 + b/q^2)) + C_1 s \mathbf{1}_{m=1} \right\}. \quad (117)$$

Here the maximum is over compatible child counts and types. To derive this form from (115), use the Cauchy identity

$$\sum_{\substack{\xi \vdash b \\ \ell(\xi) \leq q}} G_q(\xi)^2 = \binom{b + q^2 - 1}{q^2 - 1}.$$

It bounds the positive child row-carrier logarithms by $Cq^2 \log(e(1 + b/q^2))$; for $q = 1$ all carrier factors are one. Discarding the favorable parent carrier logarithm only increases the bound. Also

$$\mathcal{L} - \mathcal{L}_0 - \mathcal{L}_1 = \sum_{\text{all}} v_b + \sum_{\text{row}} (I(\eta) - I(\mu) - I(\nu)),$$

and both sums have nonnegative summands. Indeed compatibility implies that $\mu + \nu$ dominates η ; Schur concavity followed by entropy concavity gives $I(\eta) \geq I(\mu + \nu) \geq I(\mu) + I(\nu)$. Retain $-v_b$ for payload rows and replace the negative row-entropy decrement by $-\delta[I(\eta) - I(\mu) - I(\nu)]$. The unused background and regular imbalances are nonnegative and may be discarded. For a regular category the carrier ratio gives $\log D_\mu + \log D_\nu - \log D_\kappa$, and $\log \binom{b}{b_0} \leq bH_2(b_0/b)$ gives $(\gamma/2 - 1)b \log 2 + bH_2(b_0/b) \leq (1 - \delta)bH_2(b_0/b)/2$ because $\delta < 1 - \gamma$. This proves (117).

10.3 Changing alphabets and paying for the changes

The case $r = 0$ is the trivial representation and satisfies both bounds with equality. Assume $r > 0$. Embed V_λ in the module induced from its first row as background and ρ as payload. Fix large absolute A and then a much larger K . To tune a diagram of size b , take its rows longer than $K\sqrt{b}$ as one row category; in the remaining diagram take columns longer than $K\sqrt{b}$ as another row category, with orientation reversed. Put the residual diagram S in a regular category. Here is the representation containment used in this operation. If R is the selected top-row diagram and ξ the remainder, the parent is their sorted-row union, whose Littlewood–Richardson coefficient in $\text{Ind}(V_R \otimes V_\xi)$ is one. Apply the transposed version of this fact to remove the selected columns from ξ . Transitivity of induction then embeds the parent in the module of the three resulting parts. Selected rows inherit the old orientation, and selected columns reverse it. If the old orientation was twisted, the residual regular spatial diagram is S' rather than S ; $D_{S'} = D_S$, so its potential is unchanged. These conventions pass the old sign twist to every spatial factor. Thus tuning enlarges the spatial module, and its positive-power trace can only increase. Choosing a new alphabet afterward changes the multiplicity space in which that spatial module is realized. At the root only, group the selected row lengths l by $\lfloor \log_2(r/l) \rfloor$, keeping the two orientations separate. No later tuning uses this grouping.

Assign a new row category its actual oriented height q . A regular category stays regular. A row category of $q = 1$ never changes. For $q \geq 2$, retain its alphabet while $b \geq Aq^2$; when $0 < b < Aq^2$, end that phase and tune the current diagram afresh. At birth,

$$q_{\text{new}} \leq \sqrt{b}/K, \quad b_{\text{new}} \geq K^2 q_{\text{new}}^2; \quad q_{\text{new}} \leq q/L \text{ at an exit,} \quad L = K/\sqrt{A} > 1. \quad (118)$$

Thus no immediate retuning is needed and alphabet dimensions shrink geometrically along successive phases.

Let b_j be the initial payload category sizes, I_0 the sum of their row entropies, s_0 the initial regular size, and $J_{\text{in}} = \sum b_j \log(r/b_j)$. Hook products and the long-row construction give

$$\Phi_{\text{in}} \geq \delta I_0 + \frac{\delta}{2} s_0 \log s_0 - C_K r, \quad (119)$$

$$\log D_\lambda \leq \log \binom{n}{r} + I_0 + \frac{1}{2} s_0 \log s_0 + J_{\text{in}}, \quad (120)$$

$$\sum_{j \text{ row}} b_j \log q_j \leq I_0 + r \log 2, \quad J_{\text{in}} \leq \epsilon I_0 + C_\epsilon r. \quad (121)$$

Here are the dimension and entropy estimates underlying these bounds. The residual diagram has every row and column of length at most $K\sqrt{r}$, so its hooks are at most $2K\sqrt{r}$ and

$$D_S \geq s_0! / (2K\sqrt{r})^{s_0}.$$

Together with $s_0 \log(r/s_0) \leq Cr$, this proves the first line. For the dimension upper bound, induced-module containment contributes $\binom{n}{r}$ and the payload allocation multinomial. Bound these by their entropy expressions, each row-category dimension by $e^{I(\eta)}$, and the residual dimension by $\sqrt{s_0!}$. This gives (120).

Within each initial row category the positive row lengths differ by at most a factor two. Its row distribution therefore has entropy at least $\log q_j - \log 2$, proving the first assertion of (121). To prove the second, choose a uniform payload box. Its category is determined by a flag for the row, column, or residual part and, for a row or column part, a nonnegative dyadic group index Z , taking $Z = 0$ on the residual part. These data have entropy J_{in}/r . The mean of Z is at most $(I_0 + J_{\text{in}})/(r \log 2)$, because a row of length l contributes $\log_2(r/l)$ before taking the integer part. For any $\alpha > 0$, relative entropy against the geometric law proportional to $e^{-\alpha z}$ gives $H(Z) \leq \alpha \mathbb{E}Z + C_\alpha$. Adding the flag entropy, which is at most $\log 3$, and choosing α sufficiently small gives $J_{\text{in}} \leq \epsilon I_0 + C_\epsilon r$ after rearrangement. This is the reason for grouping at the root; no subsequent tuning needs that extra grouping.

We detail the amortization, because using one fixed alphabet would leave a divergent carrier cost. At an exit of mass b , the adverse potential jump is at most

$$C_2 b + C_K s_{\text{new}} + C_2 b \log_+(q^2/b). \quad (122)$$

To verify this jump estimate, denote by R, C, S the new row, column, and residual parts, of respective sizes $|R|, c_*, s$. Entropy of the part assignment gives

$$I_{\text{row}}(\eta) \leq I_{\text{row}}(R) + I_{\text{row}}(C) + I_{\text{row}}(S) + b \log 3.$$

Since the old height is at most q ,

$$I_{\text{row}}(C) - I_{\text{col}}(C) \leq c_* \log q - c_* \log(c_*/q), \quad I_{\text{row}}(S) \leq s \log q.$$

The hook bound for the residual gives

$$\phi(S) \geq \frac{\delta}{2} s \log b - C_K s - \frac{1+\delta}{2} s \log(b/s).$$

Substituting these inequalities and using $x \log(b/x) \leq Cb$ for $0 < x \leq b$ proves (122); empty parts contribute zero. The total mass ever converted to regular categories is at most r , because a regular category never becomes a row category again. We therefore charge the $C_K s_{\text{new}}$ term only once to each converted box, for a total at most $C_K r$. The linear term $C_2 b$ in (122) will be charged to the total exit mass. Only its logarithmic excess needs an imbalance saving.

Call a split balanced when both child masses are in $[b/4, 3b/4]$. During a phase with $q \geq 2$ the parent mass satisfies $b \geq Aq^2$. At an unbalanced split, $v_b \geq \theta b$, where $\theta = \log 2 - H_2(1/4) > 0$. Put

$$f(t) = \frac{\log(e(1+t))}{t}, \quad t > 0.$$

The carrier cost is $C_1 b f(b/q^2) \leq C_1 b f(A)$, since f is decreasing. If a child of mass x exits at this split, then

$$x \log_+(q^2/x) \leq q^2/e \leq b/(eA).$$

There are at most two exiting children. Choose $A \geq 4$ so large that $C_1 f(A) + 2C_2/(eA) \leq \theta/2$. The carrier cost and these logarithmic excesses are then bounded by half of the negative imbalance term. At a balanced exit, $x \geq b/4 \geq Aq^2/4 \geq q^2$, so the logarithmic excess is zero. No K -dependent constant has entered this choice of A .

It remains to sum carrier costs at balanced nodes of a phase, which may form a branching subtree. Distribute each such cost evenly among its b boxes. A box pays $C_1 f(b/q^2)$. Along its balanced ancestors, successive masses decrease by a factor at most $3/4$, even if unbalanced nodes intervene. Reversing the sequence and using $b/q^2 \geq A$ gives

$$\sum_{\text{balanced ancestors}} f(b/q^2) \leq \sum_{j \geq 0} f(A(4/3)^j) \leq \frac{C \log(e(1+A))}{A}.$$

Summing over boxes bounds the balanced cost by a constant times the phase's birth mass. The constant depends on the fixed A and is independent of K . The exit frontier has total mass at most that birth mass, so it also pays the linear exit cost $C_2 b$. By (118), a box initially in row category j encounters at most $1 + \log q_j / \log L$ phases: each retuning reduces the new alphabet by a factor L , and regular conversion ends the sequence. Each of the total birth mass and the total exit mass is consequently at most

$$\sum_{j \text{ initial payload row}} b_j (1 + \log q_j / \log L).$$

The $b^{2/3}$ term is also summable, including categories that stay small through many ambient levels. At a balanced split,

$$b_0^{2/3} + b_1^{2/3} - b^{2/3} \geq cb^{2/3}.$$

The power sum cannot decrease at any other split or conversion and its terminal value is r , so all balanced costs sum to $O(r)$. For unbalanced splits, the positive error satisfies the numerical bound

$$\sum_{\text{unbalanced}} b^{2/3} \leq \sum_{\text{unbalanced}} b \leq \theta^{-1} \sum_{\text{unbalanced}} v_b.$$

To bound the last sum, the scalar count entropy $b \log(n/b)$ at ambient size n decreases by exactly v_b at a split. A conversion increases it by $\sum_j b_j^{\text{new}} \log(b/b_j^{\text{new}})$. Its terminal value is zero, so summing all the nonnegative decrements gives

$$\sum_{\text{payload splits}} v_b = r \log(n_{\text{root}}/r) + J_{\text{in}} + \sum_{\text{exits}} \sum_j b_j^{\text{new}} \log(b/b_j^{\text{new}}). \quad (123)$$

This identity bounds the total positive error by the initial count entropy and the conversion entropies; it makes no further subtraction of the operator term $-v_b$. Each exit creates at most three categories, so its last sum is at most $b \log 3$ and is covered by the phase-mass bound. Finally the $m = 1$ term costs at most Cr .

Iterating (117) down to single sites, whose log trace is zero, now gives

$$\sigma_{\text{in}} \leq -\Phi_{\text{in}} + C_K r + C_3 \{r \log(n/r) + J_{\text{in}} + r\} + \frac{C_3}{\log L} \sum_{j \text{ row}} b_j \log q_j. \quad (124)$$

Here C_3 depends on the already fixed A , but not on K ; all K -dependent losses are in $C_K r$. Choose K after A so that $C_3/\log L$ is small compared with δ , and then choose ϵ in (121) sufficiently small. Equations (119)–(124) yield

$$\sigma_{\text{in}} \leq -\frac{\delta}{2} I_0 - \frac{\delta}{2} s_0 \log s_0 + Ch_n(r).$$

The dimension comparison (120) proves (101). Combining its fourth-root bound with (100), in proportions chosen to cancel the positive $Ch_n(r)$ term, proves the final assertion of Theorem 10.1.

For clarity, this also supplies a complete mixing consequence. Choose an absolute integer q so large that the dimension powers are absorbed in

$$\text{Tr}_{\text{reg}} F_n^q - 1 \leq \sum_{r=1}^{n-1} 2^r e^{-c_2 q h_n(r)} = o(1),$$

where $c_2 > 0$ is absolute after increasing q . The count 2^r bounds partitions of the lower diagram. Split at $r = \sqrt{n}$ to see that the sum tends to zero: below it $h_n(r) \geq \frac{1}{2} r \log n$, and above it $h_n(r) \geq r$. Schatten Hölder bounds the squared Hilbert–Schmidt norm of the nonconstant part of T_n^q by this positive moment, without requiring normality. Plancherel and translation therefore give total-variation convergence from every initial deck after q sweeps, or qd physical shuffles.

11 Core charges and a low-dimension cutoff

The previous recursions used either the first-row level or a changing tensor alphabet. Here the cost of a restriction is charged directly to boxes lying beyond the first several rows and columns of a Young diagram. The goal is again a negative dimension power for one sweep, but the intermediate statement concerns a specific projection at its midpoint: types with small product dimension have exponentially small overlap with the full sweep.

There are two bounds for each weighted cycle factor. One depends only on the diagrams, and the other depends on their actual positive traces. We choose between them while expanding the recursion. The second choice must use the traces at the current frontier; a later upper bound cannot replace them before the choice is made. We make that invariant explicit below.

Logarithms in this section are to base two, unless $\ln$ is written. For a nontrivial diagram $\lambda \vdash n$, put $F_\lambda = \log_2 D_\lambda$, $k = n - \lambda_1$, and $L = \log_2(n/k)$. For a box x let $a(x) = \min(\text{row}(x), \text{column}(x))$, with indices beginning at one. We call a type surviving if its sweep matrix is nonzero; for $n \geq 2$ it then has at most $n/2$ rows. Traces with zero contribution may be omitted throughout. For $s \geq 0$ define the core count

$$u_s(\lambda) = \#\{x \in \lambda : a(x) > s\}.$$

Thus the core is what remains after deleting the first s rows and the first s columns. Although its box coordinates shift under deletion, its remaining row lengths form a straight Young diagram.

Theorem 11.1. *There is an absolute $c > 0$ such that $\|T_n|_{V_\lambda}\|_{\text{op}} \leq 2^{-cF_\lambda}$ for every surviving nontrivial type.*

11.1 Dimension charges and the local sparse estimate

We need two elementary inputs, uniformly over surviving nontrivial types for all sufficiently large dyadic n :

$$F_\lambda \geq bk, \quad 1 \leq k \leq n^{1/100} \implies \|T_n|_{V_\lambda}\| \leq 2^{-b_1 k \log_2 n}, \quad (125)$$

where $b, b_1 > 0$ are absolute. We prove the sparse estimate locally because its range grows with n .

For the dimension estimate, write τ for the diagram below the first row. The hook formula gives

$$D_\lambda = \binom{n}{k} D_\tau / \prod_{j \leq \lambda_1} \left(1 + \frac{\tau'_j}{\lambda_1 - j + 1} \right).$$

The denominator is at most 2^k : for an integer $t \geq 0$ and positive integer a , $1 + t/a \leq 2^t$, and $\sum_j \tau'_j = k$. When $k \leq n/8$, the bound $\binom{n}{k} \geq (n/k)^k$ proves $F_\lambda \geq 2k$. For $k \geq n/8$, first suppose that a first row or first column has length $a \geq n/10$. Transpose if needed and retain that row together with $v = \lfloor n/100 \rfloor$ boxes below it. Such a subdiagram exists: after transposition its tail has at least $n/8$ boxes if the chosen row was the original first row, and at least $n/2$ if it was the first column. Tableaux of a subdiagram extend to the full diagram. The same formula on this smaller diagram gives $D_\lambda \geq \binom{a+v}{v} 2^{-v}$, exponential in n . If both the first row and first column are shorter than $n/10$, every hook is shorter than $n/5$ and Stirling's inequality gives $D_\lambda \geq (5/e)^n$. These cases prove the first estimate. They also show $F_\lambda = O(k \log n)$ on the sparse range, the upper bound following from $D_\lambda \leq \binom{n}{k} D_\tau$ and $D_\tau \leq \sqrt{k!}$.

To prove the second estimate, realize V_λ on ordered injections of k labels. A function of that exact position type is harmonic in each label slot, because the type does not occur on $k-1$ slots. For distinct input positions x_a, x_b , let h_{ab} be their largest differing bit index in the sweep order. For $J \subseteq [k]$ put

$$g_J(x, y) = \prod_{a < b: a, b \in J} \left(1 - \mathbf{1}_{\{y_a, <_{h_{ab}} = y_b, <_{h_{ab}}\}} (-1)^{y_{a, h_{ab}} + y_{b, h_{ab}}} \right).$$

The individual route from an input to an output is unique. Two labels can first share a switch only at h_{ab} , when their earlier output bits agree. Equal output bits also at h_{ab} make the prescription impossible, giving a zero factor; opposite bits give one common fair coin in place of two, giving a factor two. Otherwise the factor is one. Thus $n^{-|J|} g_J$ is exactly the prescribed subtuple transition probability, including zero for incompatible routes.

The number of shared switches among s compatible paths is at most $s \log_2 s/2$, by splitting at the last coordinate and adding $\min(s_0, s_1)$ to the two child counts. The binary entropy inequality pays this addition. Hence $0 \leq g_J \leq k^{k/2}$. On the exact position type we may replace $g_{[k]}$ by

$$g'(x, y) = \sum_{J \subseteq [k]} (-1)^{k-|J|} g_J(x, y) :$$

every omitted-slot term is annihilated by harmonicity. This sum has absolute value at most $2^k k^{k/2}$. Moreover it vanishes if some label is isolated in the potential-contact graph, whose edge ab is the event $y_{a, <_{h_{ab}}} = y_{b, <_{h_{ab}}}$; the terms with and without that isolated label then cancel in pairs.

A graph without isolated vertices contains a spanning forest without isolated vertices, and such a forest has at least $\lceil k/2 \rceil$ edges. For independent uniform, unrestricted binary strings x, y , declare that ab is not an edge when $x_a = x_b$; on distinct input pairs use the potential-contact condition above. An edge constraint in a forest costs d/n when a leaf is integrated out: the possible last differing input bit h has probability $2^{h-1}/n$, and agreement of the earlier output bits costs $2^{-(h-1)}$. Successive leaf removal therefore multiplies these costs. There are at most k^{2k+1} forests to include in the union bound. Conditioning both endpoint tuples to be injective costs at most two on $k \leq n^{1/100}$ for large n . Consequently

$$\Pr(\text{no isolated label}) \leq 2k^{2k+1}(d/n)^{\lceil k/2 \rceil} \leq 2^{-(2/5)k \log_2 n} \quad (k \geq 2).$$

The tuple kernel density relative to uniform has the additional factor $(n)_k/n^k \leq 1$. Its compressed Hilbert–Schmidt norm is therefore at most $2^k k^{k/2}$ times the square root of this probability. This is $2^{-b_1 k \log_2 n}$ for an absolute $b_1 > 0$, after increasing the lower size threshold. The operator norm is smaller, and at level one the sweep is exactly zero. This proves (125).

The charge to the deep core has the following form:

$$\sum_{x: a(x) \geq R} \{L + \log_2 a(x)\} \leq (1 + o(1))F_\lambda \quad (R \rightarrow \infty), \quad (126)$$

uniformly over surviving nontrivial diagrams of sufficiently large size. Here the $o(1)$ depends only on R , not on the diagram. For any straight diagram of size u , a box at minimum index a has hook length at most $2u/a$: each of its row and column lengths is at most u/a . Apply this to τ in the first-row hook formula above. The coordinates below the first row shift by one, changing $\log a$ by at most a constant. Stirling’s bound then gives

$$\sum_{x \text{ below row 1}} \{L + \log_2 a(x)\} \leq F_\lambda + O(k).$$

If $L > \sqrt{\log_2 R}$, the lower bound $F_\lambda \geq kL - O(k)$ absorbs that error uniformly as $R \rightarrow \infty$.

In the complementary case, remove the first $q = \lfloor \sqrt{R} \rfloor$ rows and columns. Let the remaining core have size u . Applying the additive-error bound just proved to boxes with $a > q$ gives $u = O((F_\lambda + k)/\log R) = O(F_\lambda/\log R)$. The dimension of this core is at most D_λ , by extending its tableaux after translating it to the upper left. Its hook formula therefore pays $\sum \log_2(a - q)$ up to $O(u)$. For charged boxes $a \geq R$, replacing $\log_2(a - q)$ by $\log_2 a$ costs $o(1)$ per box, and adding L costs at most $u\sqrt{\log_2 R} = o(F_\lambda)$. This proves (126).

11.2 Two bounds for a weighted cycle factor

The positive network recursion is $K_n = H(K_m \otimes K_m)H$, where H is the orthogonal average of the outer matching switches. For a positive D and $p \geq 2$,

$$\text{Tr}(HDH)^p \leq \text{Tr}(HD^{p/2}H)^2. \quad (127)$$

Indeed take an eigenbasis of HDH on the range of H . Scalar Jensen gives $\langle v, Dv \rangle^{p/2} \leq \langle v, D^{p/2}v \rangle$ for each basis vector. Squaring and summing is bounded by the Hilbert–Schmidt square of the compression. This proof uses scalar convexity and does not require operator convexity of the power function.

Resolve the two halves into types μ, θ . For the moment, let M_μ and M_θ be arbitrary positive matrices on these two types, and put $t_\mu = \text{Tr } M_\mu^2$ and $t_\theta = \text{Tr } M_\theta^2$. In the unrestricted recursion they will be $K_m^{p/2}$ on the respective types; allowing arbitrary positive matrices will also cover the midpoint projections below. Write $c(g)$ for the number of cycles of a

permutation g , including fixed points. Expansion of the matching projection gives the regular trace bound

$$\frac{\text{Tr}(EHEH)}{(2m)!} \leq 4^{-m} \sum_z |e(z)|^2 2^{c(\alpha^{-1}\zeta)}. \quad (128)$$

Here E is the element of the half-preserving group algebra whose only nonzero Fourier block is $M_\mu \otimes M_\theta$ on the selected pair of child types, and $E = \sum_{z=(\alpha,\zeta) \in S_m \times S_m} e(z)z$. The trace in (128) is the regular trace of S_{2m} . To verify it, expand each matching average as a sum over subsets of the m matched pairs. For a term to return to the half-preserving subgroup, its two switch subsets J, I must satisfy $\alpha(J) = \zeta(J) = I$. The permissible J are unions of cycles of $\alpha^{-1}\zeta$, so there are $2^{c(\alpha^{-1}\zeta)}$. The second half-preserving coefficient exchanges the two maps on J , up to inversion. This operation is a bijection on the paired terms; symmetrizing $|e(z)e(z')| \leq (|e(z)|^2 + |e(z')|^2)/2$ proves the bound. Schur orthogonality gives

$$(m!)^2 \sum_z |e(z)|^2 = D_\mu t_\mu D_\theta t_\theta.$$

The cycle factor is therefore averaged under the product of the two normalized squared-coefficient laws. Since $(2m)!/4^m \leq (m!)^2$, the normalized half-density is

$$\phi_\mu(g) = \frac{D_\mu |\text{Tr}(M_\mu V_\mu(g))|^2}{t_\mu}.$$

It is a probability density relative to uniform measure on S_m , by matrix-coefficient orthogonality; we only use it when $t_\mu > 0$. For independent α, ζ with these respective densities, define $T_{\mu\theta} = \mathbb{E} 2^{c(\alpha^{-1}\zeta)}$. The next lemma connects this regular-trace calculation to a fixed parent representation. It also identifies the quantities retained when we later stop a recursion according to its actual traces.

Lemma 11.2 (A selected child pair). *Let B_0, B_1 be positive operators in the group algebra of S_m , let $B = H(B_0 \otimes B_1)H$, and fix $\lambda \vdash 2m$ and $p \geq 2$. Let $\mathcal{I}_\lambda$ consist of the pairs (μ, θ) with $c_{\mu\theta}^\lambda > 0$, and write $N_\lambda = |\mathcal{I}_\lambda|$. For such a pair use*

$$M_\mu = B_0(\mu)^{p/2}, \quad M_\theta = B_1(\theta)^{p/2}, \quad t_\mu = \text{Tr } B_0(\mu)^p, \quad t_\theta = \text{Tr } B_1(\theta)^p$$

and form $T_{\mu\theta}$ from their coefficient-square densities above. Pairs with a zero trace are omitted. Then

$$D_\lambda \text{Tr } B(\lambda)^p \leq N_\lambda^2 \max_{(\mu,\theta) \in \mathcal{I}_\lambda} (D_\mu t_\mu)(D_\theta t_\theta) T_{\mu\theta}. \quad (129)$$

If all selected traces vanish, the left side is zero.

Proof. Put $D = B_0 \otimes B_1$, and let $E_{\mu\theta}$ be the central projection onto this child isotypic component. On V_λ , only pairs in $\mathcal{I}_\lambda$ occur. The projection includes all their Littlewood–Richardson multiplicity copies. Set $Y_{\mu\theta} = E_{\mu\theta} D^{p/2}$, which is positive because $E_{\mu\theta}$ commutes with D . Apply (127) on V_λ , followed by the Hilbert–Schmidt triangle inequality, to obtain

$$D_\lambda \text{Tr } B(\lambda)^p \leq D_\lambda \left\| \sum_{\mathcal{I}_\lambda} H Y_{\mu\theta} H \right\|_{\text{HS}, V_\lambda}^2 \leq N_\lambda^2 \max_{\mathcal{I}_\lambda} D_\lambda \text{Tr}_{V_\lambda} (H Y_{\mu\theta} H)^2.$$

The parent isotypic projection commutes with every group-algebra operator, including H . Thus the incompatible pairs have been removed before the next step: the selected positive trace satisfies

$$D_\lambda \text{Tr}_{V_\lambda} (H Y_{\mu\theta} H)^2 \leq \text{Tr}_{\text{reg}, S_{2m}} (H Y_{\mu\theta} H)^2.$$

There is no further restriction-multiplicity factor; those copies were already included in the parent trace, and the right side includes that entire trace with regular multiplicity D_λ . Cyclicity, (128), and child orthogonality bound this regular trace by

$$\frac{(2m)!}{4^m(m!)^2}(D_\mu t_\mu)(D_\theta t_\theta)T_{\mu\theta}.$$

The prefactor is at most one. Notice that the child trace exponent is still p : the matrices in the coefficient densities have exponent $p/2$, and their Hilbert–Schmidt squares restore p . $\square$

The first estimate is representation theoretic. Put

$$G_\mu = \sum_{b=0}^m \sum_{\substack{\xi \vdash b \\ \zeta \vdash m-b}} (c_{\xi\zeta}^\mu)^2.$$

Here $c_{\xi\zeta}^\mu$ is the Littlewood–Richardson multiplicity of $V_\xi \otimes V_\zeta$ in the restriction of V_μ to $S_b \times S_{m-b}$. Then

$$\log_2 T_{\mu\theta} \leq \frac{1}{2}(\log_2 G_\mu + \log_2 G_\theta). \quad (130)$$

The function $2^{c(g)}$ counts the subsets fixed by g , so it is the character of the direct sum of all subset permutation modules. For an irreducible summand V_j in that sum, its contribution under the two coefficient-square laws is $\text{Tr}(\hat{\phi}_\mu(j)^* \hat{\phi}_\theta(j))$. These Fourier matrices are positive semidefinite, because a square of a coefficient of a positive matrix is positive definite, and are contractions because the densities integrate to one.

To bound their traces, let Q_j be the j -isotypic projection in $V_\mu \otimes \overline{V}_\mu$. Fourier orthogonality gives

$$\text{Tr} \hat{\phi}_\mu(j) = \frac{D_\mu}{t_\mu D_j} \text{Tr}[(M_\mu \otimes \overline{M}_\mu)Q_j].$$

Cauchy–Schwarz bounds the last trace by the geometric mean of the traces with M_μ^2 on either tensor factor. Each partial trace of Q_j is scalar by Schur’s lemma, so this is at most the multiplicity of j in $V_\mu \otimes \overline{V}_\mu$ after the displayed normalization. Summing over subset modules, Frobenius reciprocity identifies that sum of multiplicities with $\sum_b \sum_{\xi, \zeta} (c_{\xi\zeta}^\mu)^2 = G_\mu$. Thus $T_{\mu\theta}$ is at most either G_μ or G_θ , and hence at most their geometric mean. This proves (130).

Let $u_s(\mu)$ count boxes outside the first s rows and columns, and put $k_\mu = m - \mu_1$. With $\eta = 10^{-9}$, $\gamma = 1/100$ and $s = \lfloor k_\mu^{1/2-\eta} \rfloor$, one has

$$\log_2 G_\mu \leq u_s(\mu) + O(k_\mu^{1-\eta/2}), \quad k_\mu \geq m^\gamma, \quad (131)$$

$$\log_2 G_\mu \leq k_\mu + O(\log m + \sqrt{k_\mu} \log k_\mu), \quad k_\mu < m^\gamma. \quad (132)$$

For the first, Pieri’s rule embeds V_μ in the induction of its core together with at most $2s$ one-row or one-column strips. Restrict that induced module to two half factors by recording each constituent’s size on each side. For the core, multiplication of the restriction and induction dimension bounds gives

$$(c_{\xi\zeta}^\mu)^2 \leq \binom{u_s}{|\xi|} \leq 2^{u_s}.$$

The strips restrict with multiplicity one, and each subsequent strip addition also has multiplicity at most one by Pieri. It remains to count the intermediate diagrams. A subdiagram of μ is specified by its first row and a partition of at most k_μ boxes below that row; its logarithmic count is $O(\log m + \sqrt{k_\mu} \log(k_\mu + 2))$. Including the size splits, the logarithm

of all intermediate choices is therefore $O((s+1)(\log m + \sqrt{k_\mu} \log(k_\mu + 2)))$. For $k_\mu \geq m^\gamma$ and $s = \lfloor k_\mu^{1/2-\eta} \rfloor$, this is $O(k_\mu^{1-\eta/2})$, after absorbing logarithms. Squaring and summing multiplicities changes only the implicit constant and proves (131). For the second, remove only the first row, using the whole lower diagram as the core in the same argument. The trivial type costs only $O(\log m)$.

The second cycle estimate depends on the actual trace. Put $r_\mu = (\text{Tr } M_\mu)^2/t_\mu$; then $\phi_\mu \leq D_\mu r_\mu$. The uniform cycle generating product, split at cycle count $m^{9/10}$ and evaluated at $x = m^{89/100}$, gives

$$\log_2 T_{\mu\theta} \leq O(1) + m^{9/10} + \frac{F_\mu + \log_2 r_\mu}{0.88 \log_2 m}. \quad (133)$$

The same bound holds with θ ; their average can also be used. Here is the tail optimization. For a uniform permutation g ,

$$\mathbb{E}x^{c(g)} = \prod_{j=0}^{m-1} \frac{x+j}{j+1}, \quad \log_2 \mathbb{E}x^{c(g)} = O(x \log m) \quad (1 \leq x \leq m).$$

Take $x = m^{0.89}$, $h = \lceil m^{0.9} \rceil$, and $A_m = 0.88 \log_2 m$. Since $m^{0.89} \log m = o(h \log m)$, Markov's inequality gives, for all sufficiently large m ,

$$\Pr_{\text{unif}}\{c(g) \geq j\} \leq 2^{-A_m j} \quad (j \geq h).$$

The law of $\alpha^{-1}\zeta$ has density at most $B = D_\mu r_\mu$: conditioning on ζ gives a translate of the first density, and averaging preserves this bound. Its cycle tail is therefore at most $\min(1, B2^{-A_m j})$ for $j \geq h$. Put $j_* = h + \lceil (\log_2 B)/A_m \rceil$. The tail-sum identity and $A_m > 1$ give

$$\mathbb{E}2^{c(\alpha^{-1}\zeta)} \leq 2^{j_*} + \sum_{j>j_*} 2^j B 2^{-A_m j} \leq C 2^{j_*}.$$

Taking logarithms proves (133). The argument with the two halves interchanged proves its symmetric version.

11.3 A preliminary trace and the cutoff allocation

There is a fixed p_0 such that, with $\delta = 10^{-5}$,

$$\text{Tr } K_n^{p_0}|_{V_\lambda} \leq 2^{\delta F_\lambda}. \quad (134)$$

We first specify the stopping contributions. At a nontrivial sparse node of size m , let B be its positive sweep operator, or any cut operator satisfying $0 \preceq B \preceq T_m T_m^*$. The local sparse estimate and $F_\mu = O(k_\mu \log_2 m)$ imply

$$\log_2(D_\mu \text{Tr } B(\mu)^p) + \frac{1}{2} \log_2 G_\mu \leq 2F_\mu - 2pb_1 k_\mu \log_2 m + \frac{1}{2} \log_2 G_\mu \leq 0 \quad (135)$$

for one sufficiently large fixed p , uniformly over nonzero sparse nodes. Here (132) bounds the last term by $O(k_\mu \log_2 m)$; level one is annihilated. Coordinate reversal permits the same estimate for either positive orientation. For each fixed finite set of sizes, the finite-size norm gap likewise permits one exponent making the left side of (135) nonpositive on every nontrivial type. A trivial child instead has $D_\mu = t_\mu = 1$ and $G_\mu = m + 1$; we stop it and charge its $O(\log m)$ half-cycle cost to its nontrivial parent.

Choose a large fixed lower size cutoff, and apply Lemma 11.2 at each remaining nonsparse node, using the same exponent p_0 throughout. Choose p_0 to satisfy (135) and the finite-size stopping inequalities at this cutoff. On each selected edge use (130). A stopped nontrivial

child contributes its actual $\log_2(D_\mu t_\mu)$ together with its half of this cycle bound; their sum is nonpositive. On every active frontier, Littlewood–Richardson compatibility now gives

$$\sum_v k_v \leq k, \quad \sum_v F_v \leq F_\lambda, \quad \sum_v u_s(v) \leq u_s(\lambda).$$

For the last inequality, consider one Littlewood–Richardson tableau of λ/μ with content θ . Entries in the first s rows are at most s . For each value $j > s$, column strictness permits at most one occurrence in each of the first s columns. Thus at least $(\theta_j - s)_+$ occurrences lie beyond both the first s rows and the first s columns. These skew core boxes are disjoint from the $u_s(\mu)$ boxes of the old core, and $\sum_{j>s}(\theta_j - s)_+ = u_s(\theta)$. Therefore $u_s(\mu) + u_s(\theta) \leq u_s(\lambda)$ at one split; iteration proves the frontier inequality. At nodes of size m , use the common cutoff $s_* = \lfloor \max(m^\gamma, \rho m)^\beta \rfloor$, where $\beta = 1/2 - 3\eta$ and $\rho = k/n$ is the root payload density, kept fixed throughout this unrolling. Here is the comparison with the varying cutoff in (131). Put $a_0 = \beta/(1/2 - \eta) < 1$. A nonsparse node with its own cutoff smaller than s_* can occur only when $\rho m \geq m^\gamma$, and then it has $k_v \leq C(\rho m)^{a_0}$. There are at most n/m nodes of size m , so the total level of these exceptional nonsparse nodes is at most

$$C(n/m)(\rho m)^{a_0} = Ck(\rho m)^{a_0-1} \leq Ckm^{-\gamma(1-a_0)}.$$

Their core counts are at most their levels, so this is a summable error as the bottom size cutoff increases. The sparse and finite-size stopping contributions have already been removed by (135); trivial-child costs belong to the active-parent overhead. All remaining nodes charge at most half their u_{s_*} . For a parent at level k_v , compatible child levels r, s satisfy $r + s \leq k_v$. Consequently $N_\lambda \leq \sum_{r+s \leq k_v} p(r)p(s)$, so the partition-number bound gives $\log_2 N_\lambda = O(\sqrt{k_v} + \log(k_v + 2))$. Restriction overheads and the errors in (131) are $O(k_v^{1-\eta/3})$ on nonsparse nodes, so at size m their sum is $O(km^{-\gamma\eta/3})$. Summing over dyadic sizes makes their total $o(k)$ as the bottom cutoff tends to infinity. These errors are independent of p_0 . A box of index a is charged at at most $O(1) + L + \beta^{-1} \log_2 a$ levels. By (126) the total is at most $(1/(2\beta) + o(1))F_\lambda < (1 + \delta)F_\lambda$. More explicitly, unrolling (129) gives

$$F_\lambda + \log_2 \text{Tr } K_n(\lambda)^{p_0} \leq \left(\frac{1}{2\beta} + \varepsilon(M) \right) F_\lambda, \quad \varepsilon(M) \rightarrow 0$$

as the fixed lower cutoff M tends to infinity. The stopping terms are nonpositive, and the displayed error includes the type counts and trivial-child costs. Since $1/(2\beta) < 1 + \delta/2$, choose M first so that the coefficient is at most $1 + \delta$, and then choose the single finite exponent p_0 as above. Subtracting F_λ proves (134). We may take M still larger before fixing p_0 when a smaller error margin is needed below.

We now turn the preliminary trace into the midpoint projection estimate. Split the ordered coordinates as evenly as possible, and write

$$T_n = CD,$$

where D applies the first group of coordinates and C the second. The two coordinate groups have $\lfloor d/2 \rfloor$ and $\lceil d/2 \rceil$ coordinates, where $d = \log_2 n$. Each segment is a product of independent sweeps on its parallel subcubes, of sizes $2^{\lfloor d/2 \rfloor}$ and $2^{\lceil d/2 \rceil}$, respectively. Let H_D, H_C be the corresponding direct product permutation subgroups. The argument applies equally with the two coordinate-group sizes interchanged; their middle log sizes differ from $d/2$ by at most $1/2$. A product type for either subgroup is a tuple of partitions, with product dimension and hence log dimension equal to the sum of the child log dimensions. On V_λ , let P_{lo}^D and P_{lo}^C be the central isotypic projections for H_D and H_C , respectively, onto product types whose log dimensions sum to at most

$$X = 0.50003F_\lambda. \tag{136}$$

Proposition 11.3 (Two midpoint projections). *There is an absolute $c_1 > 0$ such that, for all sufficiently large dyadic n and all surviving nontrivial $\lambda \vdash n$ with $k \geq n^{1/100}$,*

$$\max \left\{ \|T_n P_{\text{lo}}^D|_{V_\lambda}\|, \|P_{\text{lo}}^C T_n|_{V_\lambda}\| \right\} \leq 2^{-c_1 F_\lambda}. \quad (137)$$

The two projections refer to different product subgroups; no commutation between them is asserted.

Proof. We prove the first estimate; reversing the coordinate order gives the second. Consider the positive operator $T_n P_{\text{lo}}^D T_n^*$ and take exponent $p = 2p_0$. Expand its outer matching layers until reaching the middle coordinate system. At a node v , its operator has the form $B_v = A_v P_v A_v^*$, where A_v is the remaining local sweep and P_v is the inherited central cutoff for its innermost product types. In particular $0 \preceq B_v \preceq A_v A_v^*$. For a positive contribution on its selected irreducible type define

$$M_v = B_v^{p_0}, \quad t_v = \text{Tr } B_v^{2p_0}, \quad r_v = (\text{Tr } M_v)^2 / t_v.$$

These are actual cut-operator quantities. The coefficient-square and cycle bounds above apply to these positive matrices just as they do to the unrestricted matrices. At a binary restriction, allocate its cutoff by the elementary projection inequality

$$\mathbf{1}_{a+b \leq B} \leq \sum_{u+v \leq B+2, u, v \in \mathbb{Z}_{\geq 0}} \mathbf{1}_{a \leq u} \otimes \mathbf{1}_{b \leq v}. \quad (138)$$

Ceiling a, b proves the cover. These are commuting central child projections, so the inequality holds before sandwiching. If $\mathcal{J}_v$ is this finite set of budget pairs, sandwiching gives

$$B_v \preceq \sum_{(u,w) \in \mathcal{J}_v} H(B_{0,u} \otimes B_{1,w}) H, \quad B_{i,u} = A_i P_{i,u} A_i^*.$$

The child product log dimensions never exceed $m \log_2 m$, so the ceilings may be capped at $\lceil m \log_2 m \rceil$; in particular $|\mathcal{J}_v|$ is polynomial in m . Monotonicity of positive Schatten norms and their triangle inequality, followed by Lemma 11.2, give the precise cut recursion

$$D_\lambda \text{Tr } B_v(\lambda)^p \leq |\mathcal{J}_v|^p N_\lambda^2 \max_{\substack{(u,w) \in \mathcal{J}_v \\ (\mu, \theta) \in \mathcal{I}_\lambda}} (D_\mu t_{0,u,\mu})(D_\theta t_{1,w,\theta}) T_{u,w,\mu,\theta}. \quad (139)$$

Here $t_{i,u,\mu} = \text{Tr } B_{i,u}(\mu)^p$, and the cycle factor is formed from these same actual matrices $B_{i,u}(\mu)^{p/2}$. Thus allocation adds only $O_{p_0}(\log m)$ to the logarithmic cost when $p = 2p_0$. The selected cutoff and child pair remain fixed when we subsequently expand a child trace.

Finally $B_{i,u} \preceq A_i A_i^*$. Eigenvalue monotonicity bounds their powered traces by those of the unrestricted operators, whenever we need the preliminary trace bound or a sparse stopping inequality. This does not require the power function to be operator monotone.

Active nodes have size at least a constant multiple of $\sqrt{n}$. At a fixed size m , each nontrivial active node has level at least $m^{1/100}$, while their total level is at most k . There are therefore at most $km^{-1/100}$ such nodes. The $O_{p_0}(\log m)$ allocation cost and the bounded rounding loss per split sum to $O_{p_0}(km^{-1/100} \log m)$ at that size. Summing over the dyadic sizes down to the middle gives $o(k)$. Trivial children are stopped and their costs have already been assigned to the parent overhead. The terminal sum includes only branches reaching the middle: stopped nontrivial branches have already contributed nonpositive terms by (135). The successive ceilings therefore give $\sum_v F_v \leq X + o(k)$ at this terminal cut, and

$$\sum_v \log_2(D_v t_v) \leq (1 + \delta)(X + o(k)) \quad (140)$$

by (134).

11.4 The actual-trace invariant

Before estimating the total cycle cost, we specify a second stopping rule. Put $d = \log_2 n$. While descending through child log sizes between d and $0.9d$, test the actual frontier traces against

$$-\sum_v \log_2 t_v > 0.2F_\lambda. \quad (141)$$

If this inequality holds, the trace already saved at that frontier will replace the more expensive terminal budget at the middle. The following invariant explains why this decision can be made using the actual traces.

A frontier consists of the nodes at the current common coordinate depth, with previously stopped branches removed and their paid contributions retained separately. Expand every currently active node once to form a complete new generation. Test that generation before removing any of its newly sparse or trivial children. At each expansion choose a maximizing term in (139), using its actual child traces and actual weighted-cycle factor. Retain the selected matrices and the resulting number T_e . Subsequent expansion replaces only a child quantity $q_v = \log_2(D_v t_v)$, without changing that earlier density.

To account explicitly for stopped children, start an edge cost at $\Gamma_e = \log_2 T_e$. When its child w is stopped, write

$$q_w + \Gamma_e = \left(q_w + \frac{1}{2} \log_2 G_w\right) + \left(\Gamma_e - \frac{1}{2} \log_2 G_w\right).$$

The first term is nonpositive for a nontrivial stopped child by (135); for a trivial child it is the $O(\log m)$ parent overhead already recorded. Keep the second term as the new residual cost Γ_e . This scalar subtraction does not change the actual matrix or coefficient density used to define T_e . After this stopping decision, freeze Γ_e as well; expanding an unstopped child later changes only that child's frontier contribution. The two usable bounds are consequently

$$\Gamma_e \leq \log_2 T_e, \quad \Gamma_e \leq \frac{1}{2} \sum_{w \text{ unstopped child of } e} \log_2 G_w.$$

The first permits the density estimate, and the second permits the core charge without charging stopped children again. At every intermediate frontier,

$$\log_2(D_\lambda t_\lambda) \leq \sum_{e \text{ retained}} \Gamma_e + \text{overheads} + \sum_{v \text{ frontier}} \log_2(D_v t_v) + \text{paid nontrivial stopping terms}. \quad (142)$$

The last terms are nonpositive. This follows inductively from (139) and the displayed scalar decomposition. In particular no independence between later maximizing choices is needed, and a newly created frontier is tested before any of its offsets are subtracted.

It remains to show that the resulting cost leaves fixed slack below the root dimension. We first estimate what the diagram-only cycle bound would cost throughout the truncated recursion. Put $l = L/d$ and $z = \log_2 a/d$ for a charged box. Over the relevant sizes, from $d/2$ to d , its core charge is at most

$$O(1) + \frac{d}{2} [\min(1, l + \beta^{-1}z) - 1/2]_+. \quad (143)$$

We charge only boxes appearing in at least one of these cores. In particular $l + \beta^{-1}z \geq 1/2 - O(1/d)$, so their weights $w_x = d(l + z)$ are at least $\beta d/2 - O(1)$. The $O(1)$ rounding terms therefore total $O(F_\lambda/d)$ by (126). For the principal term put $q = \beta^{-1}$ and $y = l + qz$. Its ratio to w_x satisfies

$$\frac{[\min(1, y) - 1/2]_+}{2(l + z)} \leq \frac{q}{4} < 0.50001,$$

because $l + z \geq y/q$. If $z < 0.49$, set $b_z = (q - 1)z < 1/2$; the same ratio equals $[\min(1, y) - 1/2]_+ / [2(y - b_z)]$. It increases on $1/2 < y < 1$ and decreases on $y > 1$, hence is at most

$$\frac{1}{4(1 - b_z)} \leq \frac{1}{4(1 - 0.49(q - 1))} < 0.491.$$

These bounds hold on the whole range $l, z \geq 0$, and therefore on the admissible diagram range. The additional geometric relation $a(x)(a(x) - 1) \leq k$ will be used below to turn a deep box into a lower bound for k .

Choose the fixed cutoff in the preliminary proof so that its core-charge and summable-error remainders are sufficiently small, and then fix p_0 . Increasing the root size afterward makes the allocation and rounding errors small as well. We take the sum of the error terms in the following budget comparisons to be at most $10^{-5}F_\lambda$; the displayed numerical comparisons all leave more slack than this. If the total core charge is at most $0.4995F_\lambda$, combining it with (140) suffices, since

$$0.4995 + (1.00001)(0.50003) = 0.9995350003 < 1. \quad (144)$$

Otherwise split the dimension budget, rather than the incurred cycle charge, according to the value of z . Give a charged box the weight $w_x = d(l + z) = L + \log_2 a(x)$, and let $W_{\text{hi}}, W_{\text{lo}}$ be the sums of these weights over $z \geq 0.49$ and $z < 0.49$, respectively. The core-charge estimate gives $W_{\text{hi}} + W_{\text{lo}} \leq (1 + o(1))F_\lambda$; the omitted shallow boxes and the other error terms cost $o(F_\lambda)$ after the fixed bottom cutoff has been chosen large enough. The two ratios following (143) therefore imply

$$0.4995F_\lambda < 0.50001W_{\text{hi}} + 0.491W_{\text{lo}} + o(F_\lambda) \leq 0.491F_\lambda + 0.00901W_{\text{hi}} + o(F_\lambda).$$

Since $0.0085/0.00901 > 0.943$, it follows, with room for those errors, that $W_{\text{hi}} \geq 0.93F_\lambda$. In particular some box has $a \geq n^{0.49}$, so the diagram contains at least $a(a - 1)$ boxes below its first row and $k \geq n^{0.97}$ for large n . In the lower portion, from log size $d/2$ to $0.9d$, each box costs at most $0.2d$. Its cost-to-budget ratio on the high set is thus at most $0.2/0.49$, while on the low set the full bound 0.50001 still applies. The total lower cost is at most

$$\frac{0.2}{0.49}W_{\text{hi}} + 0.50001W_{\text{lo}} + o(F_\lambda) \leq \left\{ 0.93 \frac{0.2}{0.49} + 0.07(0.50001) + o(1) \right\} F_\lambda < 0.418F_\lambda.$$

In the upper portion use (133), testing (141) at each new frontier as specified above.

As long as (141) has not fired, (134) and monotonicity for cut operators give

$$\sum_v \log_2 r_v = 2 \sum_v \log_2 \text{Tr } M_v - \sum_v \log_2 t_v \leq (0.2 + 2\delta)F_\lambda.$$

At size $m \geq n^{0.9}/2$, the additive $m^{0.9}$ terms in (133), summed over at most n/m nodes, are $O(nm^{-0.1}) = O(n^{0.91})$. Summing over these dyadic sizes keeps this bound, which is $o(F_\lambda)$ because $F_\lambda \geq bn^{0.97}$. The type and allocation overheads are also $o(F_\lambda)$ by the previous bounds. The remaining inverse-log-size sum gives an upper-portion cost of at most

$$\frac{1.2 + 2\delta}{2(0.88)} \ln(1/0.9)F_\lambda + o(F_\lambda) < 0.076F_\lambda. \quad (145)$$

The factor $1/2$ is the averaging of the two child bounds; the logarithm is the integral of inverse log size across this portion. If the diagnostic never fires, the total charge is below $(0.418 + 0.076)F_\lambda = 0.494F_\lambda$. If it fires, test it on the newly created frontier before removing any newly sparse children. Previously stopped sparse nodes have already paid their edge offsets and retain nonpositive contributions. At the triggering frontier the actual sum $\sum_v \log_2(D_v t_v)$

is below $0.8F_\lambda$. Use (142) immediately with that sum. The density-cycle estimate cannot be charged on this triggering level, because its diagnostic hypothesis has failed there. Use the crude Littlewood–Richardson bound instead. Its core part charges boxes with $\log_2 a$ of order d , so (126) bounds the single-level charge by $O(F_\lambda/d) = o(F_\lambda)$. For newly sparse children, the additional bound in (132) is also $o(F_\lambda)$, since there are at most n/m children of size $m \geq 2^{.9d-O(1)}$ and $F_\lambda \geq bn^{.97}$. The total is therefore below $0.876F_\lambda + o(F_\lambda)$. Both cases leave fixed slack below one. In every case we have $\log_2(D_\lambda t_\lambda) \leq (1 - \varepsilon_0)F_\lambda$ for some fixed $\varepsilon_0 > 0$, at all sufficiently large active roots. Thus $t_\lambda \leq 2^{-\varepsilon_0 F_\lambda}$. Since $T_n P_{10}^D T_n^*$ has norm $\|T_n P_{10}^D\|^2$ and t_λ is its $2p_0$ moment, taking a $4p_0$ -th root proves (137). Its constants and size threshold are independent of the exponent used in the subsequent norm induction. $\square$

11.5 Closing the norm induction without a circular base

Write $P_D = P_{10}^D$ and $P_C = P_{10}^C$. With $T_n = CD$,

$$T_n = T_n P_D + P_C T_n (I - P_D) + (I - P_C) CD (I - P_D).$$

Proposition 11.3 bounds the first two terms by $2^{-c_1 F_\lambda}$ each. The central projections commute with their own segments, $P_D D = D P_D$ and $P_C C = C P_C$. On a high product type, the child dimension-power estimates multiply, and its product log dimension exceeds X . Thus, if all child estimates hold with exponent g , the last term has norm at most

$$\|(I - P_C)C\| \|D(I - P_D)\| \leq 2^{-2gX} = 2^{-1.00006gF_\lambda}.$$

This uses no commutation between P_C and P_D . We need a size threshold independent of the final small exponent. A crude polynomial gap supplies it. For a unit vector v , put $\varepsilon = 1 - \|T_n v\|^2$. If v_i is the vector after i matching projections, orthogonality gives $\sum_i \|v_i - v_{i-1}\|^2 = \varepsilon$. Hence v is within $\sqrt{d\varepsilon}$ of every matching-fixed space. Each edge transposition in that matching moves v by at most $2\sqrt{d\varepsilon}$. Any transposition is a product of at most $2d - 1$ cube-edge transpositions along a cube path, and any permutation is a product of at most $n - 1$ arbitrary transpositions. Telescoping therefore bounds $\|\pi v - v\|$ by $Cnd\sqrt{d\varepsilon}$. The average of πv over S_n is zero on a nontrivial irreducible, so $1 \leq Cnd\sqrt{d\varepsilon}$. Together with $F_\lambda \leq n \log_2 n$, this supplies an exponent an^{-C_0} in the bound $\|T_n\| \leq 2^{-gF_\lambda}$, for absolute $a, C_0 > 0$; dimension-one nontrivial types are annihilated.

Fix temporarily a desired exponent $0 < c \leq c_1/2$. At a parent of size n , induction and this independent gap permit a common child exponent

$$g = \max\{c, \min(c_1/2, an^{-C_0})\}.$$

Thus $g \leq c_1/2$ and $g \geq a'n^{-C_0}$ for an absolute $a' > 0$. Relative to 2^{-gF_λ} , the saving in the high-high term is $1 - 2^{-.00006gF_\lambda}$. It is bounded below by an inverse polynomial in n , whereas the two low terms have relative size at most $2 \cdot 2^{-(c_1/2)F_\lambda}$. On active roots, $F_\lambda \geq bn^{.01}$, so one absolute large-size threshold absorbs the low terms for every such c . Below that threshold choose $c > 0$ small enough to fit every finite nontrivial norm gap, and also small enough for (125) on sparse roots, where $F_\lambda = O(k \log n)$. This closes the induction without a circular choice of its base and proves Theorem 11.1.

For completeness, let μ_q be the permutation law after q sweeps and define its squared relative L^2 deviation by $\chi^2 = n! \sum_{g \in S_n} |\mu_q(g) - 1/n!|^2$. The regular Fourier consequence is

$$\chi^2 \leq \sum_{\lambda \neq (n), T_n(\lambda) \neq 0} D_\lambda^2 2^{-2qcF_\lambda} = o(1) \quad (146)$$

for a sufficiently large absolute q . There are at most $2^{O(\sqrt{k} \log(k+2))}$ types at level k , $F_\lambda \geq bk$, and $F_\lambda \rightarrow \infty$ at each fixed positive k . These facts justify the last limit and give the same bound from every starting deck by translation.

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